EUROPEAN
COMMISSION

Brussels, 18.4.2024
SWD(2024) 62 final

COMMISSION STAFF WORKING DOCUMENT

IMPACT ASSESSMENT REPORT

Accompanying the document

COMMISSION REGULATION (EU) …/…

implementing Directive 2009/125/EC of the European Parliament and of the Council as regards ecodesign requirements for local space heaters and separate related controls and repealing Regulation (EU) 2015/1188

{C(2024) 1732 final} - {SEC(2024) 124 final} - {SWD(2024) 65 final}

Table of Contents

1. INTRODUCTION: POLITICAL AND LEGAL CONTEXT 7
   1.1. EU energy and climate objectives 7
   1.2. The contribution of Ecodesign and Energy labelling 8
   1.3. Legal context of the impact assessment 9
2. PROBLEM DEFINITION 10
   2.1. What are the problems? 10
   2.1.1. The first key problem: missed in-use energy savings from local space heaters 10
   2.1.2. The second key problem: market failures due to lack of regulatory provisions on circular economy 11
   2.2. What are the problem drivers? 11
   2.2.1. Unclear scope of the LSH Regulation 11
   2.2.2. Restrictive scope of the LSH Regulation 12
   2.2.3. Inadequate definitions concerning **gas and liquid fuel local space heaters** . 13
   2.2.4. Circular economy: insufficient availability of spare parts 13
   2.2.5. Circular economy: lack of rules on upgradability of controllers 13
   2.2.6. Circular economy: incomplete information on the end-of-life of appliances . 13
   2.3. How likely is the problem to persist? 14
3. WHY SHOULD THE EU ACT? 14
   3.1. Legal basis 14
   3.2. Subsidiarity: Necessity of EU action 14
   3.3. Subsidiarity: Added value of EU action 15
4. OBJECTIVES: WHAT IS TO BE ACHIEVED? 15
   4.1. General objectives 15
   4.2. Specific objectives 15
5. WHAT ARE THE AVAILABLE POLICY OPTIONS? 16
   5.1. What is the baseline from which options are assessed? 17
   5.2. Business As Usual (BAU) scenario 17
   5.2.1. Sales and stock 17
   5.2.2. Energy consumption 18
   5.2.3. Environmental aspects 19
   5.2.4. Economic aspects 20
   5.2.5. Social impacts 20
   5.3. Description of the policy options 20
   5.3.1. Policy option 1 (BAU): Business as usual 21
   5.3.2. Policy option 2 (PO2): Clarifying and expanding the scope 22
   5.3.3. Policy option 3 (PO3): defining ecodesign limits for low power modes 23

5.3.4. Policy option 4 (PO4): circular economy 23
5.4. Options discarded at an early stage 25
5.4.1. Voluntary agreement by the industry 25
5.4.2. Mandatory third party certification 26
5.4.3. Modification of the verification tolerances 26
5.4.4. Introduction of stricter ecodesign requirements and NOx emission limits 26
5.4.5. Removing the conversion coefficient (CC) from the calculations of the energy efficiency 27
6. WHAT ARE THE IMPACTS OF THE POLICY OPTIONS? 27
6.1. Environmental impacts 28
6.2. Economic impacts 30
6.2.1. Business turnover 30
6.2.2. Consumer expenditure 30
6.2.3. Costs 31
6.3. Social impacts. Employment 32
6.4. Affordability (payback period) 33
6.5. Burden reduction 33
6.6. SMEs 34
6.7. Geographical impacts 34
6.8. Impacts of current geopolitical instability 35
6.9. Summary of impacts 35
7. HOW DO THE OPTIONS COMPARE? 36
7.1. Effectiveness 36
7.2. Efficiency 37
7.3. Coherence with other EU policies and legislative framework 38
7.3.1. Coherence with the overarching objectives of EU policies 38
7.3.2. Coherence with Article 15(5) of the Ecodesign Framework Directive 38
7.3.3. Coherence with the subsidiarity and proportionality principles 38
7.3.4. Overall coherence assessment 39
8. PREFERRED OPTION 39
8.1. Preferred policy option 39
8.2. REFIT (simplification and improved efficiency) 41
8.3. Application of the ‘one in, one out’ approach 41
8.4. Overview in the context of current energy prices 41
9. HOW WILL ACTUAL IMPACTS BE MONITORED AND EVALUATED? 42
ANNEX 1: PROCEDURAL INFORMATION 44
1. LEAD DG, DECIDE PLANNING/CWP REFERENCES 44
2. ORGANISATION AND TIMING 44
3. CONSULTATION OF THE RSB 44
4. EVIDENCE, SOURCES AND QUALITY 44

ANNEX 2: STAKEHOLDER CONSULTATION (SYNOPSIS REPORT) 45
1. REVIEW STUDY AND STAKEHOLDER CONSULTATION 45
2. CONSULTATION FORUM 45
3. STAKEHOLDER COMMENTS DURING AND AFTER THE CONSULTATION FORUM 45
4. PUBLICATION OF THE CALL FOR EVIDENCE AND STAKEHOLDERS FEEDBACK 50
5. SECOND MEETING OF THE CONSULTATION FORUM ON 10 MARCH 2022 50
ANNEX 3: WHO IS AFFECTED AND HOW? 51
1. PRACTICAL IMPLICATIONS OF THE INITIATIVE 51
2. SUMMARY OF COSTS AND BENEFITS 51
3. RELEVANT SUSTAINABLE DEVELOPMENT GOALS 53
ANNEX 4: ANALYTICAL METHODS 54
1. SALES 54
2. STOCK 56
3. TURNOVER AND EMPLOYMENT 58
4. DATA INPUT 59
4.1. Product price 60
4.1.1. On price elasticity 61
4.2. Energy data input: EU Reference Scenario 2020 (REF2020) 61
4.2.1. Fuel costs 62
4.2.2. Fuel emission factors 62
4.3. Employment factors 63
4.4. Mark-up factors 63
4.5. Use hours 63
4.6. Average heat output 65
4.7. Average efficiency 65
4.8. Material input 66
5. MAIN ASSUMPTIONS FOR THE MODELLING 68
ANNEX 5: ADDITIONAL INFORMATION ON THE IMPACTS OF THE MEASURES 70
1. FURTHER INSIGHT ON THE ANALYSIS OF THE IMPACTS ASSESSED IN SECTION 6 OF THE IA 70
1.1. Environmental impacts 70
1.1.1. In-use energy consumption 70
1.1.2. Primary energy consumption 71
1.1.3. Greenhouse gas emissions reduction 71
1.2. Economic impacts 72

1.2.1. Turnover 72
1.2.2. Consumer expenditure 73
2. RESOURCE efficiency ANALYSIS 73
2.1. Water consumption savings 73
2.2. Waste reduction 74
ANNEX 6: ANALYSIS OF CURRENT ECODESIGN REQUIREMENTS FOR LOCAL SPACE HEATERS 76
1. EFFECTIVENESS 76
1.1. Energy savings 77
1.2. NOx emissions 79
2. EFFICIENCY 79
3. Relevance 81
ANNEX 7: SOME MEASURES THAT DO NOT ENTAIL ANY DIRECT ENVIRONMENTAL OR ECONOMIC IMPACT 83
1. REPHRASING THE OVERALL DEFINITION OF LOCAL SPACE HEATERS 83
2. CHANGING THE DEFINITION OF AIR-HEATING APPLIANCES 83
3. CHANGE THE DEFINITION OF “ELECTRIC PORTABLE HEATERS” 84
4. REMOVE THE “ELECTRIC RADIANT HEATER” CATEGORY 84
5. CHANGE THE DEFINITIONS FOR LUMINOUS LOCAL SPACE HEATER AND TUBE LOCAL SPACE HEATER 85
6. UPDATE OF THE CC-FACTOR TO 2.1 85
ANNEX 8: SENSITIVITY ANALYSIS 86
1. DATA INPUTS FROM THE MIX SCENARIO 86
2. EFFECT OF FUEL PRICES: CONSUMER EXPENDITURE 87
3. EFFECT OF LOW CARBON ELECTRICITY: GHG EMISSIONS 88
4. EFFECT OF RENOVATION RATES: ENERGY CONSUMPTION, GHG EMISSIONS, AND END USER EXPENDITURE 89
ANNEX 9: SME TEST 91
1. IDENTIFICATION OF THE AFFECTED BUSINESSES 91
2. THRESHOLD QUESTION: TO WHAT EXTENT IS THE INITIATIVE RELEVANT FOR SMES? 91
ANNEX 10: MULTI-CRITERIA ANALYSIS 93
1. MULTI-CRITERIA DECISION ANALYSIS METHODOLOGY: BASIC DESCRIPTION OF SOCIAL MULTI-CRITERIA EVALUATION (SMCE) METHODOLOGICAL FRAMEWORK AND SOFTWARE TOOL 93
2. FIRST STEP: MULTI-CRITERIA ANALYSIS FOR PO2 95
2.1. Justification of the score for the qualitative criteria 96

2.2. Result of the multi-criteria analysis 96
3. SECOND STEP: MULTI-CRITERIA ANALYSIS FOR A PREFERRED POLICY OPTION COMBINING $PO2_F$, PO3 AND PO4A 103
ANNEX 11: THE ECODESIGN AND ENERGY LABELLING FRAMEWORK 108

Glossary

Term or acronym	Meaning or definition
BAU	Business-as-usual (describing a scenario without any further intervention)
CBA	Cost-benefit analysis
CC	Conversion Coefficient
CEAP	Circular Economy Action Plan
CTP	Climate Target Plan
DoC	Declaration of Conformity
EED	Energy Efficiency Directive
EPBD	Energy Performance of Buildings Directive
EPREL	European Product Registry for Energy Labelling
ESPR	Ecodesign for Sustainable Products Regulation
GAR	Gas Appliances Regulation
GHG	Greenhouse gas
IA	Impact Assessment
IPCC	Intergovernmental Panel on Climate Change
HDD	Heating Degree Days
LSH	Local Space Heater
MEErP	Methodology for the Ecodesign of Energy-related Products
MSA	Market Surveillance Authority
MTOE	Million tonnes of oil equivalent
NCEP	National and Climate Energy Plans
NGO	Non-Governmental Organisation
RED	Renewable Energy Directive
REF2020	EU Reference Scenario 2020
R&D	Research and Development
SDGs	Sustainable Development Goals
SMCE	Social Multi-criteria Evaluation
SME’s	Small and Medium Enterprises
UN	United Nations

1. INTRODUCTION: POLITICAL AND LEGAL CONTEXT

1.1. EU energy and climate objectives

The EU has longstanding objectives to increase energy efficiency and to reduce its greenhouse gas emissions. These go along with other objectives to reduce its environmental impacts. In December 2019, the Commission presented the European Green Deal[1] to strengthen these objectives and as the cornerstone of its strategy to fulfil the United Nation's 2030 Agenda for Sustainable Development[2]. In September 2020, it presented a Climate Target Plan (CTP) for 2030[3], showing the need for a higher contribution of energy efficiency and renewable energy to achieve a net 55% GHG emission reduction most cost-effectively, in line with the Paris Agreement. The European Parliament and Council subsequently agreed to achieve this level of reduction in GHG emissions. The Commission followed this by adopting the EU "Fit for 55" package[4] aiming to achieve the neceessary cut in GHG emissions.

One pillar of the CTP and subsequently the 'Fit for 55' package is energy efficiency. A revision of the Energy Efficiency Directive (EED)[5] to increase the overall ambition was proposed. This would require Member States to enhance their National Energy and Climate Plans (NECPs) to achieve at least an overall 9% reduction in EU energy consumption in 2030 compared to the reference[6]. The EED itself contains limited energy saving measures and in addition to Member State actions, additional measures at EU level are needed[7]. In this context, the ecodesign and energy labelling rules for products arise as important instruments to realise EU's energy and decarbonisation objectives[8].

Another pillar of the European Green Deal is a more circular economy. The new Circular Economy Action Plan[9] sets out steps to work towards this. It aims to reduce product environmental impacts for example through promoting longer product lives, greater resource efficiency and enhancing recycling and recycled content. Certain of these aspects can be addressed through measures under the Ecodesign Directive since decisions made at design can influence the product's life span and ease of maintenance, repair, reuse, upgrade, recyclability and waste handling. The Ecodesign Working Plan 2016-2019 identified ecodesign measures' potential to contribute to circular economy objectives and subsequent supporting studies have systematically considered resource efficiency aspects.

Reducing energy use and promoting the circular economy are also important for many other reasons[10]. Lower energy use reduces the EU's energy import dependence and improves energy security, aspects that are particularly relevant in the current context of continuous increase of energy prices. On 8 March 2022, the Commission published its Communication "RepowerEU:

---

[1] The European Green Deal, COM(2019) 640 final.
[2] Transforming our world: the 2030 Agenda for Sustainable Development.
[3] Stepping up Europe's 2030 climate ambition. Investing in a climate-neutral future for the benefit of our people (COM/2020/562 final).
[4] 'Fit for 55': delivering the EU's 2030 Climate Target on the way to climate neutrality, COM(2021) 550 final.
[5] Proposal for a Directive of the European Parliament and of the Council on energy efficiency, COM(2021) 558 final.
[6] EU reference scenario 2020 - Publications Office of the EU (europa.eu).
[7] These additional measures include not only ecodesign and energy labelling but also the EU ETS, vehicle CO_2 legislation and the Energy Performance of Buildings Directive.
[8] An explanation of the framework legislation is provided in annex 11.
[9] Circular Economy Action Plan for a more competitive Europe, COM(2020) 98.
https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52015DC0614&from=EN.
[10] https://www.odyssee-mure.eu/data-tools/multiple-benefits-energy-efficiency.html.

Joint European action for more affordable, secure and sustainable energy"[11], one of its pillars is the reduction of EU dependence on fossil fuels by energy savings and boosting energy efficiency.

Ecodesign and energy labelling contribute to improved air quality and support energy system integration and the alleviation of energy poverty. It is also expected to lead to the creation of jobs, encourage innovation and support and facilitate economic growth[12]. Both energy savings and promoting the circular economy lead to reduced environmental damage from materials extraction and resource efficiency. Most of these co-benefits are difficult to quantify, but are well known and perceived by society.

In March 2022 the Commission adopted the ESPR (Ecodesign for Sustainable Products Regulation) a proposal to replace the Ecodesign Framework Directive[13], which will expand the scope of ecodesign from energy-related products to any kind of physical good that is placed on the market, with some exceptions that are set out in the ESPR itself. Ecodesign product regulations adopted under the existing Directive would nevertheless continue in force until repealed. When the proposed legislation is eventually adopted future amendments to existing product regulations would be carried out under its framework. The adoption of the ESPR is expected to have little influence on the results of this IA, as it is further explained in section 5.

1.2. The contribution of Ecodesign and Energy labelling

There are market barriers that hamper the introduction and uptake of more energy efficient and more long lasting products. It is desirable for policies to address these barriers since they lead, from the point of view of the consumer and society, to unrealised economically viable energy savings. Behavioural failures refer to the cognitive limitations and biases that prevent consumers and investors to appreciate rationally the benefits of energy efficiency[14]. Market failures arise from the fact that many impacts and aspects of energy supply or resource use are not brought into market prices[15]. Market barriers such as lack of information and awareness or financing challenges result in economically rational energy savings not being realised or more sustainable choices not being made.

The EU Ecodesign Directive aims to address these barriers by setting performance requirements to remove the worst performing products from the market. The assessment aims to ensure that the minimum requirements are set at the level of Least Lifecycle Consumer Cost. These harmonised requirements have specifically addressed energy efficiency and have led to significant reductions in energy use of household and industrial products. Being set at EU level, they have avoided subjecting industry to multiple national rules.

Energy labelling rules complement ecodesign requirements. These provide information to consumers with the aim of encouraging them to purchase products that have a better energy performance than the minimum. This has helped consumers to reduce their energy bills by easily identifying and comparing more energy efficient appliances. Nearly 80 % of the EU public recognise the label and say it has influenced their purchase decision[16].

---

[11] EUR-Lex - 52022DC0108 - FR - EUR-Lex (europa.eu)
[12] https://combi-project.eu/2018/06/22/combi-results-overview-policy-conclusions/
[13] Directive 2009/125/EC of the European Parliament and of the Council of 21 October 2009 establishing a framework for the setting of ecodesign requirements for energy-related products (OJ L 285, 31.10.2009, p.10)
[14] DellaValle N., Bertoldi P. (2021) "Toward a more situated energy efficiency policy agenda".
[15] E.g. impacts on air pollution, biodiversity, resource use, climate change and energy security
[16] Special Eurobarometer 492. "Europeans' attitudes on energy policy report. European Commission September 2019.

The impact of these measures are estimated through the Ecodesign Impact Accounting[17]. It is estimated that in 2020 the combined ecodesign and energy labelling measures contributed a quarter of the energy savings and a third of the greenhouse gas emissions reductions to achieve the EU's 20-20-20 climate and energy policy goals.

The benefits of ecodesign and energy labelling are not limited to the EU energy and environmental goals but they also contribute to some of the 17 Sustainable Development Goals (SDGs) adopted by the UN. By reducing in-use energy consumption and the subsequent environmental impacts such as GHG emissions or acidification, ecodesign and energy labelling contribute to SDG 7 (affordable and clean energy), SDG 8 (Decent work and economig growth), SDG 10 (reduced inequalities), SDG 11 (sustainable cities and communities) and SDG 13 (climate action). In particular the following targets included in the abovementioned SDGs are addressed: target 7.3 (double the improvement of energy efficiency), target 8.4 (improve resource efficiency in consumption and production), target 10.1 (reduce income inequalities) by reducing the consumer's energy bill and notably for those in situation of energy poverty, target 11.6 (reduce the environmental impact of cities) and target 13.1 (strengthen resilience and adaptative capacity to climate-related hazards). In addition, circular economy measures under ecodesign also contribute to the following SDGs and targets: the abovementioned targets 8.4 and 11.6 by lengthening the lifetime of the product, reducing waste and increasing recyclability and SDG 12 (responsible consumption and production) and in particular its targets 12.2 (sustainable management and use of natural resources) and 12.5 (substantially reduce waste generation).

Annex 3 further elaborates on SDGs, explaining how the proposed measures contribute to them and providing a quantitative estimation of that contribution.

1.3. Legal context of the impact assessment

Following an assessment and preparatory study, local space heaters were regulated under the Ecodesign Framework Directive through Commission Regulation (EU) 2015/1188 on ecodesign requirements for local space heaters[18] (hereafter the LSH Regulation). At the time of adoption, the ecodesign and energy labelling requirements for local space heaters were estimated to achieve energy savings of approximately 3,8 Mtoe and related GHG emission reductions of 6,7 MtCO2eq by 2020.

Article 7 of the LSH Regulation, mandates a review of the Regulation, in particular regarding the possibility to set stricter requirements for energy efficiency and NOx emissions, the modification of the verification tolerances of Annex IV, the validity of the correction factors and the appropriateness of introducing third-party certification. Consequently a review of the LSH Regulation was included in the Ecodesign Working Plan 2016-2019[19]. This noted that the review would also assess aspects relevant to the circular economy. The review has been carried out through a review study[20] and this impact assessment[21].

---

[17] Ecodesign impact accounting annual report 2020

[18] Commission Regulation (EU) 2015/1188 of 28 April 2015 implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to ecodesign requirements for local space heaters (OJ L 193, 21.7.2015, p. 76)

[19] Communication from the Commission Ecodesign Working Plan. COM(2016) 773 final, Brussels, 30 November 2016. (Ecodesign Working Plan 2016-2019)

[20] Review study on Local Space Heaters, Final Report, Viegand Maagøe and Danish Technical Institute, May 2019 for the European Commission. Final report _ Local space heaters - VM15052019.pdf (applia-europe.eu).

2. PROBLEM DEFINITION

2.1. What are the problems?

2.1.1. The first key problem: missed in-use energy savings from local space heaters

The EU Reference Scenario 2020 projects energy consumption to be reduced to 1.124 Mtoe of primary energy and 864 Mtoe of final energy in 2030, which is insufficient to attain the energy efficiency target under the fit-for-55 package. As indicated above, the draft revision of the EED sets out an additional binding reduction of 9% of energy consumption in the EU from energy efficiency measures, which would bring the energy consumption down to 1.023 Mtoe of primary energy and 787 Mtoe of final energy by 2030.

The revised EED will mandate Member States to lay down measures at national level to collectively contribute to the EU binding target, thus underscoring the importance of co-responsibility among Member States to reach the global EU energy and climate goals. This approach entails the risk that some Member States may not fully realise their national objectives and therefore jeopardise the achievement of the 9% target. Ecodesign and energy labelling measures (which unlike measures in the scope of the EED have the advantage of being directly applicable across the EU) must contribute to that 9% by reducing energy consumption on top of the reductions achieved under EED.

The modelled in-use energy consumption from **local space heaters** was 15 Mtoe in 2020[22], 1,7% of EU total final energy consumption and 4% of the final energy consumption of households and services. In view of this it is desirable to maximise further cost-effective increases in the energy efficiency of local space heaters.

Section 1.5 of the review study indicates that there could be potential for further energy efficiency gains through addressing a number of aspects of the current LSH Regulation. In this respect, the review study concludes that the LSH Regulation presents barriers that impede gaining further energy savings because the legislative scope does not clearly delimit the types of local space heaters that must fulfil the ecodesign requirements.

The contribution of local space heaters to EU energy efficiency objectives can be enhanced by clarifying and moderately broadening the scope of the legislation, without technological improvement. In fact, section 4.5.3 of the review study identifies that technology is mature and unlikely to yield significant energy efficiency improvements, in particular for electric heaters. Options for achieving better efficiency all relate to fitting the heaters with suitable controllers. Further, section 6 of the review study assesses the possibilities for energy efficiency improvements, which may come from either automatic programming or control accuracy, both linked to the use of controllers but not to the improvement of the heating technology in itself. There is therefore little to

---

[21] The current review of the LSH Regulation does not consist of a back-to-back evaluation and impact assessment according to the Better Regulation Guidelines. This is because the review of the LSH Regulation started in 2017, and at that time the commonly used methodology for the reviews of the legislation on ecodesign and energy labelling consisted of a review study and a later impact assessment. The approach was known and accepted at the time, and has also been used for a number of other reviews still in the pipeline for consideration by the Board in 2022 and 2023. However, DG ENER has decided to gradually change this practice so that new reviews starting as of 2022 will align fully to the standard Better Regulation process, and will integrate what was formerly known as review studies in back-to-back evaluations and impact assessments. The first case where this new approach will be used is the review of the External Power Supply Regulation.

[22] From EU Reference Scenario 2020

do in terms of technological evolution, and priority should be given to removing the abovementioned legislative barriers rather than on setting new energy efficiency limits, to ensure that current heaters are coupled with the appropriate controllers.

Besides preventing the realisation of energy savings, the lack of clarity of the scope results in unnecessary burden for manufacturers and Market Surveillance Authorities (MSAs). For manufacturers, the unclear scope generates competitive disadvantages between manufacturers whose products are compliant with the ecodesign rules and those whose products are not, resulting in extra costs for compliant manufacturers and hindering access to the market in equal conditions. These excess costs are quantified in section 6.5. On the side of MSAs, there is loss of time and resources on the selection of the products to be controlled because of the uncertainty about the inclusion or not in the scope of the LSH Regulation of specific product categories like underfloor heaters. Both industry and Member States declare that the current scope of the LSH Regulation is a source of confusion among manufacturers on the one hand, which are not sure about whether their products must or not fulfil the ecodesign rules, and national authorities on the other hand, which are frequently engaged in discussions with manufacturers about whether their products are in the scope of the LSH Regulation and would consequently be subject to market surveillance.

2.1.2. The second key problem: market failures due to lack of regulatory provisions on circular economy

The lack of regulatory provisions on circular economy lead to negative environmental effects that are not internalised in the price of local space heaters. This means that industry does not receive sufficiently strong signals to incorporate resource efficiency criteria in the design and manufacture of their products. In addition, circular economy and climate and energy policies like energy efficiency, have not always been well integrated. For instance, the IPCC's Summaries for Policy Makers hardly mentioned circularity or resource efficiency as a decarbonisation option, and the first Circular Economy Action Plan of 2015 vaguely mentioned climate change while acknowledging that "to date ecodesign requirements have mainly targeted energy efficiency". Nowadays, scientific literature has demonstrated the connection between circular economy, energy savings and climate change, thanks to studies like Material Economics' and Sitra's, entitled *The circular economy – A powerful force for climate mitigation* (2018)[23], from the UN's International Resources Panel[24][25] and the Commission's analysis for the Long-Term Strategic Vision to reduce GHG emissions. Now these interlinkages are acknowledged at both scientific and policy levels, notably under the European Green Deal and its initiatives (e.g. Renovation Wave, EPBD revision).

2.2. What are the problem drivers?

2.2.1. Unclear scope of the LSH Regulation

Slave heaters are defined in the LSH Regulation as local space heaters that **are not capable of autonomous operation** and need to receive signals from an external master controller that is not part of the product. As explained in section 1.5.2 of the review study, slave heaters are explicitly

[23] Material Economics and SITRA (2018) *The circular economy – A powerful force for climate mitigation.* Stockholm, Material Economics Sverige AB. In: https://media.sitra.fi/2018/06/12132041/the-circular-economy-a-powerful-force-for-climate-mitigation.pdf

[24] UNEP (2017) *Resource Efficiency: Potential and Economic Implications*. Nairobi: UN Environment. In: http://www.resourcepanel.org/reports/resource-efficiency

[25] IRP (2020) *Resource Efficiency and Climate Change: Material Efficiency Strategies for a Low-Carbon Future*. Hertwich, E., Lifset, R., Pauliuk, S., Heeren, N. A report of the International Resource Panel. Nairobi: United Nations Environment Programme.

excluded from the scope of the LSH Regulation based on the assumption that they would be connected to general heating systems (e.g. for an entire building) by means of central controllers that incorporate the necessary functions to make the full system compliant with the LSH Regulation. Nevertheless, the exclusion of slave heaters from the scope of the LSH Regulation is a source of confusion and conflict between manufacturers and MSAs.

First, the review study points out in section 1.5.2 that stakeholders have different opinions on whether **underfloor heaters** are slave heaters and should therefore be excluded from the scope of the LSH Regulation. Contrary to the definition of slave heaters, many underfloor heaters such as heating cables, do not get any signal from the controller because they are simple resistors that do not have the capacity to receive and interpret any electronic signal, but are just connected to a thermostat that autonomously switches on and off the current. These underfloor heaters are therefore not slave heaters, but they are placed on the market as such and are therefore exempted from the ecodesign measures.

Second, there are also doubts about whether **electric towel heaters**, also called towel rails, are in the scope of the LSH Regulation. This is because the LSH Regulation considers local space heaters as devices whose purpose is to ensure the thermal comfort of human beings. As explained in section 1.5 of the review study, towel rails are not only intended to heat towels but also to heat the space in which they are placed. In reality, most towel heaters are designed with enough power to easily heat most bathrooms without any additional heat source. There are national differences, but particularly in Member States with high shares of electric heating like France, towel heaters fulfil this double purpose of heating towels and room at the same time. However, only around 10%[26] of towel rails are compliant because the majority of products are declared as intended exclusively to heat towels and therefore do not fulfil the ecodesign rules. Apart from the missed energy savings, section 1.5.2 of the review study indicates that this situation creates a distortion between those manufacturers that consider the product in the scope and those for whom towel rails are outside of the scope.

Finally, many local space heaters are placed on the market separately from the controller just to escape the application of the LSH Regulation.

2.2.2. Restrictive scope of the LSH Regulation

The LSH Regulation excludes tube heaters[27] above 120 kW heat output power. As explained in section 1.5.3 of the review study there is a historic reason for this exclusion, namely that the LSH Regulation follows the same approach as the EN 416 standard, which also excluded tube heaters above 120 kW before it was amended. Nevertheless, the latest version of the standard now includes tube heaters between 120 and 300 kW, whose potential energy savings are significant enough to include those devices in the scope of the LSH Regulation, ensuring also coherence with the tests and results set in the standards for these products.

The LSH Regulation is also restrictive with respect to low power modes. The formula for the calculation of the seasonal space heating energy efficiency of local space heaters includes a correction factor called F(4), which accounts for electric heater energy consumption in standby

[26] Estimation from the review study based on stakeholder inputs

[27] Tube heaters are defined in the LSH Regulation as a local space heater using gaseous or liquid fuel, which is equipped with a burner; which is to be installed above head level, near the subjects to be heated, which heats the space primarily by infrared radiation from the tube or tubes heated by the internal passage of products of combustion and of which the products of combustion are to be evacuated through a flue duct.

mode. Nevertheless, according to section 6.5 of the review study, most local space heaters rarely have a mode that can be identified as "standby" as defined in the LSH Regulation, leading the manufacturer to declare F(4) as zero even if many local space heaters feature low power modes other than standby, that are not defined in the LSH Regulation but consume energy anyway and lead to concealed energy consumption. This is the case of **idle mode**, **off mode** and **networked standby mode**[28]. The restriction of the LSH Regulation to standby mode therefore leaves low power modes unregulated.

*2.2.3. Inadequate definitions concerning **gas and liquid fuel local space heaters***

As explained in section 1.5.3 of the review study, some stakeholders have claimed that the terms used to define the different types of gas and liquid heaters are not aligned with the market. In this respect, the LSH Regulation differentiates between "open fronted" and "closed fronted" as being equivalent to open combustion and closed combustion respectively. But they are not the same. A heater can be closed fronted but feature open or closed combustion, depending on whether the air necessary for combustion is taken from the heated room or from the open air through a flue duct. In fact, the definition of open fronted given in the LSH Regulation has led some manufacturers to place on the market as open fronted, heaters that are in reality closed fronted/open combustion, whose energy efficiency requirements should be more stringent The inadequacy of the definitions results therefore in lost energy savings.

2.2.4. Circular economy: insufficient availability of spare parts

A 2016 study carried out for DG Environment[29] shows that the availability and prices of spare parts are among the major barriers to reparability. Spare parts are usually not available or it takes a long time to deliver them, creating the incentive to get rid of the product and replace it instead of repairing it. Some producers deliver spare parts only to their partners, thus hindering competition among the latter and independent repairers.

2.2.5. Circular economy: lack of rules on upgradability of controllers

Used but well-functioning heaters may be unnecessarily replaced because their controllers and functions are not upgradeable. Redesigning products whose controllers can be replaced by more functional and efficient ones, would diminish the number of old devices that are scrapped and would also improve the energy efficiency of those old devices. No advances in the heater technology itself are expected, but future efficiency gains could be realised due to more efficient controllers, if it was possible to upgrade the controller.

2.2.6. Circular economy: incomplete information on the end-of-life of appliances

Analyses carried out during the review study show benefits of providing recyclers with sufficient and good quality information related to the recycling and disposal of heaters, in order to maximise material recovery and therefore reduce the costs of recycling.

According to the review study, local space heaters are rather simple devices, and their material composition consists mostly of different types of metals. Metals are highly recyclable compared to other materials, which gives a high recycling rate for local space heaters, above 87%. In spite of

---

[28] As defined in commission Regulation (EU) No 801/2013, *OJ L 225, 23.8.2013, p. 1–12*

[29] Monier, V., et. al., Study on socioeconomic impacts of increased reparability Final report, Deloitte the EC DG ENV, 2016.

being relatively easier to recycle than other appliances, local space heaters are becoming gradually more complex in terms of electronics, and detailed information for the recovery of electronic components should be provided. Electronic components are particularly interesting for recycling because they contain high added value materials.

2.3. How likely is the problem to persist?

Further energy savings from local space heaters do not depend on breakthrough technological improvements that would naturally replace the existing stock, but rather on reducing energy consumption by fitting heaters with suitable controllers that already exist in the market. In the absence of an amendment to the LSH Regulation, in-use energy savings will stagnate because out-of-scope products will continue to be fitted with unsuitable controllers impeding further reduction of energy consumption.

As far as circular economy, the repair activity of local space heaters is currently inexistent according to stakeholder information and there is no driver that could reverse this situation in the coming years.

It can therefore be concluded that the problems will persist in the absence of any intervention.

3. WHY SHOULD THE EU ACT?

3.1. Legal basis

Article 114 of the Treaty on the Functioning of the European Union (TFEU) is the legal base for measures related to the functioning of the internal market and therefore constitutes the legal base for the Ecodesign Framework Directive[30] and the derived legislation such as the LSH Regulation.

In addition, the Ecodesign Framework Directive lays down a set of criteria to identify energy-related products that may be subject to ecodesign measures. According to the review study, local space heaters are eligible on the basis of the potential energy savings that can still be achieved through the introduction of additional cost-effective measures.

3.2. Subsidiarity: Necessity of EU action

The LSH Regulation hinders the level playing field for manufacturers across the EU by creating a competitive disadvantage between those manufacturers that fulfil the ecodesign rules in force and those who do not. The good functioning of the internal market for local space heaters needs to be assured through EU action by modifying the scope of the LSH Regulation. Otherwise Member States could make different interpretations of that scope and could hinder the placing of specific products on their respective national markets on the basis of their diverging interpretations.

Further, the contribution of ecodesign to the Green Deal objectives, in particular to the achievement of carbon neutrality by 2050 and the mobilisation of industry for a clean and circular economy, would be undermined if collective EU action is not taken on the efficiency of energy-related products such as local space heaters.

[30] Directive 2009/125/EC of the European Parliament and of the Council of 21 October 2009 establishing a framework for the setting of ecodesign requirements for energy-related products (OJ L 285, 31.10.2009, p.10)

3.3. Subsidiarity: Added value of EU action

Ecodesign measures contribute to the global EU energy and climate objectives by setting requirements that influence the characteristics of energy related products. In this respect, the approach from ecodesign requirements is different than the approach from other EU energy and climate legal acts such as EED, EPBD[31] and RED[32], which are based on the implementation by Member States of solutions based on existing technologies, but do not directly require the development of the technologies themselves. Direct EU action through ecodesign rules reinforce therefore the actions taken at national level. Ecodesign measures can only be addressed at EU level since they act directly on the product, where uniformity needs to be assured to avoid different national rules undermining the EU internal market for that product.

In addition, the modification of the LSH Regulation can enhance the efficiency of MSAs. Ifthe scope of the LSH Regulation is clarified, MSA's will spend less time selecting the products subject to inspection and will not need to discuss with industry the application of the ecodesign rules to specific product sub-groups.

4. OBJECTIVES: WHAT IS TO BE ACHIEVED?

4.1. General objectives

The general objective is to contribute to the EU's energy and environmental objectives set in the TFEU. There are various possible synergies within this overall objective. The TFEU states that the aim of the EU energy policy includes ensuring security of energy supply and promoting energy efficiency and energy saving. Lower energy consumption is usually accompanied by a reduction in other environmental impacts including GHG and pollutant emissions as well as lower resource use and extraction.

4.2. Specific objectives

The specific objectives are aimed at correcting the problems identified in section 2:

1) Capture in-use cost-efficient energy savings from local space heaters ;
2) Contribute towards a circular economy by addressing premature obsolescence of local space heaters;
3) Reduce administrative burden in relation to ecodesign of local space heaters.

There are possible trade-offs between these objectives. The main ones are:

- Measures to increase energy efficiency might require more materials to be used for the products or materials that have bigger impacts. These trade-offs will be captured by the EcoReport tool[33]

---

[31] Directive (EU) 2010/31/EU of the European Parliament and of the Council of 19 May 2010 on the energy performance of buildings (OJ L 153, 18.6.2010, p.13)

[32] Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018 on the promotion of the use of energy from renewable sources (OJ L 382, 21.12.2018, p.82)

[33]The Ecoreport tool is a Commission tool developed in Excel by VHK specifically intended to calculate the environmental impacts of the production and use of energy-related products on the basis of the materials that they incorporate and their quantities
https://www.eup-network.de/fileadmin/user_upload/Produktgruppen/Methodology_prep_study/MEErP_study_by_vhk/20110819_Ecoreport_2011_MEErP.xls

and presented in Annex 5. The effect will probably be small since the use phase dominates the impacts for these appliances.

- Measures to reduce the material consumed in the manufacture of local space heaters might lead to more energy and material consumption in the use phase if they result in a longer lifetime of less efficient local space heaters. However, this effect is likely to be relatively small since there is limited potential for increasing the in-use energy efficiency of these appliances. This is explained in section 7.1.
- Measures to extend the scope of appliances covered by requirements flowing from objective 1 or 2 may lead to a greater administrative burden. Similarly, additional requirements responding to these objectives might add to the burden.

5. WHAT ARE THE AVAILABLE POLICY OPTIONS?

The available policy options focus on the achievement of the general and specific objectives through the implementation of measures that maximise the contribution of local space heaters to energy efficiency without imposing significant burden to manufacturers or costs to end-users, as set out in Article 15(5) of the Ecodesign Framework Directive. *Figure 5-1* shows the overall **intervention logic** and the links between drivers, problems, objectives and measures.

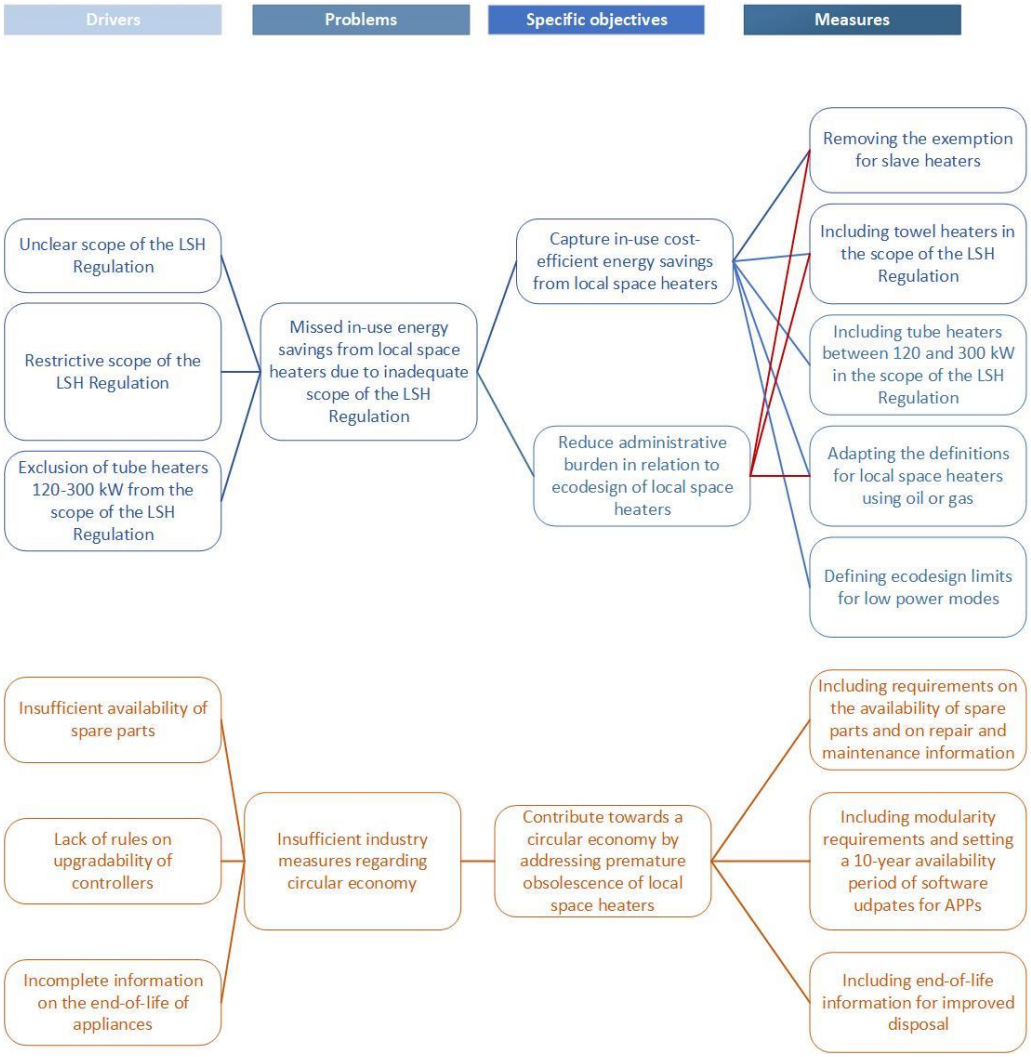


Figure 5-1. Schematic overview of drivers, problems, objectives and measures and their interlinkage

The policy options have been designed on the basis of the input provided in section 7 of the review study, and notably section 7.1.1 on scope clarifications and section 7.1.3 on changing the calculation of the correction factor (F4) accounting for the presence of low power modes. The main focus is on the correction of deficiencies of the current legal framework by eliminating loopholes currently used to sidestep the application of the ecodesign rules. For circular economy measures the input provided in section 6.7 of the review study on resource efficiency options has been taken into account.

5.1. What is the baseline from which options are assessed?

The electricity prices, carbon intensity and building renovation rate that are used as input to this impact assessment are taken from the EU Reference Scenario 2020 (REF2020). REF2020 projects EU and Member States energy developments up to 2050, on the basis of current global and EU market trends and incorporating the effect of the "Clean Energy for All Europeans Package"[34]. Policies that are not legally implemented are not part of REF2020. More details about further input data and analytical models are provided in Annex 4.

As in the review study, it has been assumed that all products in the scope of the LSH Regulation will be compliant with the energy efficiency limits as from the date of application, namely June 2025, but efficiencies will not exceed those minimum requirements. This approach has been confirmed by stakeholders participating in the IA, and is consistent with the expectations for the next years of small efficiency gains per product unit due to the maturity of the current technology.

As indicated in section 1.1, it is expected that the effect of the future ESPR on the baseline will be modest. First, the new products in scope of the ESPR do not consume energy during their use, and therefore the increase in energy in scope will primarily relate to embedded energy (manufacture, distribution and EoL of products), which is only 0,7% of the energy use during the lifecycle of local space heaters. Second, the embedded energy will not all be consumed in the EU but will be shared between EU and non-EU countries depending on the split of production. Third, EU producers are subject to ETS pricing and other regulatory measures that mean that they are likely to have less room for improvement compared to non-EU producers. Finally, the time frame to agree the ESPR, develop and adopt secondary legislation and for it to come in force will mean that substantial impacts could not be expected for the next ten years. All these factors mean that any changes caused by the ESPR to the EU energy baseline for local space heaters can be expected to be minimal. Major differences from the BAU explained in section 5.2 will come from the current situation of high energy prices rather than from the new scope of the ESPR.

5.2. Business As Usual (BAU) scenario

5.2.1. Sales and stock

Under the baseline, sales and stock of **electric** heaters remain rather stable from 2020, around 17 million units and 251 million units respectively. Sales and stock of **gaseous and liquid fuel** heaters have already dropped due to BREXIT and they are expected to go down further due to increasing sales of wood burning stoves, which are outside the scope of the LSH Regulation, as well as many homes replacing gas-fired local heaters with gas central heating. The influence of gaseous and liquid fuel heaters is not very significant because they make up only 0,7% of the stock and 0,4% of sales. The stability of the projected electric heater sales means that business turnover will remain stable until 2050.

[34] Clean Energy for all Europeans Package

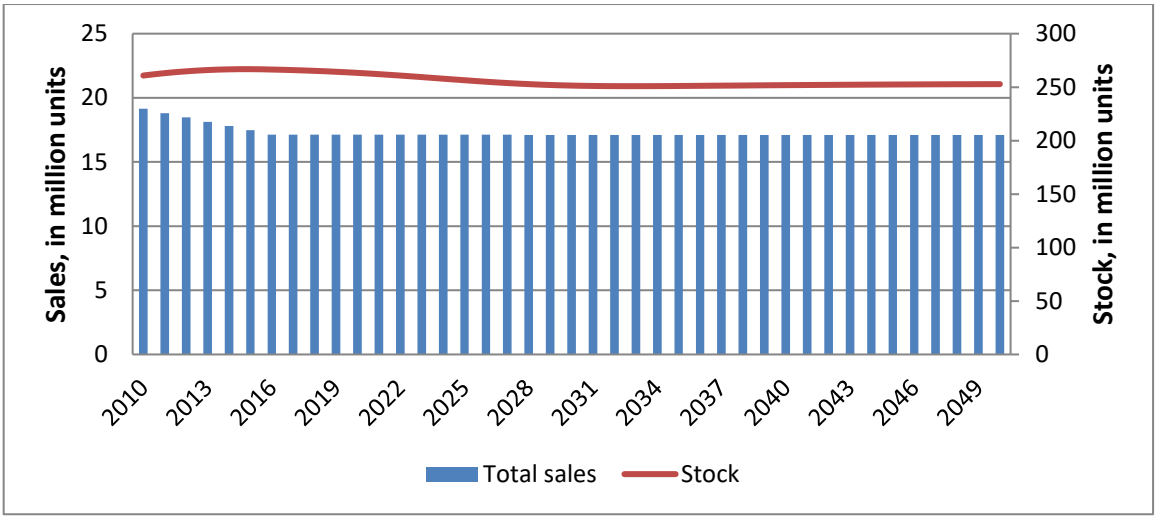


Figure 5-2. Total sales and stock development for all local space heaters from 2010 to 2050

5.2.2. Energy consumption

In-use energy consumption of local space heaters is the energy consumed during the use of the appliance. Under the baseline, in-use energy is expected to decrease 34% between 2020 and 2050, split in two phases. From 2016 to 2030 reduction of energy consumption is largely due to energy efficiency requirements brought by the current LSH Regulation. However, the continued savings after 2030 are primarily caused by a decrease in heating demand as result of the rise of temperatures due to climate change and the better insulation of buildings. The heating load is projected to decrease around 19% from 2030 to 2050, half due to decreased Heating Degree Days (HDDs)[35] and half due to increased average building insulation[36]. The HDD reduction is not expected to modify the number of units sold, but rather the time that heaters are turned on.

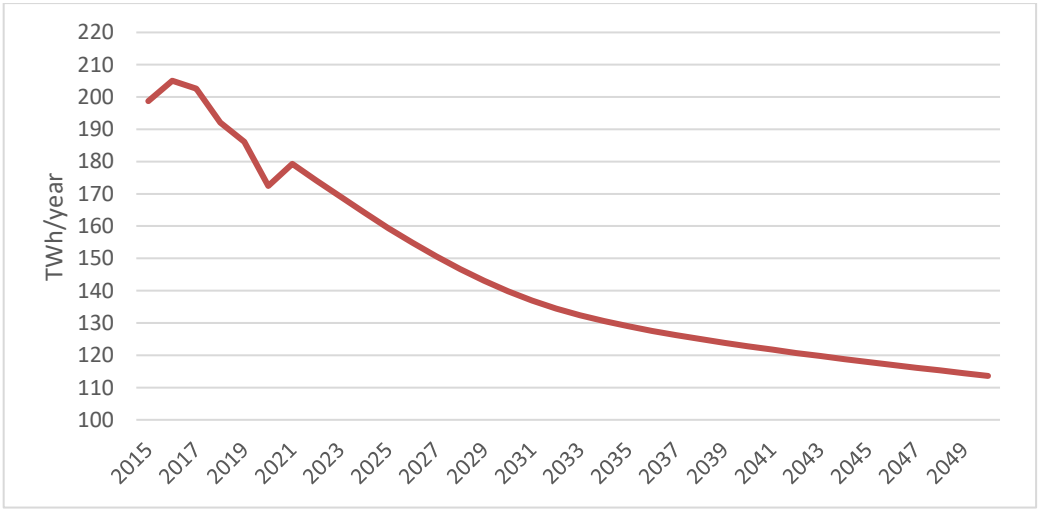


Figure 5-3. In-use energy consumption for all local space heaters in the BAU scenario

Primary energy in turn, accounts for both in-use and embedded energy (energy used during production, distribution and EoL processing). For local space heaters, primary energy follows the same trend as in-use energy, since embedded energy is 0,7% of total energy and barely changes under BAU. In absolute numbers primary energy is approximately twice the in-use energy, as the

[35] Based on Eurostat data: Heating degree days by country - annual data [nrg_chdd_a]
[36] Based on the EU Reference 2020 scenario, Sheet: RESID_Renovation

energy consumption from electric heaters, which accounts for more than 99% of the stock, must be multiplied by a CC=2,1 to account for the energy losses in the conversion process from primary energy to electricity.

In-use energy consumption (TWh/year)				Primary energy consumption (embedded + in-use) in Mtoe/year			
2020	**2030**	**2040**	**2050**	**2020**	**2030**	**2040**	**2050**
172	140	123	114	30	25	22	20

Table 5-2. Expected final and primary energy consumption for BAU

5.2.3. Environmental aspects

5.2.3.1.Greenhouse gas emissions

GHG emissions shown in table 5-2 are for the full lifecycle and include embedded emissions from production, distribution and end-of-life. Under BAU, GHG emissions reduce by 43% from 2020 to 2030 and by 79% to 2050. GHG emissions reductions follow therefore a steeper downward curve than energy reduction commented in 5.2.2, due to the expected reduction in GHG intensity of the electricity mix. GHG emissions reductions level off by 2040 as the major share of carbon-free electricity generation has already been implemented, in contrast with energy consumption that continues reducing.

	GHG emissions embedded and during use [MT CO_2-eq./year]			
	2020	**2030**	**2040**	**2050**
BAU	37	21	11	8

Table 5-2. Expected GHG emissions for BAU

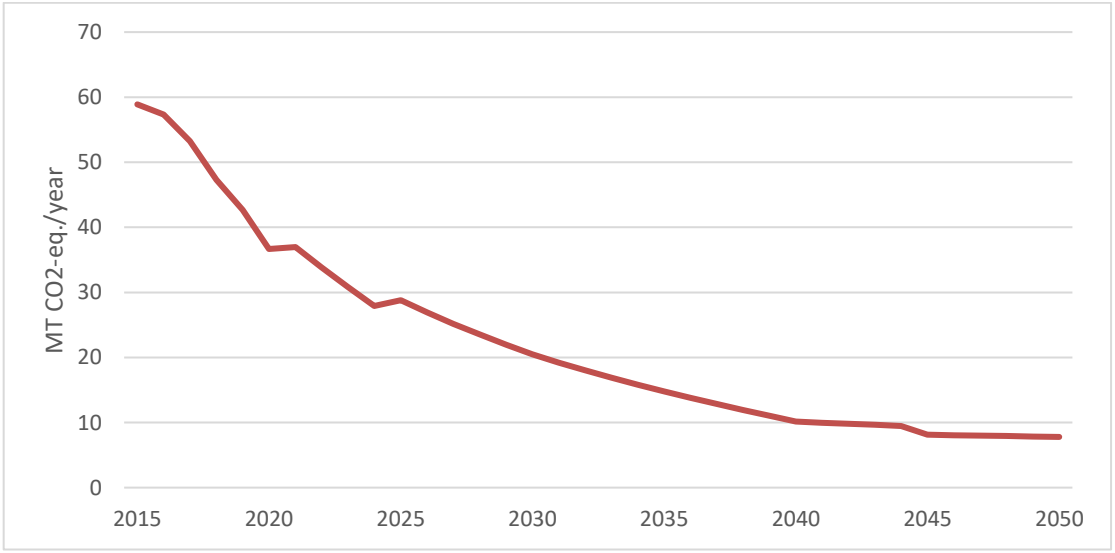


Figure 5-4. GHG emissions from the use phase of local space heaters in the BAU scenario

5.2.3.2.Acidification

As for GHG emissions, acidification impacts cover both the use phase and manufacturing, distribution and end of life. The current level of acidification emissions (2020 data) is about 165 kt SO_{2eq} per year. As shown in table 5-3, under BAU this is expected to have decreased by 43% by 2030. The acidification emissions have a similar trend to GHG-emissions, since they are primarily linked to fossil electricity production.

	Acidification [kt SO_2eq./year]			
	2020	**2030**	**2040**	**2050**
BAU	165	94	48	38

Table 5-3. Expected acidification for BAU

5.2.4. Economic aspects

5.2.4.1.Business turnover

The yearly turnover of the various business sectors related to local space heaters are expected to be relatively static under BAU, since as explained in section 5.2.1, sales will remain stable throughout the analysis period.

Turnover (M€/year)												
	Manufacturer			**Wholesale**			**Retail/repair & maintenance**			**Total**		
	2020	**2030**	**2050**	**2020**	**2030**	**2050**	**2020**	**2030**	**2050**	**2020**	**2030**	**2050**
BAU	1.043	1.051	1.051	1.999	2.014	2.014	2.299	2.316	2.316	5.340	5.382	5.382

Table 5-4. Expected business turnover per year for BAU

5.2.4.2.Consumer expenditure

Consumer expenditure includes purchase costs (which in turn incorporate end-of-life costs that are normally paid by the user through a levy at purchase), energy consumed during use, and repair and maintenance costs. The consumer expenditure under BAU is expected to decrease by 10% by 2030 as shown in the table due to energy efficiency improvements, higher outdoor temperatures and better insulation of buildings, as explained in section 5.1.

	Consumer expenditure [M€/year]			
	2020	**2030**	**2040**	**2050**
BAU	34.161	30.657	27.360	26.079

Table 5-5. Expected consumer expenditure for BAU

5.2.5. Social impacts

5.2.5.1.Employment

During the period after the financial crisis, from 2009 to 2014, the turnover for electric local space heaters remained constant but employment, dropped at a rate of 3%. This is assumed to result from productivity improvements. This trend is assumed to continue during the period under analysis. Therefore even if the turnover of the sector under BAU remains stable, there is projected to be a 23% decrease in employment between 2020 and 2030.

5.2.5.2.Health and safety

There are no specific health and safety aspects related to BAU or the options assessed. Only gas and liquid heaters raise concerns related to pollutant emissions and safe handling of fuel. However, the options assessed in this impact assessment do not entail any relevant impacts.

5.3. Description of the policy options

Policy options have been established through the review study. During this study communication with stakeholders and market surveillance authorities identified a number of loopholes and other

needs for scope clarification. The study explored these issues further and the resulting options are presented here.

The policy options will mostly affect manufacturers' activity, who will need to adapt their products to the new ecodesign requirements. Around 60 manufacturers are estimated to produce local space heaters in the EU, 96% SMEs. Since no specific feedback on the policy options has been received from SMEs during the elaboration of the review study and the IA, it has been considered that the position of SMEs as regards the policy options is represented by the sector associations participating in the process that are enumerated in section 5.4.1. Some big companies have provided their opinion as individual entities, although most of the industry feedback comes from the associations.

For the purpose of the analysis, all policy options are considered to apply in mid-2025, two years after the expected date of publication of the amendments to the LSH Regulation. Placing the date of application in the middle of the year is a request from industry, which claims that the winter period is more problematic for regulatory changes as it falls in the middle of the sales season. A two-year transitional period will give time to industry to adapt to the new requirements, in particular for those products that are currently out of the scope of the LSH Regulation but that will need to comply with it after the legal amendment applies.

Problems	Policy options	Measures
Problem 1: missed in-use energy savings from local space heaters due to inadequate scope of the LSH Regulation	PO2: clarifying and expanding the scope	- PO2a. Removing the exemption for slave heaters - PO2b. Including towel heaters in the scope of the LSH Regulation - PO2c. Including tube heaters between 120 and 300 kW in the scope of the LSH Regulation - PO2d. Adapting the definitions for local space heaters using oil or gas
	PO3: ecodesign measures on low power modes	- Defining ecodesign limits for low power modes
Problem 2: insufficient industry measures as regards of circular economy	PO4: circular economy	- PO4a. Including requirements on the availability of spare parts and on repair and maintenance information - PO4b. Including modularity requirements and setting a 10-year availability period of software updates for Apps - PO4c. Including end-of-life information for improved disposal

Table 5-6. Correspondence between problems, policy options and measures

5.3.1. Policy option 1 (BAU): Business as usual

BAU corresponds to the baseline scenario explained in point 5.2.

5.3.2. Policy option 2 (PO2): Clarifying and expanding the scope

PO2 aims at fixing loopholes present in the LSH Regulation through the following measures:

- PO2a. Removing the exemption for slave heaters, so that slave heaters come in the scope of the LSH Regulation. This measure will bring about energy savings by broadening the scope of the legislation and very importantly, by removing the legal loophole whereby heaters and the corresponding controllers are sold separately in order to sidestep the application of the LSH Regulation. For slave heaters and controllers sold separately, this measure must be complemented with the mandatory **provision of information** to easily identify heaters and controllers that can be installed together in order to comply with the ecodesign limits. Heaters should therefore include product information about which controllers can be fitted to the slave heater and which functions those controllers should feature in order to maximise energy efficiency and likewise, controllers should include information about the type of local space heater they are suitable for and their specific functions. This information should not only be included in the user's manual but also be visibly displayed on the packaging of the product and as part of the product information provided on the Internet.
- PO2b. Including towel heaters in the scope of the LSH Regulation: specific ecodesign limits will apply to towel heaters, which should be nevertheless different for those with heat output ≥ 250 W and those with heat output < 250 W. Towel heaters with capacities ≥ 250 W are powerful enough to heat large bathrooms as the only heat source, hence stricter requirements should be set for these large heaters. For those below 250 W, as their capacity only allows for heating smaller and well-insulated bathrooms and the cost of controllers needed to fill the higher efficiency requirements could be too high compared to the current purchase cost of the heaters, more lenient efficiency requirements should be set.
- PO2c. Including tube heaters between 120 and 300 kW in the scope of the LSH Regulation.
- PO2d. Adapting the definitions for local space heaters using oil or gas: a modification of the definitions for open and closed-fronted local space heaters to better reflect the terminology used by the industry and in the standards, thus reducing the risk of misinterpretation of the LSH Regulation and ensuring that all the products inside the defined categories fulfil the ecodesign criteria. Furthermore, by defining three instead of two categories for gas and liquid fuel local space heaters, stricter and more fitting energy efficiency requirements could be set for each group and consequently reduce energy consumption.

Stakeholders' views

Stakeholders generally support option PO2a, although manufacturers think that it is important to reduce the wording of the complementary information about heaters and controllers to avoid burdensome translations when exporting to different Member States. They also request the LSH Regulation to spell out the information to be displayed on the product. Producers of heating cables and mats expressed doubts about the benefit of mandating information on heaters sold without controllers, as they already do it in the absence of a legal obligation. A large company doubted the effectiveness of heater-controller combinations when they each come from a different manufacturer, and requested the legislation to mandate manufacturers to develop a complete solution to be placed on the market. Some stakeholders also suggested to adapt the scope of the LSH Regulation so that controllers are specifically included, while others suggest just to add a definition of controller. This is however not necessary because removing the exemption for slave heaters already ensures that the whole system (heater + controller) fulfils the ecodesign requirements, and because the calculation of the energy efficiency of a local space heater already includes the effect of controllers.

Stakeholders, including industry associations and individual companies, agree with PO2b, PO2c and PO2c. There are diverging views on the threshold for distinguishing small and large towel heaters. A few NGOs argue that the limit should be 100 or 150W instead of 250W, because heaters less powerful than 250W are capable of heating small, well-insulated bathrooms. Some industry members, on the other hand, oppose including any requirements for towel heaters below 300 W. A minority of stakeholders object to including towel heaters at all, arguing that the heat output is enough to dry towels but not sufficient to maintain a level of human comfort for safety reasons related to possible electric overloads. This opinion is not supported by market data, which show output powers sufficient to heat a medium-sized room. The compromise generally accepted by stakeholders is to set a thresthold at 250W, with more lenient requirements for heaters below that threshold.

5.3.3. Policy option 3 (PO3): defining ecodesign limits for low power modes

In order to ensure an energy efficient behaviour of local space heaters in low power modes, minimum values for standby (0,5W), off (0,2W), idle (0,1W) and network standby (2W) modes are included and local space heaters would need to comply with those values in order to be placed on the market. This is in line with other ecodesign regulations, like the washing machine regulation, where similar adaptations have been made. At the same time, the F(4) factor should be removed from the formula to calculate the seasonal energy efficiency of local space heaters because it becomes irrelevant.

Stakeholders' views

No stakeholders oppose PO3. A large company has suggested setting a new F(4) applying to low power modes additional to standby, but a majority support removing F(4) and setting instead new minimum requirements for different additional low power modes which reflect more faithfully the real behaviour of local space heaters.

5.3.4. Policy option 4 (PO4): circular economy

- 4a. Including requirements on the availability of spare parts and on repair and maintenance information

Given the broad range of local space heaters and their different prices, the LSH Regulation could include different lists of spare parts to be availablefor each of the following product sub-groups:

- portable heaters;
- fixed heaters, towel heaters and underfloor heaters;
- storage appliances;
- electric visibly glowing radiant heaters;
- gas and liquid fuelled heaters.

The benefits of reparability for local space heaters need to be very carefully weighed. Section 6.7 of the review study already concluded that resource-saving requirements might be counterproductive, since:

- Local space heaters lifetime l is already long for many sub-groups at around 15 years for electric heaters (excluding the cheaper portable devices that average 9 years) and 20 years for gas fuelled.

- If spare parts have to be available for a long period, for instance 20 years or more, manufacturers may have to store spare parts that may never be used.
- Consumers tend not to repair cheaper products, which are precisely the ones that feature lower lifetimes, of around 9 years for portable heaters.

The lists of spare parts will therefore be limited to components whose replacement may be considered by the user on the basis of the costs involved and the age of the product when it breaks down. For safety reasons, the lists will also specify which parts should only be available for professional repairers and which ones could also be available for users.

Due to the above restrictions, the lists of spare parts have been developed in close cooperation with APPLiA (Home Appliance Europe), the main representative of the sector.

The listed spare parts should be available for 10 years (estimated average lifetime for the entire product group) after the production of the model stops and have a maximum delivery time of 15 working days. Further, spare parts shall be easily removable with common available tools and without permanent damage to the appliance.

In addition, repair and maintenance information shall be provided by suppliers to independent repairers.

- 4b. Including modularity requirements and setting a 10-year availability period of software updates for APPs

Redesigning products so that controllers can be replaced by more functional and efficient ones, would diminish the number of old devices discarded and improve their energy efficiency. It would also reduce repair costs because users could change the controller themselves.

In addition, in line with the update of controllers, any Apps that are linked to them should also be updated. Software update requirements can ensure that new versions of an App remain compatible with older products, thus ensuring a long lifetime. Software updates should therefore be available for 10 years after the production of a model stops.

- 4c. Including end-of-life information for improved disposal

Information regarding dismantling the product and access to any of the components referred to in point 1 of Annex VII of the WEEE Directive[37], must be available on a free access website. This information shall include the sequence of dismantling steps as well as the tools and technologies needed to remove the targeted components. The end-of-life information shall be available until at least 15 years after the placing on the market of the last unit of a product model.

Stakeholders' views

Member States and NGOs support the availability of spare parts. Manufacturers did not oppose this but raised the following issues:

[37] Directive 2012/19/EU of the European Parliament and of the Council of 4 July 2012 on waste electrical and electronic equipment (WEEE) (OJ L 197, 24.7.2012, p. 38)

- The delivery time for spare parts may be long due to the supply chain: consumers normally ask an installer to deliver the part, who in turn needs to get if from the wholesaler, and the latter will need to ask for it to the factory. All these steps add to the delivery time.

- The lists of spare parts should inform about which parts are available for professional repairers and which ones can be fitted by the end user.

- Most portable local space heaters are in the low-cost range (below 100 €). Repairing such devices will exceed the cost of the product, making it counterproductive to provide spare parts. In case of failure during the warranty period the whole product is returned and scrapped.

- Several components are integrated and soldered and should not be considered as reparable spare parts – e.g. electronic displays and temperature sensors are integrated into thermostats.

No comments have been provided by stakeholders regarding end-of-life information.

APPLiA strongly contests the modularity option because it increases the complexity of housing parts and connectors in order to fit controllers that will probably be used in only a few cases. Furthermore, end users might replace their heater for many other reasons than the wish to get enhanced control functions, e.g. design, general house renovation, or installing central heating to replace local space heating. APPLiA points out that the costs of modularity can be very high and thus significantly increase the price of the product.

Finally, Member States and NGOs' supported requirements for software updates. Industry did not directly oppose this but opinions ranged from those stating that software updates are not a problem to those that were more cautious, since the communication is often programmed with proprietary protocols and, at some point in the lifetime of a heater, it becomes expensive to update a system if it is being phased out.

5.4. Options discarded at an early stage

5.4.1. Voluntary agreement by the industry

According to the Ecodesign Framework Directive, voluntary agreements must be given priority provided they meet the objectives in a faster and more cost-effective manner than implementing acts. Further, Annex VIII of the Ecodesign Framework Directive sets out some indicative criteria that self-regulatory initiatives must incorporate to be alternative to an implementing measure. One of those criteria is representativeness, which means that industry and its associations taking part in a self-regulatory action must represent a large majority of the relevant economic sector. Main associations of the sector are APPLiA, EHI (European Heating Industry), HKI (Industrial Association of House, Heating and Kitchen Technology), AIE (European Association of Electrical Contractors), CEFACD (European Committee of Manufacturers of Domestic Heating and Cooking Appliances), EUHA (Electric Underfloor Heating Appliances) and ELVHIS (European Leading Association of Radiant Gas Heaters Manufacturers). The large amount of industry associations is the consequence of the heterogeneity of the sector in terms of product groups. None of them have shown willingness to undertake the leadership for a self-regulatory initiative in the different comments and position papers submitted about the amendment to the LSH Regulation. It is therefore not likely that a self-regulatory action would ever be operational.

Further, the heterogeneity of the sector and the presence of a large number of SMEs would undermine the effectiveness of self-regulation. While an industry agreement would be suitable for sectors dominated by a small number of companies, or by companies located in a relatively reduced

geographical area, an EU-wide agreement on local space heaters covering the totality of companies, many of them SMEs, would be complex to govern and would imply a high level of coordination between associations, national authorities and Commission equipped with the appropriate resources.

On the basis of the foreseen complexity of implementation and lack of initiative by the private sector to undertake such initiative, a voluntary agreement has been ruled out.

5.4.2. Mandatory third party certification

The evaluation of the appropriateness to introduce third party certification for conformity assessment is a legal mandate set out in Article 7 of the LSH Regulation. This evaluation has been carried out in section 1.5.4 of the review study and is briefly explained here.

The relevance of third party testing depends on the type of energy used by the heater. For electric heaters, third-party certification seems unnecessary and disproportionate in the view of the additional costs that it entails, because the energy efficiency tests do not leave room for interpretation and deviations for testing that can be misused by the manufacturer. For gas and liquid fuel heaters on the contrary, the large number of parameters to be adjusted (fuel quality, air flow and flue duct settings) makes it easier for the manufacturer to adjust the test in order to get the best possible result. Third party testing could thus be relevant for gas and liquid fuel heaters from a technical point of view. Nevertheless, the application of third party conformity assessment just for gas and liquid fuel heaters would entail the implementation of a burdensome procedure that would not apply to electric heaters, and would therefore undermine the level playing field between both types of devices by increasing the costs of non-electric heaters. Accreditation costs related to the services provided by certification authorities (laboratories) are estimated to be between 2.500 and 3.000 €. Further, given the very low market share that gas and liquid fuel appliances represent, the benefits of third party certification in terms of energy and GHG emission savings would be minimal.

Given the background above, third party certification has been ruled out.

5.4.3. Modification of the verification tolerances

The tolerances have been found to be in line with the accuracy of the current test methods and should therefore remain.

5.4.4. Introduction of stricter ecodesign requirements and NOx emission limits

The objective of both PO2 and PO3 is to reduce in-use energy consumption. An alternative to both policy options would have been to set stricter ecodesign limits by reducing the values of correction factors F(2) and F(3), which are applied to the formula used for the calculation of the energy efficiency from local space heaters. Section 7.1.2 of the review study discusses the possible introduction of stricter ecodesign measures making it more difficult to reach the minimum energy efficiencies set out in the legislation. Nevertheless, the review study concluded in section 7.2.2 that a reduction of the correction factors should only be made when there is enough evidence about how the current factors have performed. This was not the case when the review study was conducted because by that time the current limits had been in force for an insufficient period, to carry out the necessary assessment. Moreover, setting more demanding ecodesign requirements without first removing the LSH Regulation's loopholes would have reduced the effectiveness of the measure, leading to wasted resources because the investments carried out by industry would not have been fully productive and the burden costs calculated in section 6.5 would have remained or even

increased. It seemed therefore reasonable to focus on alternative measures that would not imply setting new limits. For the same reason, stricter pollutant emission requirements have also been discarded.

A few NGO's supported setting stricter energy efficiency and NOx emission requirements. The rest of stakeholders, including industry associations and individual companies, proposed to keep the current values and to assess the topic at the next review of the LSH Regulation.

5.4.5. Removing the conversion coefficient (CC) from the calculations of the energy efficiency

The CC is used to express final energy in terms of primary energy, which allows to compare heaters using electricity to heaters using oil or gas. It has been argued that the CC adds complexity to the calculation of the energy efficiency without really being representative of real efficiency, and therefore it has been considered to remove the CC and focus exclusively on the consumption of final energy.

Even though it is always desirable to reduce the complexity of the LSH Regulation, removing the CC would make it impossible to compare electric heaters with oil and gas heaters, with a view to the future possible inclusion of electric heaters in the energy label for local space heaters (currently only fossil fuel heaters are in the scope of the energy label). Moreover, the CC is used in other EU regulations and is widely accepted. The decision on its removal is therefore outside of the scope of the revision of the LSH Regulation and, if adopted, should cover a number of regulations and not only local space heaters.

6. WHAT ARE THE IMPACTS OF THE POLICY OPTIONS?

This section describes the impacts of each policy option from the environmental, economic and social points of view. Modelling principles follow the MEErP (Methodology for Ecodesign of Energy-related Products)[38]. The analytical model, described in detail in Annex 4, has been specifically developed for impact assessments by the external consultant Viegand Maagøe. The assumptions provided in Annex 4 have been systematically crosschecked with stakeholders, in particular with the industry. Future energy prices and financial rates follow the EU Reference Scenario 2020.

It is necessary to note that the data used for the modelling of the different scenarios present some limitations, which are indicated here and explained in detail in Annex G of the review study and in Annexes 1 and 4 of this IA.

First, there is a lack of field data about the energy performance of local space heaters. Annex G of the review study assumes that the efficiencies of all products in the scope of the LSH Regulation will not exceed the minimum requirements. This assumption has also been taken in this IA. There is no statistical information on products that perform better than needed and therefore the minimum threshold has been taken for the estimation of the impacts. This seems to be an adequate assumption because, as already pointed out, in the absence of technological improvements the in-use energy savings per product unit are assumed to be rather modest.

[38] Material-efficiency Ecodesign Report and Module to the Methodology for the Ecodesign of Energy-related Products (MEErP) PART 1: MATERIAL EFFICIENCY FOR ECODESIGN - Final report to the European Commission - DG Enterprise and Industry 5 December 2013 (https://op.europa.eu/en/publication-detail/-/publication/b7650397-32f1-436c-82c4-df39aef297a3)

Second, the sales data that have been purchased from a data provider are not broken down by product category. Disaggregated sales figures were estimated by the consultant on the basis of feedback from industry representatives, and in particular APPLiA. Disaggregated price data are also not available. This shorcoming has been addressed by discussing the disaggregated prices in the 2015 impact assessment with stakeholders and correcting the price-elasticity formula explained in Annex 4.

Third, there is little data about the impacts of circular economy measures. In view of this lack of solid evidence, it has been assumed that resource efficiency measures increase the lifetime of a number of product subgroups by 5%, that 20% of the products in these subgroups are repaired only once over their lifetime and that the additional repair costs would be 20% of the product price. For instance, some studies indicate a very large variation in the rates charged for spare parts and repair services between different manufacturers and Member States, making it very difficult to determine an average repair cost across the EU[39]. The assumptions are therefore based on estimations made by the consultant on the basis of the characteristics of local space heaters, existing bibliography and stakeholders input. In any case, changes in these assumptions have a negligible effect on total energy figures, because resource efficiency measures mostly affect embedded energy, and this only accounts for 0,7% of the total local space heater energy consumption .

Fourth, there is no specific data on production costs because industry does not make them available. Compliance costs have therefore not been calculated on the basis of production costs but on the basis of manufacturer turnover, by subtracting from the manufacturer turnover for a specific policy measure and year, the manufacturer turnover under BAU in the same year. This would imply the assumption that local space heater market develops as a perfectly competitive market where profits are zero. Although this is a simplification, it seems acceptable in the view of the high level of competition in the sector (around 60 manufacturers with 96% SMEs as indicated in section 5.3) and the low level of innovation, which tends to reduce product differentiation.

The fifth and last important limitation is that PO4c impacts have not been quantified due to the lack of data on how the recycling process would improve as result of better information on recoverability and recyclability. The impact of PO4c will be anyway very small, first because local space heaters are nowadays highly recyclable, and second because embedded energy accounts only for 0,7% of the total energy use, as pointed out above. The measure is mostly intended to improve recoverability of electronic components, which will be increasingly present in the appliance sector as result of technological updates to the product.

6.1. Environmental impacts

This section addresses the variation of energy consumption due to the application of the different measures and the subsequent change in GHG and acidification emissions. Other environmental impacts are addressed in Annex 5.

The measures in PO2 and PO3 reduce only in-use energy and not embedded energy.

PO2a, PO2b and PO2c measures result in the type of heaters addressed by the respective measures to be included in the scope of the LSH Regulation. PO2d results in a redistribution of the existing products that are already in the scope towards categories with higher energy efficiencies. The main difference in terms of impacts between these four measures is due to the number of products affected rather than the energy efficiency gained per unit of product, since the policy options are

[39] See Annex 6, Section 6.2.2

not targetting the increase of energy efficiency per unit. In this respect, PO2a is the option that would have the biggest impact from all four, as it includes 63% of the products affected by the measures. PO2b would include 11%, PO2c 0,02% and PO2d 0,3%.

PO3 lays down minimum values for low power modes that local space heaters would need to comply with. PO3 applies to all electric heaters and therefore to 99% of the total sales. However, as the consumption of low power modes is very low in comparison with active modes (0,5% of energy efficiency increase has been considered for PO3 in comparison with the 3,75% for PO2a, PO2b and PO2c, as indicated in section 5 of Annex 4), PO3 generates a relatively modest yearly reduction of in-use energy consumption.

PO4a is expected to have no effect on certain types of heaters that are either too cheap to be repaired such as portable heaters, or would only be retired for a major cause such as a house renovation, like for instance electric underfloor heaters or open to chimney gas heaters. For other product subgroups, the availability of spare parts is expected to moderately extend the lifetime by 5%. This will result in a small reduction of sales and production, and therefore in the embedded energy needed to produce the heaters. Because the benefits of spare parts availability are only realised in the long-term, when old devices need to be replaced by new ones, the abovementioned reductions only occur from 2040 onwards.

Finally, PO4b decreases both in-use and embedded energy consumption. This is because modularity reduces in-use energy by fitting better performing controllers and also reduces embedded energy by extending the lifetime of the products.

Tables 6-1 and 6-2 show the main environmental impacts for all the measures.

	In-use energy savings vs BAU[40]						**Primary energy savings vs BAU**					
	Yearly savings (Twh/year)			**Cumulative savings (Twh)**			**Yearly savings (Mtoe/year)**			**Cumulative savings (Mtoe)**		
	2030	**2040**	**2050**	**2030**	**2040**	**2050**	**2030**	**2040**	**2050**	**2030**	**2040**	**2050**
PO2a	1,99	4,09	4,69	7,94	40,63	85,25	0,359	0,739	0,847	1,434	7,337	15,393
PO2b	0,48	0,93	0,88	1,89	9,53	18,65	0,086	0,167	0,159	0,341	1,721	3,367
PO2c	0,03	0,06	0,07	0,11	0,59	1,29	0,002	0,005	0,006	0,010	0,051	0,111
PO2d	0,06	0,12	0,14	0,22	1,17	2,56	0,005	0,011	0,012	0,019	0,101	0,220
PO3	0,19	0,33	0,32	0,81	3,62	6,89	0,035	0,059	0,058	0,147	0,654	1,244
PO4a	0,00	0,00	0,00	0,00	0,00	0,00	0,000	0,005	0,005	0,000	0,005	0,052
PO4b	0,02	0,03	0,03	0,07	0,36	0,71	0,003	0,008	0,008	0,013	0,072	0,150

Table 6-1. Energy savings for each measure

	Greenhouse gas emissions (embedded + in-use)						**Acidification**					
	Yearly savings ($MtCO_2eq$/year)			**Cumulative savings ($MtCO_2eq$)**			**Yearly savings (tSO_2eq/year)**			**Cumulative savings (tSO_2eq)**		
	2030	**2040**	**2050**	**2030**	**2040**	**2050**	**2030**	**2040**	**2050**	**2030**	**2040**	**2050**
PO2a	0,284	0,308	0,286	1,245	4,583	7,528	1.255	1.365	1.267	5.512	20.284	33.318
PO2b	0,068	0,070	0,054	0,295	1,078	1,682	300	309	238	1.306	4.769	7.445
PO2c	0,006	0,012	0,014	0,022	0,117	0,254	25	54	62	98	517	1.126
PO2d	0,012	0,025	0,029	0,046	0,240	0,523	51	112	128	202	1.062	2.314
PO3	0,027	0,025	0,020	0,128	0,418	0,634	121	109	86	567	1.850	2.807
PO4a	0,000	0,012	0,012	0,000	0,012	0,132	0	68	69	0	68	750
PO4b	0,003	0,007	0,006	0,011	0,057	0,123	12	35	33	50	269	608

Table 6-2. Savings on GHG emissions and acidification for each measure

[40] In this table, primary energy includes both in-use energy and embedded energy from production, distribution and end-of-life phases, i.e. all life cycle phases.

6.2. Economic impacts

6.2.1. Business turnover

To calculate the business turnover it is assumed that the costs of adaptation of the product to the legislation will be fully passed on to consumers as higher product prices. Business turnover is obtained as follows: first, the 2020 prices per product subgroup estimated as explained in the introductory part of section 6, are entered in the price-efficiency elasticity formula explained in Annex 4, to obtain the evolution over time of the purchase cost per unit for each product subgroup. The elasticity formula makes purchase costs dependent on the energy efficiency of each product, as higher efficiencies will enable higher prices. The unitary price per year and product subgroup is then multiplied by the expected sales each year to get a total purchase cost per year and per product subgroup. Then the purchase costs for all subgroups are added in order to obtain a total purchase cost, which will be the turnover for the respective year. The last step is to multiply the turnover by a mark-up factor depending on whether the turnover corresponds to the manufacturer, the wholesale agent or the retailer. Mark-up factors are given in section 4.4 of Annex 4.

Applying this methodology results in the turnover for the different measures displayed in table 6-3. PO2 and PO3 increase turnover in the short term due to in-use energy efficiency improvements being realised as from the date of application of the LSH Regulation. This increase of turnover is one-off, because it is due to the new products that enter the scope of the legislation by its date of application, and levels off afterwards because sales remain steady. PO4a in turn does not have any effect in the short term but it decreases turnover in the long term, as old products are being updated or repaired. This impact is to some extent compensated by an increase in repair activities. PO4b presents a hybrid behaviour, since the turnover increases in the short term due to the need to adapt the products design to modularity requirements, but it decreases in the long term because the fitting of upgraded of controllers extends the lifetime of the equipment.

Quantitatively, the substantial adaptations of the product required by PO4b produce a very large increase in turnover, but will also result in much higher user expenses as addressed in the following section. The change in turnover due to PO2a is also large, because of the number of units affected and the cost of fitting better controllers. The rest of the options present less significant changes.

	Turnover (M€)													
	Manufacturer			**Wholesale**			**Retail/Repair & Main.**			**Total**			**Change vs (BAU)**	**Change vs (BAU)**
	2020	**2030**	**2050**	**2020**	**2030**	**2050**	**2020**	**2030**	**2050**	**2020**	**2030**	**2050**	**2030**	**2050**
PO2a	1.043	1.106	1.106	1.999	2.119	2.119	2.299	2.437	2.437	5.340	5.662	5.662	280	280
PO2b	1.043	1.066	1.066	1.999	2.043	2.043	2.299	2.349	2.349	5.340	5.458	5.458	76	76
PO2c	1.043	1.051	1.051	1.999	2.014	2.014	2.299	2.317	2.317	5.340	5.382	5.382	0	0
PO2d	1.043	1.056	1.056	1.999	2.024	2.024	2.299	2.327	2.327	5.340	5.407	5.407	25	25
PO3	1.043	1.055	1.055	1.999	2.021	2.021	2.299	2.324	2.324	5.340	5.400	5.400	18	18
PO4a	1.043	1.051	1.016	1.999	2.014	1.948	2.299	2.316	2.240	5.340	5.382	5.204	0	-178
PO4b	1.043	1.136	1.127	1.999	2.177	2.161	2.299	2.503	2.485	5.340	5.816	5.773	434	391

Table 6-3. Total turnover and turnover changes for each measure

6.2.2. Consumer expenditure

As indicated in section 6.2.1, it has been assumed that the level of cost pass-through from manufacturer to end-user is 100%. Consequently, consumer expenditure per year for each measure has been calculated by summing the yearly purchase costs calculated in section 6.2.1 to the repair and maintenance costs, and subtracting the in-use energy savings.

The impacts of the different measures on consumer expenditure are displayed in Table 6-4. The large PO2a savings are due to the large amount of products included in the scope, whose in-use energy savings will offset the initial purchase costs. PO4b on the contrary, does not offset the important product price increase with energy savings, resulting in a significant net loss for the consumer. This is also true for PO2d, although in this case the reduction in consumer savings is less marked. The rest of the measures yield consumer savings that are smaller than PO2a, either due to a more limited scope as it is the case of PO2b, or to both a more limited scope and smaller savings per unit as in the case of PO2c and PO3.

	Consumer savings						
	Increase of purchase costs over BAU in 2025 (%)	Savings vs BAU [M€/year]			Cumulative savings vs BAU [M€]		
		2030	2040	2050	2030	2040	2050
PO2a	5,81	270	704	849	630	5.963	13.850
PO2b	1,65	60	153	148	110	1.302	2.815
PO2c	0,04	1	3	3	3	22	53
PO2d	0,56	-9	-3	-1	-77	-135	-149
PO3	0,35	30	58	58	102	589	1.177
PO4a	0,00	0	18	18	0	18	199
PO4b	8,06	-222	-301	-301	-1.337	-3.934	-6.943

Table 6-4. Consumer savings for each measure

6.2.3. Costs

6.2.3.1.Compliance costs

As indicated in the introductory part of section 6, total compliance costs in a specific year have been calculated by subtracting the manufacturer turnover for BAU from the manufacturer turnover for that specific year and policy measure. Compliance costs per unit result from dividing the compliance costs by the yearly sales.

	Compliance costs (M€/year)		Unitary compliance costs (€/unit per year)	
	2030	2050	2030	2050
PO2a	54,73	54,73	3,20	3,20
PO2b	14,87	14,87	0,87	0,87
PO2c	0,09	0,09	0,01	0,01
PO2d	4,99	4,99	0,29	0,29
PO3	3,63	3,63	0,21	0,21
PO4a	0,00	-34,64	0,00	-2,02
PO4b	84,75	76,44	4,95	4,47

Table 6-5. Compliance cost for each measure

6.2.3.2.Administrative costs

Administrative costs are defined as those costs borne by businesses, citizens, civil society organisations and public authorities as a result of administrative activities performed to comply with administrative obligations included in legal rules.

There will be extra although minor administrative costs linked to the adoption of the proposed measures for the products in the scope. Administrative costs are split among the costs of updating the Declaration of Conformity (DoC) to the new requirements, which would apply to models currently in the scope, and costs of making a new DoC, which would apply to models that would fall in scope for the first time. For the former 0,50 hours effort has been estimated , and for the latter 1 hour. Those times are multiplied respectively by the estimated number of models already in

the scope[41] and by the estimated number of new models in the future scope for each measure. The two results are multiplied by the EU average hourly labour cost for 2021 (32€/h, the latest available) and summed[42]. These will be one-off costs incurred in 2025, the year of application of the Regulation, as the DoC will only need to be issued once.

All models in the current scope will need a DoC update regardless of the measure considered, as it will at least be necessary to refer to the new legal background. Only PO2a, PO2b and PO2c add new models to the scope and the administrative costs therefore increase compared to the remaining measures.

	New models in scope	Models in current scope	Cost issuing new DoC for new models in scope(€)	Costs update DoC for models in current scope (€)	Total (€)
PO2a	11.013	45.112	352.416	721.792	1.071.208
PO2b	6.174	45.112	197.568	721.792	919.360
PO2c	1	45.112	32	721.792	721.824
PO2d	0	45.112	0	721.792	721.792
PO3	0	45.112	0	721.792	721.792
PO4a	0	45.112	0	721.792	721.792
PO4b	0	45.112	0	721.792	721.792

Table 6-6. Administrative costs by measure

6.2.3.3.Other additional costs

For PO4c, other additional minor costs for manufacturers will be to make available repair and maintenance information, as well as information on recycling. Manufacturers will however be able to offset the additional cost by charging a proportionate fee to repairers in compensation for the provision of information.

For PO4a, it will also be necessary to update the information on the manufacturers' or importers' websites regarding availability of spare parts. The LSH Regulation already requires certain information to be provided on websites and in technical documentation, therefore it is assumed that the additional information could be provided at negligible extra cost.

6.3. Social impacts. Employment

A good understanding of the employment effects of the measures would require general equilibrium economic modelling. In any case, the likely effect is small enough to be difficult to identify with any certainty at the economy-wide level. The dominant economic impact is expected to arise from the avoided expenditure on energy by consumers due to the improved performance of the products, resulting in a net saving that will be spent in other areas of the economy. Although it cannot capture many of the likely interactions, a first indication of the scale of the effect is given by taking account of the differing employment intensity of the energy supply sectors and the general economy. It is assumed for the purpose of the assessment that these do not change in the coming

[41] The number of models have been worked out from EPREL data. EPREL is currently restricted to non-electric local space heaters, but considering that the latter mean around 1% of the total sales, by extrapolation it is deduced that the approximately 600 models of non-electric heaters that are currently registered in EPREL are equivalent to 62.300 models among all product subgroups. The number of new models that come in the scope of the LSH Regulation per product subgroup is obtained by multiplying the share of products originally out of scope as given in section 5 of Annex 4 (30% for fix heaters, 90% for underfloor heaters and 90%for towel rails), by the sales of the respective product subgroups in 2025, which gives the number of new units in scope. The results for the different product subgroups are summed to get the total units in the scope and divided by the total sales of local space heaters, which gives the share of new units in the scope from the total of units sold. The result is finally multiplied by 62.300. This procedure assumes therefore proportionality between number of models and number of units sold.

[42] Wages and labour costs - Statistics Explained (europa.eu)

decades. Estimates of employment intensity for the relevant sectors calculated from Eurostat data are:

Economic sector	Employment per €m turnover
Electricity, gas ,steam and air conditioning supply – NACE D	0,9
Whole economy	5,2

Table 6-8. Employment per m€ turnover

In addition, it can be estimated the employment in the local space heaters sector alone based upon its employment intensity. The net impact on employment would take account of all the three factors as shown in table 6-9.

	Job gains/year in local space heaters sector (a)			Jobs in the overall economy (jobs)						Net impact (a+c-b)		
				Job losses/year in electricity supply due to better energy efficiency (b)			Job gains/year in overall economy due to the reduction of user expenditure (c)					
	2030	2040	2050	2030	2040	2050	2030	2040	2050	2030	2040	2050
PO2a	1.062	799	601	243	633	764	1.404	3.661	4.415	2.223	3.827	4.252
PO2b	323	243	183	54	138	133	312	796	770	581	901	820
PO2c	1	1	1	1	2	3	5	14	17	5	13	15
PO2d	57	57	57	-8	-3	-1	-47	-16	-5	18	44	53
PO3	79	59	45	27	52	52	156	302	302	208	309	295
PO4a	0	-589	-459	-	-	-	0	93	93	0	682	552
PO4b	1.729	1.161	873	-200	-271	-271	-1.154	-1.565	-1.565	775	-133	-421

Table 6-9. Employment on the local space heaters sector, on electricity supply and on the rest of the economy

6.4. Affordability (payback period)

The unit price is expected to increase between 0,04% and 8%, depending on the measure. For PO2a, PO2b, PO2c and PO3, the higher price is outweighed by the energy consumption savings, giving a payback time between 2 to 5 years, significantly below the lifetime of 9 years for portable local space heaters, which is the shortest of all product subgroups. For PO4b the energy savings generated by modularity, which would allow to replace an existing controller with a new and more efficient one, cannot compensate in a reasonable timeframe the increase of purchase cost of adapting the product design to the requirements. PO2d also has a large payback time. For PO4a payback time is zero because there is neither price increase nor in-use energy savings against 2030, which is the year taken as reference.

	Increase in average unit price vs BAU		Annual in –use saving vs. BAU until 2030		Pay-back time (years)
	€/unit	%/unit	€/year	%/year	
PO2a	7,05	5,81	1,65	1,50	4,3
PO2b	1,92	1,65	0,40	0,36	4,8
PO2c	0,01	0,04	0,005	0,004	2,4
PO2d	0,64	0,56	0,02	0,02	37,9
PO3	0,47	0,35	0,16	0,14	2,9
PO4a	0,00	0,00	0,00	0,00	0
PO4b	10,92	8,06	0,02	0,01	699

Table 6-7. Payback periods for each measure

6.5. Burden reduction

As explained in section 2.1.1, the lack of clarity of the LSH Regulation is a source of burden with negative consequences on the activity of manufacturers and market surveillance authorities. For manufacturers, the burden is due to the unlevelled playing field between manufacturers compliant

with the ecodesign rules and those who are non-compliant, which increases administrative and compliance costs of the former with respect to the latter. Excess administrative costs for compliant manufacturers would be 547.995 € in total, 526.042 € for PO2a and 21.953 € for PO2b[43]. As far as the excess compliance costs, they amount to 49,4 M€, 47,7M€ for PO2a and 1,7M€ for PO2b[44].

Is it not possible to quantify the burden reduction for MSAs because there are no data about how burden lengthens the time for authorities to select the appropriate products, including the deliberations with manufacturers about the possible inclusion of their products in the scope. Nevertheless, the reduction of the burden can be qualitatively estimated from the number of models affected per product category. This is provided in table 7-1 in section 7, where it is shown that the biggest burden reduction comes from PO2a, which includes the largest number of models, followed by PO2b and PO2d. The rest of the measures are neutral in this respect.

6.6. SMEs

Around 57 SMEs will need to adapt their products to the new measures. It is expected that PO2 measures do not entail any significant additional burden for SME's because no stricter energy efficiency limits are set. Manufacturers making products that will come in the new scope, will need to fit a suitable heater-controller combination or to inform about that combination in the product packaging, but will not need to develop different technological solutions to fulfil the legislation. Only PO4b would add significant costs because it will make it necessary to redesign some products in order to meet the modularity requirements, and this impact would be more important for SME's because of their limited financial capabilities.

An SME test is carried out in Annex 9.

6.7. Geographical impacts

France is the major user of fix electric heating in the EU (including fixed heaters, storage heaters, visibly glowing and towel rails) with around 80% of the total of units sold in Europe, followed by Spain and Sweden, with 245.000 and 150.000 units respectively[45]. Measures PO2a (slave heaters will become fixed local space heaters when included in the scope of the legislation) and PO2b on towel rails, will subsequently be felt with most intensity in France, and moderately in Spain and Sweden. This is also the case for PO4a, as it has been assumed in the study according to Annex 4, that the lifetime of electric fixed, visibly glowing and towel heaters will be extended, which are again mostly used in France. Other Member States with a significant amount of fixed heaters and

[43] Excess administrative costs have been calculated by multiplying the number of models of slave heaters and towel heaters originally in the scope by 1 hour for drawing up the DoC and by the estimated hourly labour cost for 2021 (32€/h). The number of models originally in the scope of the LSH Regulation per product subgroup is obtained by multiplying the share of products in the scope before the application of the new measures as given in section 5 of Annex 4 (70% for fix heaters, 10% underfloor heaters and 10% towel rails), by the sales of the respective product subgroups in 2025, which gives the number of units originally in scope. The units of fix heaters, underfloorh eaters and towel rails are then summed and the result divided by the total sales of local space heaters, which gives the share of units originally in the scope from the total of units sold. The result is multiplied by 62.300 to obtain the number of models originally in the scope.

[44] Excess compliance costs have been calculated by dividing the increase of manufacturer turnover per year between 2024 and 2025 (year of application of the amendment to the LSH Regulation) with the share of non-compliant units for both slave heaters and towel heaters, in order to obtain what the total increase of the turnover would be if all the products were out of the scope and were to be brought in the scope, and this total is multiplied by the share of units already in the scope to get the excess of turnover (hence compliance costs) corresponding to products already in the scope. Note that, as explained in section 6.2.2, costs are directly calculated from manufacturer turnover.

[45] Own-elaborated information from data provided by industry.

where the abovementioned measures will have a notable impact are Finland, Germany, Poland, Denmark and Austria.

There is some more uncertainty about the geographical impact of PO3 and PO4b due to the lack of data on the distribution of portable heaters per country, which account for 42% of local space heaters sales and are also affected by these two measures. Nevertheless, according to the modelling the most significant energy savings from PO3 and PO4b will be achieved from big fixed appliances rather than small portable ones, which are less powerful and also less prone to innovation, so the savings are expected again to concentrate on the above-mentioned countries.

There is also uncertainty about the geographical impact of PO2c and PO2d due to the lack of data on market share of the affected product subgroups per Member State. However, EPREL data indicate that fossil fuel local space heaters manufacturers are predominantly located in Italy, and to a lesser extent in Germany and France. Assuming that manufacturers are established in places that are relatively close to the market, it seems reasonable to deduce that also Italy, Germany and France, in this order, will be the Member States most affected by PO2c and PO2d.

6.8. Impacts of current geopolitical instability

In view of the sharply increasing energy prices due to the present political instability, the use of the energy prices from REF2020 for the projections raises the question of whether the analysis presented in the previous sections is realistic. The REPowerEU communication makes clear that there will be structural changes in EU energy supply that are likely to result in continuing higher energy prices. Higher energy prices would increase consumer expenditure during the use phase and therefore amplify the consumer savings from the proposed measures. The REPowerEU communication also underscores the need to step up energy efficiency investments in the light of the changed geopolitical situation. All these factors point to the likelihood that the assessment as described insufficiently values the energy savings from PO2 and PO3 and therefore gives too low an efficiency score to them.

Section 8.4 includes an affordability analysis to understand how the considered options would be affected in a situation of escalating energy prices. In this analysis, the payback time for higher energy prices have been calculated assuming the adoption of ecodesign measures which are more ambitious than those assessed in this IA and that would provide an additional 2% increase in the energy efficiency of local space heaters (thus doubling the energy savings of the preferred policy option in this IA). In addition, Annex 8 includes a sensitivity analysis based on the Climate Target Plan MIX scenario[46], with higher energy prices, lower carbon intensity and higher building renovation rates vis-a-vis REF2020. However, the prices from this scenario are lower than current prices. There is currently no PRIMES scenario that has been adapted to the new situation.

6.9. Summary of impacts

Table 6-10[47] shows a qualitative summary of the impacts of the different measures, based on the quantitative results presented in the previous sections 6-1 to 6-8. Further insight on the impacts of the different policy options, including graphs, are provided in Annex 5.

[46] Policy scenarios for delivering the European Green Deal (europa.eu)

[47] Where no impact is assessed this is shown as a '0' in the table. Any improvement in any of the aspects as a result of the option is shown using '+'. The number of '+' is indicative of the scale of the impacts but these should be treated with care since the magnitude of the differences in many cases is large. Similarly any deterioration in an aspect as a

	In-use energy savings			Primary energy savings			GHG savings			Acidification			Consumer savings			Employment			Payback time
	2030	2040	2050	2030	2040	2050	2030	2040	2050	2030	2040	2050	2030	2040	2050	2030	2040	2050	
PO2a	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	++++	+
PO2b	++	++	++	++	++	++	++	++	++	++	++	++	++	++	++	++	++	++	+
PO2c	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	++
PO2d	+	+	+	+	+	+	+	+	+	+	+	++	-	-	-	+	+	+	--
PO3	+	+	+	+	+	+	+	+	+	++	+	+	+	++	++	+	++	++	++
PO4a	0	0	0	0	+	+	0	+	+	0	+	+	0	+	+	0	+	+	0
PO4b	+	+	+	+	+	+	+	+	+	+	+	+	---	---	---	++	-	-	---

Table 6-10. Summary of the impacts of the different measures

7. HOW DO THE OPTIONS COMPARE?

The options identified are individually assessed against the objectives set out in Section 4. Each option is assessed for its effectiveness in achieving those objectives, its efficiency in doing so and its coherence.

7.1. Effectiveness

Effectiveness refers to the extent to which the proposed options would achieve the objectives set out in Section 4. Effectiveness has been assessed from three angles: energy savings (Effect1), which will also lead to a proportionate reduction of GHG emissions and acidification, resource efficiency (Effect2) in terms of reduction of fuel (coal/lignite, petroleum products, natural gas and biomass/waste) and material for the production of heaters, and reduction of administrative burden (Effect3). Besides the impacts considered in Effect2, further impacts related to resource efficiency/circular economy are addressed in Annex 5. The values displayed in table 7-1 are cumulative over the periods 2025 (year of implementation of the preferred policy option) - 2030, 2025 - 2040 and 2025 - 2050. The reason why it is preferred to use cumulative values over a period instead of values for a single year is explained in section 7.2.

	Energy savings - Effect1							Resource efficiency - Effect 2							Reduced burden - Effect3 (from section 6.4)			
	In-use energy savings (Twh)			Embedded energy savings (Twh)			Score	Fuel savings (Mtons)			Material saved for manufacturing (Tons)			Score	MSAs	Manufacturers		Score
																Excess Administrative costs that disappear (k€)	Excess compliance costs that disappear (M€)	
	2025-2030	2025-2040	2025-2050	2025-2030	2025-2040	2025-2050		2025-2030	2025-2040	2025-2050	2025-2030	2025-2040	2025-2050		2025			
PO2a	7,7	40,4	84,0	0,0	0,0	0,0	+++	1,78	9,13	18,72	0	0	0	+++	+++	526,04	47,7	+++
PO2b	1,8	9,5	18,6	0,0	0,0	0,0	++	0,43	2,15	4,11	0	0	0	++	++	21,95	1,7	++
PO2c	0,1	0,6	1,3	0,0	0,0	0,0	+	0,03	0,13	0,28	0	0	0	+	NA	NA	NA	0
PO2d	0,2	1,2	2,6	0,0	0,0	0,0	+	0,05	0,26	0,56	0	0	0	+	+	+	+	+
PO3	0,8	3,6	6,8	0,0	0,0	0,0	++	0,18	0,81	1,51	0	0	0	++	NA	NA	NA	0
PO4a	0,0	0,0	0,0	0,0	0,1	0,6	+	0	0	0	0	2.003	22.197	+	NA	NA	NA	0
PO4b	0,1	0,4	0,7	0,0	0,1	0,3	+	0,02	0,08	0,16	0	2.566	9.663	+	NA	NA	NA	0

Table 7-1. Effectiveness of the different measures

result of the option is shown using '-', being the number of '-' is indicative of the scale of the impacts. The majority of the measures lead to in-use consumer cost savings and relatively low payback times. Two of the options, PO2d and PO4b, do not have offsetting cost savings and are expected to result in an overall negative impact on consumers and employment.

Table 7-1 shows that all measures lead to a reduction of energy consumption (Effect1), with PO2a achieving the greatest savings with around 73% of the total. PO4 measures develop their full potential in the long term. Only measures PO4a and PO4b bring some small long-term savings of embedded energy due to the extension of the lifetime of the products.

Effect2 follows a similar behaviour as Effect1 regarding fuel saved, since fuel consumption decreases as in-use energy consumption does. Further, savings in embedded energy generated by circular economy measures are also reflected in a reduction of the use of materials.

As far as Effect3, PO2a and PO2b reduce the burden for manufacturers as explained in section 6.5. Also PO2d reduces burden by aligning the LSH Regulation with the standards, although in this case it is not possible to monetise the burden reduction of the alignment.

Possible trade-offs

PO4a and PO4b extend the lifetime of the products, for which reason a trade-off could be expected whereby the extension of the lifetime of old equipment would slow down the purchase of new devices, resulting in a final increase of in-use energy. Nevertheless, in the case of the measures under analysis the extension of the lifetime of the heaters does not really delay the purchase of more efficient ones: product subgroups that come in the scope for the first time are considered to increase their efficiency one-off when they come in the scope, but their efficiency will remain stable afterwards. The energy efficiency per unit remains therefore the same throughout the whole period of analysis, what means that a replaced heater will have the same efficiency as a new one belonging to the same product subgroup, as long as the former was already in the scope of the LSH Regulation at the moment of purchase. Circular economy measures will therefore reduce embedded energy without increasing in-use energy consumption.

7.2. Efficiency

This section assesses the benefit generated per monetary unit spent by the consumer for each policy option with respect to BAU, or in other words, the efficiency of consumer expenditure in generating the benefits addressed in section 7.1. Four aspects are assessed as regards their efficiency, namely in-use energy savings (Effic1), embedded energy savings (Effic2), fuel savings (Effic3) and material savings (Effic4).

It is clear from the analysis performed in section 6.1 that the reduction of in-use energy consumption follows an investment made by the consumer in the purchase of new equipment (in the case of PO2 and PO3) or on the upgrade of controllers (in the case of PO4b). The reduction of embedded energy in turn is either the consequence of the upgrade of controllers (in the case of PO4b) or an investment on repair costs (in the case of PO4a). Consequently, the consumer investment considered for Effic1 and Effic3 will be the costs of purchase of new equipment (in the case of PO2 and PO3) or the investment in new controllers (in the case of PO4b) whereas for Effic2 and Effic4 the consumer investment will correspond to either the investment in new controllers (PO4b) or the repair costs (for PO4a).

The efficiency is based on the analysis of time periods instead of yearly values, because yearly values would just provide the ratio benefits/investments for one single year and might not be representative of the real benefits produced by an investment, the effects of which are spread over several years. The periods of analysis are 2025 (year of implementation of the policy option) - 2030, 2025 - 2040 and 2025 - 2050.

	Effic1	Effic2	Effic3	Effic4

	in-use energy savings/consumer costs (purchase or modularity) (Kwh/€)			Score	embedded energy savings/repair costs (Kwh/€)			Score	fuel savings/consumer costs (purchase or modularity) (ton/M€)			Score	material savings/repair costs (ton/M€))			Score
	2025-2030	2025-2040	2025-2050		2025-2030	2025-2040	2025-2050		2025-2030	2025-2040	2025-2050		2025-2030	2025-2040	2025-2050	
PO2a	10,1	20,5	26,8	++	-	-	-	0	2.335	4.636	9.500	++	-	-	-	0
PO2b	8,7	17,6	21,5	++	-	-	-	0	2.023	3.980	7.615	++	-	-	-	0
PO2c	90,2	189,3	257,4	+++	-	-	-	0	20.945	42.825	91.040	+++	-	-	-	0
PO2d	3,1	6,4	8,8	+	-	-	-	0	714	1.456	3.101	+	-	-	-	0
PO3	15,9	27,9	32,9	++	-	-	-	0	3.693	6.322	11.807	++	-	-	-	0
PO4a	-	0	0	0	-	0,93	0,91	++	-	0	0	0	-	34,1	33,6	++
PO4b	0,1	0,1	0,2	0	-	0,02	0,06	+	15	28	53	0	-	0,9	3,3	+

Table 7-2. Efficiency of the different measures

Table 7-2 shows that PO2c gets the best score on Effic1 and Effic3, followed by PO3, PO2a, PO2b and PO2d. This is because the benefits generated by PO2c and PO3 require reduced consumer investments. Nevertheless, this high efficiency is nuanced by the small effectiveness shown in section 7.1, particularly for PO2c, which would make this option almost irrelevant if applied as a standalone measure. In fact, only PO2a and to a lesser extent PO2b and PO3 are able to produce tangible environmental benefits.

As far as Effic2 and Effic4, table 7-2 shows that PO4a is more efficient than PO4b, as the latter generates embedded energy savings at the expense of a very significant increase in consumer investment. The rest of the options do not generate embedded energy savings.

7.3. Coherence with other EU policies and legislative framework

7.3.1. Coherence with the overarching objectives of EU policies

From the analysis in section 6, it is clear that the proposed options aim to contribute to achieving Treaty objectives in relation to energy and the environment, and are fully consistent with EU policy goals such as set out in the Green Deal. In particular, the options contribute explicitly to *the EU's climate ambition for 2030 and 2050* and *mobilising industry for a clean and circular economy* as discussed in that Communication. The options have also been designed to ensure as much as possible a simplification of requirements in line with Better Regulation goals.

7.3.2. Coherence with Article 15(5) of the Ecodesign Framework Directive

Article 15(5) of the Ecodesign Framework Directive requires implementing measures to fulfil a number of criteria, against which all measures must be assessed. Only the criteria on "no significant negative impact on consumers in particular as regards affordability and life-cycle costs" is not fulfilled by some of the measures. In particular, PO2a and PO4b raise the price of the product by 5,8% and 8% respectively, as shown in table 6-7. However, PO2a leads to offsetting energy savings in use that rapidly repay the additional purchase cost. In contrast, the additional costs of PO4b are not recovered in the lifetime period of a local space heater. Likewise, PO2d increases the price of the product and is not able to offset it with energy savings. However, this measure still yields environmental benefits, as shown in table 7-1.

7.3.3. Coherence with the subsidiarity and proportionality principles

Section 3 already assesses subsidiarity. All policy options have implications on the design and characteristics of the product and are therefore better addressed at EU than national level in order to ensure the correct functioning of the internal market.

As far as proportionality, most of the measures deliver energy savings with limited product adaptations. The clarification of the coverage of the LSH Regulation through PO2a does not require any new developments but merely the application of a suitable heater-controller combination (slave heaters). The extensions of the scope of the LSH Regulation through PO2b and PO2c will not force development of new technologies. Most of the energy savings will be achieved by applying the ecodesign measures to product sub-groups that are today partially compliant (tube heaters and towel heaters). In the case of towel heaters, it is estimated that 10% of the devices in the market are already compliant with current ecodesign rules for fixed heaters. For tube heaters the measures concern only 8% of the units, since the remaining 92% is below 120 kW.

There will be additional but very limited administrative costs because there is no modification of the certification procedure. These have been calculated in section 6.2.3.

Concerning MSAs, there is no modification of the current framework as regards market surveillance and therefore no additional burden linked to the proposed policy options. On the contrary, it is expected that there will be efficiency improvements as indicated in section 6.5.

7.3.4. Overall coherence assessment

The factors assessed in this section have all been taken into account in summarising the coherence of the options. The overall assessment of coherence is shown in table 7-3.

PO2a	PO2b	PO2c	PO2d	PO3	PO4a	PO4b
+	++	++	0	++	++	--

Table 7-3. Overall assessment of coherence for the different measures

8. PREFERRED OPTION

8.1. Preferred policy option

In the view of the analysis carried out in section 7, the preferred policy option lumps together PO2a, PO2b, PO2c, PO2d, PO3, PO4a and PO4c. PO4b is ruled out due to its very low efficiency. This will result in one of the problem drivers not being addressed, but the figures clearly show that modularity is not cost-efficient. The efficiency of PO2d is not good either and its payback is of almost 38 years, which means that it is not fully coherent with Article 15(5) of the Ecodesign Framework Directive. Nevertheless, PO2d has been included because, in spite of the long payback period, the price increase per unit is low and it still yields some societal benefits in the long term. In addition, PO2d corrects a regulatory loophole whereby closed fronted open combustion devices are declared as open fronted, increasing energy consumption and creating an unlevelled playing field for manufacturers of these products.

As explained in the introductory part of section 6, no quantitative analysis has been carried out for PO4c due to lack of data, but it has been included in the preferred option because it will barely entail any cost for manufacturers and is expected to contribute to a better material recovery and recycling. Except PO2d, the measures grouped in the preferred policy option show between good and very good efficiencies and, put together, address both in-use energy savings and circular economy challenges.

Tables 8-1 to 8-5 summarise the impacts of the preferred policy option PO2+PO3+PO4a+PO4c. It is to note that, since PO2 extends the scope of the LSH Regulation, PO3 and PO4a will apply to more products and therefore their benefits will be leveraged with respect to the figures provided in section 6 for each measure assessed separately.

	2025-2030	2025-2040	2025-2050
Cumulative in-use energy savings (Twh)	10,8	56,2	116,4
Cumulative primary energy savings (Twh), including both in-use and embedded energy	22,3	116,2	241,0
Cumulative fuel saved (Mtons)	2,5	12,7	25,7
Cumulative material saved (tons)	0	2.563	28.397
Cumulative GHG emissions savings (Mton CO_2eq)	1,8	6,6	11,0
Cumulative acidification savings (ton SO_2eq)	7.775	29.065	48.793
Cumulative additional business turnover (M€)	2.617	6.554	9.122
Cumulative decreased consumer expenditure (M€)	748	7.893	18.323
Cumulative increased direct employment (number of jobs)	10.758	24.011	34.000

Table 8-1. Summary of cumulative values for the preferred policy option (PO2+PO3+PO4a+PO4c)

The administrative costs for this option are 1.272 k€, which are one-off and take place in 2025.

	Energy savings - Effect1							Resource efficiency - Effect 2							Reduced burden - Effect3 (from section 6.4)			
	In-use energy savings (Twh)			Embedded energy savings (Twh)			Score	Fuel savings (Mtons)			Material saved for manufacturing (Tons)			Score	MSAs	Manufacturers		Score
																Excess Administrative costs that disappear (k€)	Excess compliance costs that disappear (M€)	
	2025-2030	2025-2040	2025-2050	2025-2030	2025-2040	2025-2050		2025-2030	2025-2040	2025-2050	2025-2030	2025-2040	2025-2050		2025			
PO2a	7,7	40,4	84,0	0,0	0,0	0,0	+++	1,78	9,13	18,72	0	0	0	+++	+++	526,04	47,7	+++
PO2b	1,8	9,5	18,6	0,0	0,0	0,0	++	0,43	2,15	4,11	0	0	0	++	++	21,95	1,7	++
PO2c	0,1	0,6	1,3	0,0	0,0	0,0	+	0,03	0,13	0,28	0	0	0	+	NA	NA	NA	0
PO2d	0,2	1,2	2,6	0,0	0,0	0,0	+	0,05	0,26	0,56	0	0	0	+	+	+	+	+
PO3	0,8	3,6	6,8	0,0	0,0	0,0	++	0,18	0,81	1,51	0	0	0	++	NA	NA	NA	0
PO4a	0,0	0,0	0,0	0,0	0,1	0,6	+	0	0	0	0	2.003	22.197	+	NA	NA	NA	0
PO4b	0,1	0,4	0,7	0,0	0,1	0,3	+	0,02	0,08	0,16	0	2.566	9.663	+	NA	NA	NA	0
Preferred option	10,8	56,2	116,4	0,0	0,1	0,8	++++	2,50	12,73	25,65	0	2.563	28.397	++++	+++	548	49,4	++++

Table 8-2. Effectiveness analysis preferred policy option (PO2+PO3+PO4a+PO4c) compared to the rest of options

	Greenhouse gas emissions (embedded + in-use)						Acidification					
	Yearly savings ($MtCO_2$eq/year)			Cumulative savings ($MtCO_2$eq)			Yearly savings (tSO_2eq/year)			Cumulative savings (tSO_2eq)		
	2030	2040	2050	2025-2030	2025-2040	2025-2050	2030	2040	2050	2025-2030	2025-2040	2025-2050
PO2a	0,284	0,308	0,286	1,245	4,583	7,528	1.255	1.365	1.267	5.512	20.284	33.318
PO2b	0,068	0,070	0,054	0,295	1,078	1,682	300	309	238	1.306	4.769	7.445
PO2c	0,006	0,012	0,014	0,022	0,117	0,254	25	54	62	98	517	1.126
PO2d	0,012	0,025	0,029	0,046	0,240	0,523	51	112	128	202	1.062	2.314
PO3	0,027	0,025	0,020	0,128	0,418	0,634	121	109	86	567	1.850	2.807
PO4a	0,000	0,012	0,012	0,000	0,012	0,132	0	68	69	0	68	750
PO4b	0,003	0,007	0,006	0,011	0,057	0,123	12	35	33	50	269	608
Preferred option	0,403	0,464	0,424	1,757	6,563	10,974	1.782	2.073	1.808	7.775	29.065	48.793

Table 8-3. Environmental analysis preferred policy option (PO2+PO3+PO4a+PO4c) compared to the rest of options

	Effic1				Effic2				Effic3				Effic4			
	in-use energy savings/consumer costs (purchase or modularity) (Kwh/€)			Score	embedded energy savings/repair costs (Kwh/€)			Score	fuel savings/consumer costs (purchase or modularity) (ton/M€)			Score	material savings/repair costs (ton/M€)			Score
	2025-2030	2025-2040	2025-2050		2025-2030	2025-2040	2025-2050		2025-2030	2025-2040	2025-2050		2025-2030	2025-2040	2025-2050	
PO2a	10,1	20,5	26,8	++	-	-	-	0	2.335	4.636	9.500	++	-	-	-	0
PO2b	8,7	17,6	21,5	++	-	-	-	0	2.023	3.980	7.615	++	-	-	-	0
PO2c	90,2	189,3	257,4	+++	-	-	-	0	20.945	42.825	91.040	+++	-	-	-	0
PO2d	3,1	6,4	8,8	+	-	-	-	0	714	1.456	3.101	+	-	-	-	0
PO3	15,9	27,9	32,9	++	-	-	-	0	3.693	6.322	11.807	++	-	-	-	0
PO4a	-	0	0	0	-	0,93	0,91	++	-	0	0	0	-	34,1	33,6	++
PO4b	0,1	0,1	0,2	0	-	0,02	0,06	+	15	28	53	0	-	0,9	3,3	+

Preferred option	9,6	20,2	32,8	++	-	0,89	0,88	++	2.221	4.565	7.226	++	-	32,9	32,5	++

Table 8-4. Efficiency analysis preferred policy option (PO2+PO3+PO4a+PO4c) compared to the rest of the options

Increase in average unit price vs BAU, sale		**Saving vs. BAU in annual cost of energy consumption**		**Pay-back time (years)**
%/unit	€/unit	%	€/year	
7,6%	10,3	2%	3,1	4,5

Table 8-5. Affordability of the preferred policy option

The pay-back period of the preferred policy option is 4,5 years, half of the average lifetime for the shortest-lived product sub-group, portable heaters.

8.2. REFIT (simplification and improved efficiency)

The proposed policy option is in line with the REFIT objectives of simplification and reduction of red tape. The clarification and extension of the scope resulting from PO2 sets out a level playing field for industry and for MSAs, simplifying their work by laying down a clearer distinction between products in and out of the scope of the legislation. MSAs have reported to the Commission that confusing scope, in particular as regards of slave heaters, is the origin of loss of time for selection of the products to be inspected, discussions with manufacturers and additional red tape. With the planned modification MSAs will be able to spend more time on actual enforcement. In addition, by broadening the scope of the LSH Regulation there will be less risk that Member States take individual legislative action to regulate the range of products currently out of the scope and therefore prevents the rise of cost of compliance with different national rules.

Additionally, as discussed in section 5.4.2, there is no need to include third party certification and the corresponding costs will be saved.

8.3. Application of the 'one in, one out' approach

There will be around 1,3 M€ administrative costs linked to the need to update the DoC's for the products currently in the scope of the LSH Regulation and to issue a new DoC for the products that will come in the scope after the adoption of the measures. No potential cost savings in administrative activities performed by businesses were identified that could be included in the current revision of the legislative framework on local space heaters.

8.4. Overview in the context of current energy prices

Energy prices are sharply increasing due to the current situation of political instability. This strong increase, which was not foreseen when the review study was carried out, could have encouraged the introduction of additional ecodesign measures because higher energy costs would make those measures to be cost-efficient. For instance stricter ecodesign limits by reducing the correction factors F(2) and F(3), which is an option ruled out at an early stage as explained in section 5.4, could have been included among the policy options under assessment.

It is not possible at this stage to make a fully-fledged impact assessment considering current energy prices because the available scenarios have not been updated with those prices. It is however possible to carry out an analysis of the impact on affordability (payback period) of introducing additional ecodesign measures, for different energy prices. Affordability is a key parameter for this analysis because other important impacts such as energy consumption reduction and GHG emission savings can only improve with the inclusion of additional ecodesign measures.

This analysis will consider a reduction of F(2) and F(3) that will double the impact of the preferred option PO2+PO3+PO4a+PO4c and will therefore bring about a further 2% reduction of energy consumption. This reduction will be added to the savings achieved under the preferred policy option. The affordability analysis will be carried out for the preferred policy option + the abovementioned 2% of additional savings with both REF2020 energy prices and the last EU prices (1 semester of 2021) made available by Eurostat[48]. The latter are assumed to remain constant over the period of study, from 2025 to 2050. Energy prices will in reality vary with time, but as said above it is not possible to predict that variation in the absence of a reliable projection. Nevertheless, fixed energy prices can still give an idea of the influence on payback of stricter ecodesign rules. Stock and sales will remain the same because stricter ecodesign measures are considered not to have any influence on those two parameters, neither in terms of global numbers nor as regards the share of the market for each of the types of local space heaters.

The affordability analysis casts the following result:

	Energy prices	**Increase in average unit price vs BAU, sale**		**Saving vs. BAU in annual cost of energy consumption**		**Pay-back time (years)**
		%/unit	**€/unit**	**%**	**€/year**	
Preferred PO	REF2020	7,6%	10,3	2%	2,27	4,5
Preferred PO + 2% additional savings	REF2020	10,3%	14,0	2,8%	3,1	4,5
Preferred PO + 2% additional savings	1 semester 2021	10,3%	14,0	3,4%	3,8	3,7

Table 8-4. Affordability under higher energy prices

The conclusion from this analysis is that inclusion of stricter ecodesign measures would be cost-efficient even under REF2020 energy prices. It is nevertheless necessary to take into account that the full review study and impact assessment were conducted in a context very different to the current one. As explained in Annex 6, the review study concluded that the current LSH Regulation had been effective but less effective than initially expected. This was not attributed to the lack of technical ambition, but to the existence of loopholes that hindered the maximisation of the energy savings. Focus in this IA has then been set on the removal of those loopholes. In addition, it seemed reasonable to keep the existing ecodesign limits because they entered into application on 1 January 2018, this is after the review study had started, and a reliable assessment of the limits would have only been possible some years after their date of application. A new reduction of the ecodesign limits without appropriate verification would have undermined the support to the legislative amendment from the industry and from some Member States.

9. HOW WILL ACTUAL IMPACTS BE MONITORED AND EVALUATED?

The main tool to verify compliance with ecodesign and energy labelling requirements is the market surveillance carried out by national authorities. Further, homogeneity of market surveillance actions at EU level is guaranteed by the Administrative Cooperation Groups, under which MSAs and the Commission meet several times per year with the objective to ensure efficient, comprehensive and consistent market surveillance across the Union.

In addition, surveillance activities are supported by the Commission through EEPLIANT3, which is a Horizon 2020-funded project with a budget of €6,85 million[49], involving a consortium of MSAs.

[48] Electricity price statistics - Statistics Explained (europa.eu)
[49] https://cordis.europa.eu/project/id/832558

The project runs from June 2019 until November 2023 and includes a work package on heaters[50], among them local space heaters. In the frame of EPPLIANT3, MSAs carry out market surveillance activities on their national markets and make recommendations that can be used later by national authorities in their regular enforcement activities. Those activities are focused on document inspections, analysis of the content of web-shops, verification of EPREL data and testing of appliances.

On local space heaters, each MSA will have to screen at least 15 models and select at least 10 models for document inspection. In total, 207 models are expected to be screened, broken down in three product subgroups: gas-fuelled, biomass-fuelled and electric. Out of those 207, 143 models will be selected for document inspection, including product information stored in EPREL, and 143 for inspection of online labels. In addition, 54 products shall be tested for compliance with ecodesign rules at lab facilities.

The project is ongoing until mid-2023 and the results are not yet available, but it is expected that EPPLIANT3 is able to estimate the overall level of compliance of the products in the market with respect to the ecodesign rules. MSAs will need to take the EPPLIANT3 conclusions into account in order to optimise efforts by focussing on those specific issues where level of compliance is deficient. This will also apply to the enforcement of the upcoming TD Ecodesign and Energy Labelling Regulations.

Finally, the revised TD Ecodesign and Energy Labelling Regulations will include a review clause that will mandate, inter alia, the evaluation of the effectiveness and regulatory efficiency and relevance of the revised regulations 5 years after their entry into force. The results of this evaluation, which will involve the acquisition of independent market data as well as extensive stakeholder consultation, should be presented to stakeholders and Member States in the Ecodesign and Energy Labelling Consultation Forums. In addition, the European Commission will be regularly analysing the latest updated data of the database EPREL, in order to monitor the evolution of the market as regards of energy efficiency, condensation efficiency, refrigerants used in the appliances and level of noise. This will allow the Commission to early detect possible technical improvements that can be assessed in future revisions.

[50] EEPLIANT - Heaters

ANNEX 1: PROCEDURAL INFORMATION

1. LEAD DG, DECIDE PLANNING/CWP REFERENCES

DG ENER, PLAN/2019/5387, Commission Ecodesign and Energy Labelling Working Plan 2016-2019 (COM/2016/0773 final).

2. ORGANISATION AND TIMING

The revision of the Ecodesign Regulation on local space heaters has been included in the Ecodesign and Energy Labelling Working Plan 2016-2019.

A draft of the impact assessment report was submitted on 2 March 2022 to the members of the Ecodesign and Energy Labelling Interservice Steering Group, involving the following DGs: SG, AGRI, BUDG, CLIMA, CNECT, COMM, COMP, DEFIS, EAC, ECFIN, ECHO, EMPL, ENV, ESTAT, FISMA, GROW, JRC, JUST, IDEA, MOVE, REFORM, REGIO, RTD, SANTE, SJ, TAXUD. A meeting took place on 28 March. There were no major comments. DGENV asked to make sure that the proposed measure on information about end of life does not overlap with the WEEE Directive. SJ submitted comments in writing after the meeting that have been considered in their entirety.

3. CONSULTATION OF THE RSB

The RSB was consulted on 11 May 2022. Prior to that, an upstream meeting took place on 21 March, in which the Board provided several recommendations to DG ENER, highlighting in particular the need to ensure coherence with related initiatives, the importance to consider scenarios in line with the "fit-for-55" package, the relevance of the cost-benefit analysis as basis for the adoption of choices on the policy options and the inclusion of sensitivity analysis on crucial parameters.

4. EVIDENCE, SOURCES AND QUALITY

The departing point for this impact assessment is the review study carried out by the consultant Viegand Maagoe, finished and handed over to the Commission in September 2019. The review study provides some of the input data used for the baseline scenario and the modelling of the different policy options, namely sales, stock and product prices, all of them split by product type. Sales data have been purchased to GfK, a reputed company in the field of data provision and analysis for the consumer goods industry. Stock data have been derived from sales and average lifetime for each product family. Average lifetime has been extensively discussed with APPLiA, the main industry interlocutor at EU level.

As it will be further explained in Annex 4, the main challenge encountered as regards of data accuracy is that the existing data made available by GfK and in PRODCOM do not have enough level of granularity as to match the different sub-categories of local space heaters. For this reason, it has been necessary to carry out a detailed crosscheck of the available information with industry representatives in order to break down the data by product category.

Electricity prices, heating degree days and building renovation rate have been obtained from REF2020, in order to ensure coherence and consistency of the impact assessment baseline scenario with existing energy policies.

ANNEX 2: STAKEHOLDER CONSULTATION (SYNOPSIS REPORT)

1. REVIEW STUDY AND STAKEHOLDER CONSULTATION

The review study was launched in October 2017 and was completed in September 2019. It was developed in an open process, taking into account inputs from relevant stakeholders including manufacturers and their associations, environmental NGOs, consumer organisations and Member States. A dedicated open public website was made available from October 2017 in order to facilitate the monitoring of the status of the study, get related information and track the meetings taking place.

During the study, two open stakeholder consultation meetings were organised at the Commission premises in Brussels on 22 June 2018 and 6 February 2019, with the following outcome:

- At the first meeting on 22 June 2018, the draft interim report, containing tasks 1 and 2 of the MEErP was discussed. The report was made available to stakeholders ahead of the meeting on 25 May, and comments were received after the meeting until 3 August 2018. A revised report was made available on 22 October 2018, taking into account all comments received during and after the meeting.

- At the second meeting on 6 February 2019 the draft final report, containing all 7 of the MEErP tasks was discussed. The report was made available to stakeholders on 21 December 2018, and comments were received until 28 February 2019.

Following the second meeting, the final report was made available on 15 May 2019, incorporating the additional comments received during and after that meeting.

In addition to the two stakeholder consultation meetings, the study team held contacts with several stakeholders via e-mail or phone as well as a few face-to-face meetings.

2. CONSULTATION FORUM

The Consultation Forum consists of a balanced representation of Member State representatives, industry associations and NGO's, in line with Article 18 of the Ecodesign Framework Directive.

A Consultation Forum specifically on local space heaters was held on 10 September 2019, where a working document consisting of a legislative proposal based on the results of the review study was discussed. The minutes, working documents and stakeholder comments received in writing before and after the Consultation Forum meeting were posted on CIRCABC[51]. Around 90 written comments were received from stakeholders, including 12 Member States.

A second Consultation Forum was held on 11 March 2022, in order to discuss the following aspects of the working documents: date of application, convenience of introducing requirements on modularity and list of spare parts. A summary is presented in section 5 of this Annex.

3. STAKEHOLDER COMMENTS DURING AND AFTER THE CONSULTATION FORUM

The comments on the Working Document can be summarized as follows:

[51] ecodesign - Library (europa.eu)

- Date of application: five stakeholders commented that the application of the new ecodesign measures should be changed from 1 January (as suggested in working documents) to 1 July (3 stakeholders) or 1 April (2 stakeholders), in order to avoid its application in the middle of the heating season. This will facilitate to manufacturers stock handling and update of brochures. The Commission agrees and this will be taken into account for setting the application date of the implementing measures.
- Scope: most stakeholders agreed with the proposed extension of the scope to towel heaters and slave heaters, although some industry associations (EUHA and EIHA) opposed to the inclusion of slave heaters. There were different opinions on the level of requirements and sizes of towel heaters to be included in the scope, namely whether the towel heaters requirements should be applicable to towel heaters over 100, 150, 200, 250 or 300 Watts, and which control factors to be applied and their values. In the view of the stakeholders' input, it has been decided that all towel heaters will be included in the scope, although requirements for heaters below 250 Watts will be more lenient.
- Pilot flame: it was discussed if permanent pilot flames in heaters using gas and oil fuels should be banned, and to limit the duration time of non-permanent pilot flames to what is strictly needed to the pre-heating function that they provide. Given the lack of agreement among stakeholders and the possibility that banning specific features can create a significant burden in the market, it has been decided to keep the current approach of having a correction factor F(5) accounting for the negative contribution of pilot flames to energy consumption.
- Availability of spare parts: NGO's suggested to extend the availability of spare parts beyond 10 years and according to the average lifetime for each product group. The proposal in this impact assessment includes 10 years for all product groups. This is because it has been considered that the consumer will have little incentive to repair space heaters, in particular the cheapest ones, and that the availability of spare parts for very long times would be a cumbersome requirement for manufacturers of these devices.
- Information on controllers and heaters to be sold separately: eight stakeholders made comments on this topic. Various suggestions were put forward, namely that information should only be included for those controllers and heaters sold separately, and that the information in the packaging should be limited to the most general aspects, since it is cumbersome to translate long messages in all EU languages. A Member State also suggested to set up a classification of controllers according to their contribution to energy efficiency. In the view of this feedback, the Commission will propose to include in the legislative act information requirements which are simple, easily understandable for the consumer and not cumbersome for display and translation.
- Stricter requirements on energy efficiency: some Member States and NGO's suggested to remove the bonuses provided by F(2) and F(3) (which account for the increase of energy efficiency due to functionalities of the controllers) for functionalities that in practice do not bring about significant energy savings, such as manual adjustments or mechanic thermostats. A Member State also suggested to make mandatory some functionalities, such as the detection of open window. Stricter ecodesign requirements have nevertheless been ruled out as explained in the main report of this impact assessment.
- Low power modes: stakeholders generally agreed to incorporate new low power modes that are specific to local space heaters.
- Modular design: NGOs and some Member States proposed to mandate the modular design of local space heaters so that controllers can be manually detached from the heater. This option has been ruled out because of the possible negative impact of that requirement on prices of local

space heaters, in particular the cheapest ones, and the doubts about consumer's reaction to this concept.

- Gas appliances: several comments targeted household gas appliances, requesting to clarify the definitions of the subcategories defined in the LSH Regulation and the appertaining limit values. Following this information, definitions of subcategories and limit values will be adapted accordingly.

An overview of the stakeholders and their participation is given in **Table A2-1. Overview of stakeholder participation in the Review study.**

Stakeholder Organisation	Participation in 1st stakeholder meeting	Participation in 2nd stakeholder meeting	Participation in CF	Written comments draft interim report	Written comments draft final report
ADAX	x			x	x
Agoria	x	x		x	
Alke BV (Agricultural local space heaters)				x	
ANEC	x	x	x		
APPLiA	x	x	x	x	x
ASSOBAGNO	x	x			x
Association of Heating Appliances Producers and Importers in Poland		x			
AT Energy Agency			x		
Barbas Bellfires BV	x	x			x
BE (Federal Ministry for Economy and FPS Health, Food Chain Safety and Environment)	x		x		
BEAMA	x	x			
BEUC	x	x			
BG Ministry of Economy and Energy			x		
CEFACD	x	x		x	x
Cenelec TC59X/WG12		x			
CLASP			x		
CZ Ministry of Industry and Trade			x		
DE (Bundesanstalt für Materialforschung und –prüfung (BAM) , Ministry of Environment, Climate Protection and the Energy Sector , UBA)	x	x	x		x
De'Longhi Appliance	x	x			x
Deltacalor					x
UK Department for Business Energy and Industrial Strategy	x				

Stakeholder Organisation	Participation in 1st stakeholder meeting	Participation in 2nd stakeholder meeting	Participation in CF	Written comments draft interim report	Written comments draft final report
DK Danish Energy Agency			x		x
DRU Verwarming B.V.	x				
ECOS	x	x	x		x
EEB			x		
EGEC		x			
EHI - European Heating industry	x	x	x		x
EHPA			x		
ELVHIS	x	x		x	
ENSTO				x	
EPEE			x		
EUHA	x		x		x
Eurovent	x	x	x	x	x
FI Finnish Ministry of Environment	x		x		
Fenix				x	
Fondital Spa				x	
FR Ministère de l'Ecologie, du Développement Durable et de l'Energie (Ministry for the Ecological and Inclusive Transition)		x	x		
GAS.BE - Testing house	x				
Gaz Réseau Distribution France					x
Gazco Ltd	x	x			
GIFAM	x				
Glen Dimplex	x				x
GRDF		x			
Groupe Atlantic	x	x		x	x
Groupe MULLER		x			
HHIC	x	x			x
HKI Industrieverband e. V.	x	x		x	x
Home Appliance Europe		x			
IE Enterprise Ireland			x		
International Network for Sustainable Energy (INFORSE) - Europe	x	x	x		x
IR Focus ApS		x			x
IRSAP				x	
IT ENEA			x		
JRAIA			x		
KÜBLER GmbH (Industrial Heating Systems)		x			

Stakeholder Organisation	Participation in 1st stakeholder meeting	Participation in 2nd stakeholder meeting	Participation in CF	Written comments draft interim report	Written comments draft final report
LCIE	x	x			x
Liquid Gas Europe			x		
LU ILNAS			x		
LV Consumer Rights Protection Centre			x		
Marcogaz	x				
Marsan industrial s.a. Haverland	x				
Muller Group	x				
Netherlands Enterprise Agency	x		x	x	
NO The Norwegian Water Resources and Energy Directorate, NVE	x	x	x	x	x
nVent (member of the EUHA)	x	x			x
ORGALIM			x		
PL Ministry of Energy			x		
PT Directorate General for Energy and Geology			x		
Rettig ICC	x	x			x
SE Swedish Energy Agency			x		
SK Ministry of Economy			x		
SPF santé publique, sécurité de la chaine alimentaire et environnement	x				
SPIUG					x
Stiebel Eltron GmbH & Co. KG [also CENELEC TC 59X WG12]	x	x			x
Taelek Oy					x
Trade Association - HEATING AND HOTWATER INDUSTRY COUNCIL	x				
Vailland Group	x	x			
ZEHNDER	x	x			
Weber Energieconsult				x	x
University of California					x

Table A2-1. Overview of stakeholder participation in the Review study

4. PUBLICATION OF THE CALL FOR EVIDENCE AND STAKEHOLDERS FEEDBACK

For this IA, a derogation has been granted and therefore no 12-weeks Open Public Consultation has been necessary. A Call for Evidence has nevertheless been published at the Have your Say Portal[52] on 21 January 2022, which feedback period finished on 18 February 2022. A total of 14 comments were received from the following stakeholders: FLA, GRDF, APPLiA, EdEn, ECOS, ELVHIS, ANEC/BEUC, EIHA, CEFACD, IRSAP, EUHA, EHI, Eurofuel and Fondital Spa. The comments provided are summarised as follows:

- Inclusion of slave heaters: EUHA and EIHA oppose to the inclusion of slave heaters in the scope of the LSH Regulation, claiming that the loophole is not in the definition but in the interpretation of that definition provided in the FAQ section at the Commission's website.

- Inclusion of towel heaters: FLA opposes to the inclusion of towel heaters below 300W.

- Inclusion of tube heaters above 120 kW: industry representatives such as GRDF and ELVHIS commented in favour of this measure.

- Modularity of controllers: whereas ECOS supports modularity, APPLiA expresses a frontal opposition.

- Availability of spare parts: generally supported by industries and environmental NGO's, although with different levels of ambition.

- Ecodesign and pollutant emission requirements: ECOS and ANEC/BEUC support setting more stringent requirements on both aspects.

The feedback to the call for evidence does not deviate therefore from the opinions provided during the review stucy.

5. SECOND MEETING OF THE CONSULTATION FORUM ON 10 MARCH 2022

Since the last meeting of the Consultation Forum for local space heaters was held on September 2019, a second Consultation Forum took place on 10 March 2022, with the main objective of updating the group on the progress of the IA and on getting feedback from stakeholders about some specific issues that needed some additional discussion. These issues related to the list of available spare parts, the definitions of gas and oil heaters, modularity and low power modes. Stakeholders generally agreed to the proposed list of spare parts and clarified some aspects related to the definitions of gas and oil heaters. As regards modularity and low power modes, stakeholders kept their positions already expressed at the 2019 session.

[52] Energy efficiency – ecodesign requirements for local space heaters (review) (europa.eu)

ANNEX 3: WHO IS AFFECTED AND HOW?

1. PRACTICAL IMPLICATIONS OF THE INITIATIVE

The LSH Regulation will apply to manufacturers, importers and authorised representatives of local space heaters in the scope of the LSH Regulation, which will need to comply with the ecodesign requirements summarised in Table A3-1.

Who	What	When
Manufacturers, importers and authorised representatives	Fulfilment of the ecodesign requirements by the new products in the scope	2025
Manufacturers, importers and authorised representatives	Fulfilment of the ecodesign requirements on low power modes	2025
Manufacturers, importers and authorised representatives	Availability of spare parts during the lifetime of the products	2025

Table A3-1. Summary of new requirements to be fulfilled

2. SUMMARY OF COSTS AND BENEFITS

I. Overview of Benefits (total for all provisions) – Preferred Option		
Description	***Amount***	***Comments***
Direct benefits		
Cumulative primary energy savings (see table 8-1)	10,9 Twh by 2030 and 117,2 Twh by 2050	
Cumulative GHG emissions savings (see table 8-3)	1,8 Mt CO_2eq by 2030 and 11 Mt CO_2eq by 2050	
Cumulative acidification reduction (see table 8-3)	7.775 ton SO_2eq by 2030 and 48.793 ton SO_2eq by 2050	
Cumulative economic savings due to the reduction of externalities (GHG and acidification)	335 M€ by 2030 and 2.076 M€ by 2040	CO_2 emission savings have been monetised at a price of 100€/tonCO_2eq, and acidification at a price of 20€/KgSO_2eq, which are common values used by the Commission in its IAs.
Cumulative fuel savings (see table 8-1)	2,5 Mton by 2030 and 25,7 Mton by 2050	
Cumulative material savings for manufacturing (see table 8-1)	0 ton by 2030 and 28.397 ton by 2050	Material and water savings take place by means of circular economy measures, which start to be effective after 2030. Therefore, no savings can be expected by 2030 for these three parameters.
Cumulative water savings (see Annex 5)	0 Mlitres by 2030 and 247 Mlitres by 2050	
Cumulative waste savings (see Annex 5)	36.387 ton by 2030 and 453.813 ton by 2050	Waste savings are majorly due to waste saved in the production of fuel for powering the electric local space heaters, and a very small share to the waste generated during the manufacturing process.
Cumulative additional business turnover (see table 8-1)	2.617 M€ by 2030 and 9.122 M€ by 2050	
Cumulative consumer expenditure savings (see table 8-1)	748 M€ less by 2030 and 18.323 M€ less by 2050	
Cumulative direct increased	10.758 extra jobs by 2030 and 34.000 by	

employment (see table 8-1)	2050	

Table A3-2. Overview of Benefits for $PO2_f$+PO3+4a as compared to BAU

II. Overview of costs – Preferred option							
		Citizens/Consumers		**Businesses**		**Administrations**	
		One-off	**Recurrent**	**One-off**	**Recurrent**	**One-off**	**Recurrent**
Costs of increasing the energy efficiency of local space heaters	Direct adjustment costs [53]				80 M€/year in 2030 and 34 M€/year in 2050		
	Direct administrative costs[54]						
	Direct regulatory fees and charges						
	Direct enforcement costs						
	Indirect costs						
Costs of repair	Direct adjustment costs				0 M€ in 2030 and 7 M€/year in 2050		
	Direct administrative costs						
	Direct regulatory fees and charges						
	Direct enforcement costs						
	Indirect costs						
Costs of updating the DoC for models in the scope and issuing a DoC for new models in the scope	Direct adjustment costs						
	Direct administrative costs			1,3 M€			
	Direct regulatory fees and charges						
	Direct enforcement costs						
	Indirect costs						

[53] Direct adjustment costs are the compliance costs, and have been calculated by subtracting the 2025 manufacturer turnover for the preferred policy option from the 2025 manufacturer turnover for BAU. For the preferred policy option, the manufacturer turnover in 2025 is 1.137 M€, and therefore compliance costs are 1.131 – 1.047 (BAU value in 2025) = 90 M€. The adjustment cost per unit are obtained by dividing the direct adjustment costs by the expected sales in 2030 (17.113.230).

[54] Direct administrative costs for the preferred policy option have been calculated from the values given in section 6.2.3.2, by summing the costs of drawing up a new DoC for PO2a, PO2b and PO2c and the general cost of updating the existing DoCs (352.416 + 197.568 + 32 + 721.792 = 1.271.808)

Costs related to the 'one in, one out' approach							
Total	Direct adjustment costs						
	Indirect adjustment costs						
	Administrative costs (for offsetting)			1,3 M€			

Table A3-3. Overview of costs for PO2 +PO3+PO4a+PO4c

3. RELEVANT SUSTAINABLE DEVELOPMENT GOALS

III. Overview of relevant Sustainable Development Goals – Preferred Option(s)		
Relevant SDG	**Expected progress towards the Goal**	**Comments**
SDG no. 7 - Affordable and clean energy, target 7.3 (double the improvement of energy efficiency)	Expected increased in energy efficiency of local space heaters will save 10,8 Twh of in-uses energy by 2030 and 116,4 Twh by 2050, contributing positively to reduce the energy bill in households.	
SDG no.8 – Decent work and economic growth; target 8.4 (improve resource efficiency in consumption and production)	The preferred policy option will save both fuel used for powering the heaters (2,5 Mtons by 2030 and 25,7 Mtons by 2050) and material used during the manufacturing process (28.397 tons by 2050).	The fuel saving will mostly correspond to the reduction of in-use energy consumption by measures included in PO2 and PO3, whereas material savings will be due to PO4a.
SDG no. 10 – Reduced inequalities; target 10.1 (reduced income inequalities)	Energy efficiency measures will yield direct cumulative consumer savings of 748 M€ by 2030 and 18.323 M€ by 2050, thus increasing the available income by the same amount.	
SDG no. 11 – Sustainable cities and communities; target 11.6 (reduce the environmental impact of cities)	There is no data about the distribution of local space heaters between cities and rural areas, but it is expected that the vast majority of energy savings take place in densely populated areas, thus helping to reduce the environmental impact of cities.	
SDG no. 12 – Responsible consumption and production; targets 12.2 (sustainable management and use of natural resources), 12.5 (substantially reduce waste generation)	Availability of spare parts will save 0,77 Twh of accumulated embedded energy by 2050 (target 12.2) and will generate material savings up to an amount of 28.397 tons in 2050 (target 12.5) thus reducing waste at EoL.	- The effects of the measures are realised in the long-term and therefore no impacts are foreseen by 2030.
SDG no. 13 – Climate action, target 13.1 (strengthen resilience and adaptative capacity to climate-related hazards)	Expected accumulated GHG emissions savings of 1,8 MtonCO2-eq. in 2030 and 11 MTtonCO2-eq. in 2050, contributing to the deceleration of climate change.	The decrease in energy GHG emissions is faster during the first years, and levels off over time due to the progressive decarbonisation of the energy grid.

Table A3-4. Relevant sustainable development goals for $PO2_f$+PO3+4a

ANNEX 4: ANALYTICAL METHODS

This Annex describes in detail the data and assumptions the models are based on.

1. SALES

Current and past sales figures are based on data provided by GfK. The main difficulty for the development of the sales forecast has been that the sales figures provided by GfK, PRODCOM and other sources do not provide enough granularity as to fit the specific product categories. For this reason the study team combined the latest data from GfK (and to a lesser extend PRODCOM) with the distribution between various base cases in the previous preparatory study and impact assessment, which was then presented at the two stakeholder meetings held during the review study. Stakeholder inputs to this data were then collected and projections were adapted. This process was reiterated in the second stakeholder meeting, until no further comments were received. The main outcomes from this process were the following:

- The sales of **electric space heaters** will be stable between 2016 and 2050. Different stakeholders do not agree among themselves, and some propose and increase while others suggest a decrease in future sales. Stable sales was the compromise most could agree to.
- The sales numbers of domestic **gas appliances** estimated in the 2015 impact assessment turned out to be too high, since the market has been declining for years. Therefore the current sales of domestic gas appliances were corrected on the basis of the data from the two industry associations CEFACD and HHIC[55], which together represent around 80 % of the total market. As a result, it has been estimated that the sales of domestic gas appliances will decrease 30 % towards 2030 and remain stable afterwards.
- Regarding liquid fuel heaters, no stakeholders have supplied any information, but it is known that the sales of liquid fuel heaters are roughly 10 % of the sales of gas heaters, for which reason it has been assumed that the share will remain constant over the considered period.

During the review process and dialogue with stakeholders, it also became apparent that the category "radiant heaters" (electric) did not represent any products on the market in the way it was defined in the LSH Regulation. This category was therefore left empty in all calculations, and removed from the definitions in the proposal for a legislative review.

Table A4-1 displays the results of the projection.

	2010	2020	2030	2050
Electric (1000 units)				
Electric portable local space heaters	6.787	7.144	7.144	7.144
Electric fixed local space heaters with a nominal heat output above 250 W	6.894	5.018	5.018	5.018
Electric fixed local space heaters with a nominal heat output equal or below 250 W	1.724	1.254	1.254	1.254
Electric storage local space heaters	311	269	269	269
Electric underfloor local space heaters	1.225	1.272	1.272	1.272
Electric visibly glowing radiant local space heaters with a nominal heat output above 1.2 kW	144	151	151	151
Electric visibly glowing radiant local space	62	65	65	65

[55] CEFACD https://www.cefacd.eu/ and HHIC https://www.hhic.org.uk/

heaters with a nominal heat output equal or below 1.2 kW				
Towel heaters	1.885	1.885	1.885	1.885
Total	19.033	17.058	17.058	17.058
Gas fuel (1000 units)				
Gaseous luminous local space heaters (non-residential)	5	4	3	3
Gaseous tube local space heaters (non-residential)	5	4	3	3
Gaseous tube local space heaters 120-300 kW (non-residential)	0,4	0,3	0,3	0,3
Gaseous fuel open combustion/open fronted with exhaust restriction	10	10	7	7
Gaseous fuel open combustion/closed fronted	28	19	15	15
Gaseous fuel balanced flue	51	28	21	21
Gaseous flueless heaters	0,0	0,0	0,0	0,0
Gaseous open to Chimney heaters	0,0	0,0	0,0	0,0
Total	100	66	50	50
Liquid fuel (1000 units)				
Liquid tube local space heaters (non-residential)	0,5	0,4	0,3	0,3
Liquid tube local space heaters 120-300 kW (non-residential)	0,0	0,0	0,0	0,0
Liquid fuel open combustion/open fronted with exhaust restriction	1	1	1	1
Liquid fuel open combustion/closed fronted	3	2	1	1
Liquid fuel balanced flue	5	3	2	2
Liquid flueless heaters	0,0	0,0	0,0	0,0
Liquid open to Chimney heaters	0,0	0,0	0,0	0,0
Total	10	6	5	5

Table A4-1. Estimated EU-27 (2020) sales from 2010-2050

The UK sales share has been removed from the dataset due to Brexit, considering that it remains constant for the entire modelling period.

Product	**UK share**
Electric portable	5.7%
Electric fixed heat output >250 W	5.7%
Electric fixed heat output <250 W	5.7%
Electric storage	5.7%
Electric underfloor	5.7%
Electric visibly glowing l heat output >1.2 kW	5.7%
Electric visibly glowing heat output < 1.2 kW	5.7%
Towel heaters	5.7%
Gas luminous	80.1%
Gas tube	80.1%
Gas tube 120-300 kW	80.1%
Gas open combustion, open fronted	83.8%
Gas open combustion/closed fronted	83.8%
Gas balanced flue	41.7%

Gas flueless	100.0%
Gas open to chimney	100.0%
Liquid tube	80.1%
Liquid tube 120-300 kW	80.1%
Liquid open combustion, open fronted	83.8%
Liquid open combustion, closed fronted	83.8%
Liquid balanced flue	41.7%
Liquid flueless	100.0%
Liquid open to chimney	100.0%

Table A4-2. UK sales shares for local space heaters

2. STOCK

The stock of local space heaters in the EU has been estimated on the basis of the sales and the expected product lifetimes considered in the preparatory study, the latter adjusted according to stakeholders' inputs. The lifetime of local space heaters ranges between 7 years for gas heaters to 40 years for electric underfloor heating systems.

In the current stock model, the statistical standard deviation from the middle lifetime value is assumed to be 2 to 6 years (depending on the type of heater), so that a normal lifetime distribution is calculated by adding and subtracting the deviation from the middle lifetime value and applying a Gaussian distribution.

Product	**Average lifetime**	**Standard deviation on lifetime**
Electric portable	9	2
Electric fixed heat output >250 W	15	3
Electric fixed heat output <250 W	15	3
Electric storage	15	6
Electric underfloor	40	2
Electric visibly glowing l heat output >1.2 kW	9	2
Electric visibly glowing heat output < 1.2 kW	9	2
Towel heaters	15	3
Gas luminous	15	3
Gas tube	20	4
Gas tube 120-300 kW	20	4
Gas open combustion, open fronted	20	4
Gas open combustion/closed fronted	20	4
Gas balanced flue	20	4
Gas flueless	7	2
Gas open to chimney	7	2
Liquid tube	20	4
Liquid tube 120-300 kW	20	4
Liquid open combustion, open fronted	20	4
Liquid open combustion, closed fronted	20	4
Liquid balanced flue	20	4
Liquid flueless	7	2
Liquid open to chimney	7	2

Table A4-3 Estimated lifetimes for local space heaters

Sales figures and lifetime are combined to obtain the stock of each product sub-group year by year. For each given year, the sales of a product sub-group on that year are combined with the abovementioned Gaussian lifetime distribution to calculate the share of those sales that will still be in stock for the remaining years until the end of the product lifetime. By adding the stocks estimated for the sales of different years, the total yearly stock is derived.

	2010	2020	2030	2050
Electric (1000 units)				
Electric portable local space heaters	61.709	66.955	67.808	60.664
Electric fixed local space heaters with a nominal heat output above 250 W	101.743	92.004	78.795	77.698
Electric fixed local space heaters with a nominal heat output equal or below 250 W	25.436	23.001	19.699	19.425
Electric storage local space heaters	4.441	4.426	4.220	4.158
Electric underfloor local space heaters	38.743	44.338	48.142	51.353
Electric visibly glowing radiant local space heaters with a nominal heat output above 1.2 kW	1.307	1.418	1.436	1.436
Electric visibly glowing radiant local space heaters with a nominal heat output equal or below 1.2 kW	560	608	615	615
Towel heaters	23.786	28.310	29.194	29.194
Total	257.724	261.060	249.909	244.543
Gas fuel (1000 units)				
Gaseous luminous local space heaters (non-residential)	67	71	63	52
Gaseous tube local space heaters (non-residential)	86	92	86	69
Gaseous tube local space heaters 120-300 kW (non-residential)	7	7	7	6
Gaseous fuel open combustion/open fronted with exhaust restriction	210	212	192	152
Gaseous fuel open combustion/closed fronted	796	591	416	304
Gaseous fuel balanced flue	1.701	1.092	644	439
Gaseous flueless heaters	0	0	0	0
Gaseous open to Chimney heaters	0	0	0	0
Total	2.867	2.065	1.408	1.021
Liquid fuel (1000 units)				
Liquid tube local space heaters (non-residential)	9	9	9	7
Liquid tube local space heaters 120-300 kW (non-residential)	1	1	1	1
Liquid fuel open combustion/open fronted with exhaust restriction	21	21	19	15
Liquid fuel open combustion/closed fronted	80	59	42	30
Liquid fuel balanced flue	170	109	64	44
Liquid flueless heaters	0	0	0	0
Liquid open to Chimney heaters	0	0	0	0
Total	280	199	134	97

Table A4-4. Stock of local space heaters 2010-2050

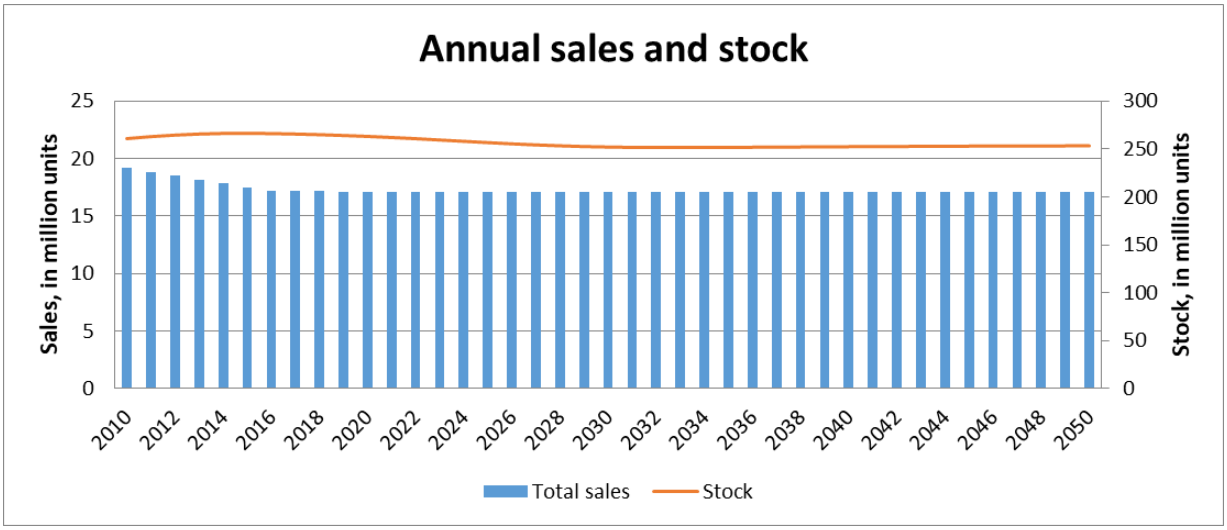


Figure A4-1. Annual sales and stock estimates of electrical local space heaters 2010-2050

3. TURNOVER AND EMPLOYMENT

The modelling of the turnover has been calculated on the basis of the sales evolution and product price explained in sections 1 and 4.1 of this Annex respectively, and applying the mark-up factors in point 4.4 of this Annex.

In addition, it has been estimated from stakeholder inputs that 60% of manufacturers of electric heaters are located inside EU (90-100% outside EU for entry level products, but 100% inside EU for mid-to-high level products), while 100% of manufacturers of gas and liquid heaters are located inside the EU .

The **employment** has been determined by considering that its evolution is proportional to the business turnover, corrected negatively by an increase in labour productivity. For **electric local space heaters**, the correction has been drawn from Eurostat data for "Manufacture of electric domestic appliances" [56] between 2008 and 2014. Unfortunately, no data beyond 2014 are available. The graph below shows turnover versus number of employees, and it is seen that while turnover stays roughly stable in the period after the financial crisis, i.e. 2009-2014, employment dropped at a rate of 2,8%/year, reduction that is attributed to the improvement in production efficiency in the sector. Therefore the employment model used in this IA for the baseline has been calculated as the evolution of the turnover minus 2,8% points.

[56] Eurostat, Annual detailed enterprise statistics for industry (NACE Rev. 2, B-E) Year 2016 [sbs_na_ind_r2]

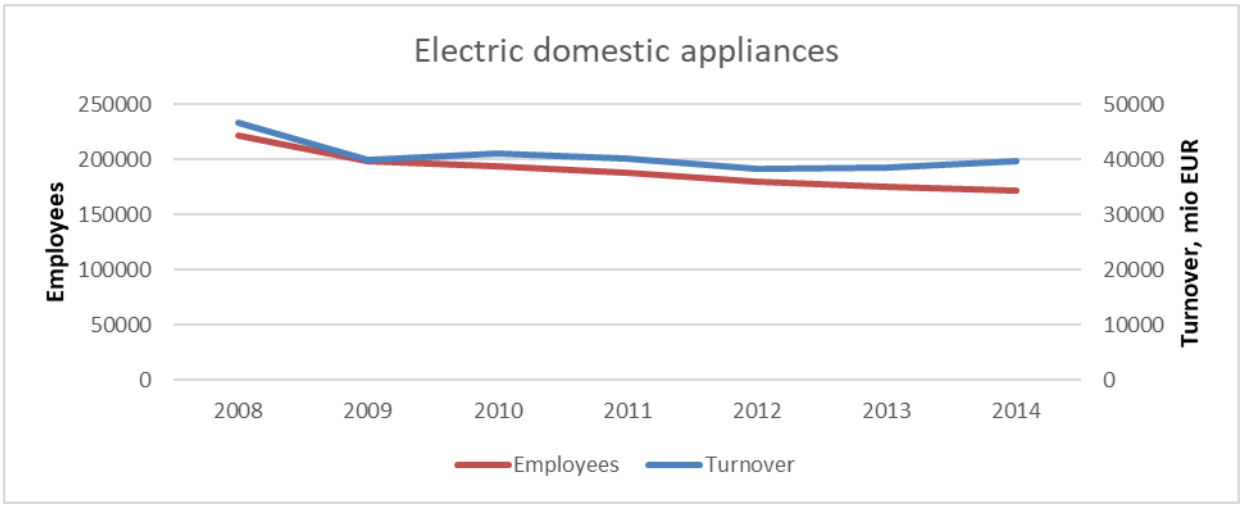


Figure A4-2. Turnover and jobs electric domestic appliance manufacture 2008-2014. Source: Eurostat 2021

For **gas and liquid heaters**, information comes from the Eurostat data for "Manufacture of ovens, furnaces and furnace burners", which makes data available until 2018. Based on this data an average correction factor of 0,72 was found for negative change in turnover vs. the change in employees, meaning that for each 1% decrease in turnover, employment decreased only 0,72%. The data also showed that for each 1% that the turnover increased, the employment increased only 0,47%.

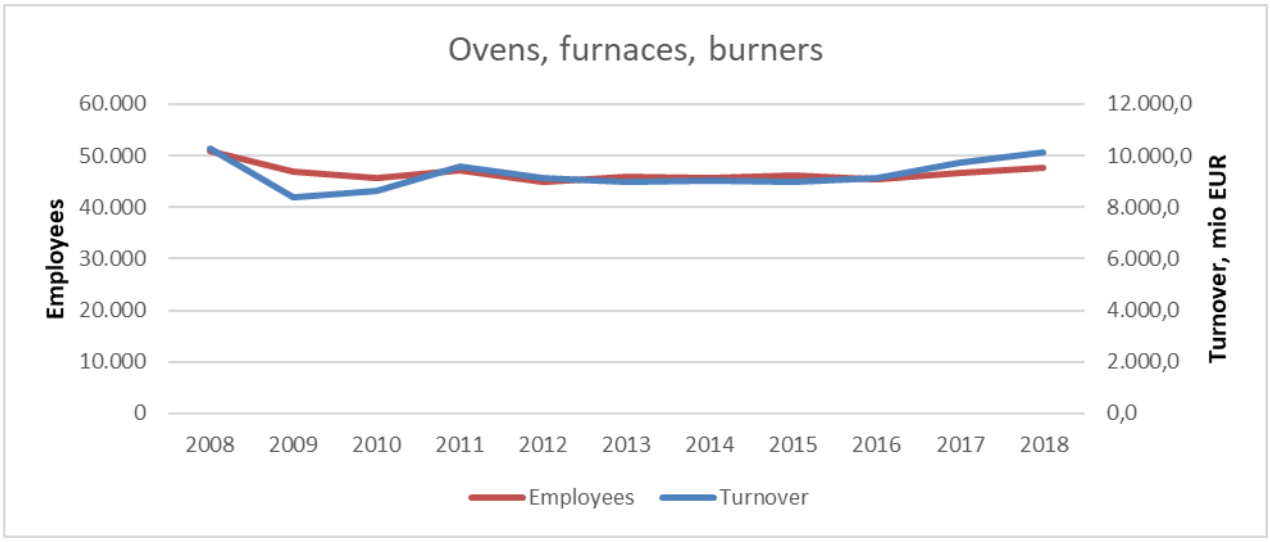


Figure A4-3. Turnover and jobs ovens, furnaces and furnace burner manufacture 2008-2018. Source: Eurostat 2021

The modelling of the employment up to 2050 is therefore calculated by applying the correction factors to the turnover. To differentiate between manufacturing, wholesale and retail, the total employment figure has been multiplied by the employment factors set out in point 4.3 of this Annex.

4. DATA INPUT

Table A4-5 shows the inputs that are used in the modelling work to estimate the evolution of the BAU scenario and the policy options. The way in which their values have been obtained or calculated are explained in the next sections.

Description	Unit

Product price	€ / unit
Fuel costs	€ / kWh
Repair and maintenance costs	€ / unit
Fuel emission factors	CO_2-eq./kWh
Employment factors	mln. € / employee
Mark-up factors	%
Conversion coefficient (CC factor)/Primary Energy Factor (PEF)	2.1
Use hours	h / year
Average capacity	kW
Average efficiency	%
Material input	kg / unit

Table A4-5. Inputs to the model

4.1. Product price

The consumer purchase price is based on the 2015 impact assessment and the input from stakeholders, as well as some online research.

The main difficulty for the calculation of the product price lies on the lack of disaggregated information for local space heaters. The 2015 impact assessment correlates purchase price and level of efficiency according to the price elasticity formula explained below, so that higher efficiency levels correspond to higher prices. The resulting prices from the elasticity formula for 2020 were compared to information available online and submitted to stakeholders for feedback, and their values adjusted if necessary. Once that the prices per product sub-group for 2020 were set, their future evolution has been calculated from the same elasticity formula (corrected as indicated in section 4.1.1) on the basis of the expected energy efficiency improvements under the different policy options.

Product	**Purchase price € in 2020**
Electric portable	33
Electric fixed heat output > 250 W	271
Electric fixed heat output < 250 W	151
Electric storage	815
Electric underfloor	192
Electric visibly glowing l heat output > 1.2 kW	37
Electric visibly glowing heat output < 1.2 kW	27
Towel heaters	251
Gas luminous	1.302
Gas tube	1.228
Gas tube 120-300 kW	7.407
Gas open combustion, open fronted	1.175
Gas open combustion/closed fronted	1.243
Gas balanced flue	1.790
Gas flueless	1.001
Gas open to chimney	1.001
Liquid tube	1.228
Liquid tube 120-300 kW	7.407
Liquid open combustion, open fronted	851
Liquid open combustion, closed fronted	2.486

Liquid balanced flue	1.790
Liquid flueless	1.001
Liquid open to chimney	1.001

Table A4-6. Consumer purchase prices for local space heaters incl. VAT without installation costs in 2020 prices

4.1.1. On price elasticity

The development in purchase prices over time is directly related to the efficiency of the appliances. The correlation between price and efficiency is described as an exponential equation, where:

$$Price = a*Exp(b*efficiency)$$

The constant *a* has been updated to fit 2020 prices in table A4-7, on the basis of the feedback from stakeholders as mentioned above.

a EXP(b*efficiency)	**a**	**b**
Electric portable	13	1
Fixed > 250	136	0,75
Fixed < 250	80	0,75
Electric storage	16	4
Electric underfloor	23	2,5
Electric visibly glowing l heat output >1.2 kW	15	1
Electric visibly glowing heat output < 1.2 kW	11	1
Towel heaters	136	0,75
Gas luminous heaters	211	2
Gas tube heaters	274	2
Gas tube 120-300 kW	1.827	2
Gas open fronted	314	3
Gas closed fronted	100	4,2
Gas Balanced flue	80	4,2
Gas flueless	840	1
Gas OTC	840	1
Liquid tube	274	2
Liquid tube 120-300 kW	1.827	2
Liquid Open fronted	134	4,2
Liquid closed fronted	200	4,2
Liquid Balanced flue	80	4,2
Liquid flueless	840	1
Liquid OTC	840	1

Table A4-7 constants for the exponential equation relating the efficiency of local space heaters to purchase price

4.2. Energy data input: EU Reference Scenario 2020 (REF2020)

For modelling the impacts of the different policy options, REF2020 input data for fuel costs, fuel emission factors and renovation rates have been used. REF2020 projects EU and Member States development in energy, transport and GHG emissions up to 2050, given current global and EU

market trends, such as population growth, macroeconomic, oil price developments, technology improvements and public policies[57].

The policies that have been included in REF2020 are those adopted at EU and Member State level, in the fields of energy, transport, and climate until December 2019, which is the cut-off date. It is assumed that these policies are implemented and intensified until 2030 and afterwards, but that no additional measures apply between 2030 and 2050. REF2020 does not reach the long-term 2050 decarbonisation objective because it does not include all the policies for reaching these targets.

4.2.1. Fuel costs

Fuel prices have been drawn from the Commission PRIMES model using REF2020 and are displayed in the following tables.

Electricity prices (€/Mwh)	**2010**	**2015**	**2020**	**2025**	**2030**	**2035**	**2040**	**2045**	**2050**
Households	168,5	175,9	192,9	200,8	209,2	207,4	207,7	208,5	212,1
Services	146,8	157,1	177,4	184,3	192,3	190,6	190,4	190,3	192,6

Table A4-8. Electricity prices from EU Reference scenario 2020

Natural gas prices (€/Mwh)	**2010**	**2015**	**2020**	**2025**	**2030**	**2035**	**2040**	**2045**	**2050**
Households	67,8	74,1	50,8	63,6	73,5	76,6	80,4	82,5	82,8
Services	54,6	61,0	41,3	51,4	59,9	62,7	66,1	67,9	68,3
Industry	37,7	35,5	22,9	28,4	34,6	37,1	41,0	43,3	43,8

Table A4-9. Natural gas prices from EU Reference scenario

LPG prices (€/Mwh)	**2010**	**2015**	**2020**	**2025**	**2030**	**2035**	**2040**	**2045**	**2050**
Households	86,2	77,0	72,2	84,7	95,8	103,5	112,5	123,9	124,1
Services	72,5	62,4	60,4	77,8	88,9	91,4	92,4	104,2	103,4
Industry	84,0	70,7	71,5	89,9	100,4	103,1	104,2	104,2	101,9

Table A4-10. LPG (for liquid local space heaters) prices from EU Reference scenario

4.2.2. Fuel emission factors

The emission factor for electricity comes also from PRIMES REF2020 (Power generation per MWh_e+MWh_{th}), and emission factors for natural gas and fuel oil from the EcoReport tool.

MT CO2-eq./TWh	**2010**	**2015**	**2020**	**2025**	**2030**	**2035**	**2040**	**2045**	**2050**
Electricity	0,318	0,305	0,214	0,179	0,143	0,109	0,075	0,061	0,061

[57] "*Building an EU Reference scenario is a regular exercise by the Commission. It is coordinated by DGs: ENER, CLIMA and MOVE in association with the JRC, and the involvement of other services via a specific inter-service group. Member States were consulted throughout the development process through a specific Reference scenario expert group.*" Source: https://eur-lex.europa.eu/resource.html?uri=cellar:56466305-b7f6-11e6-9e3c-01aa75ed71a1.0001.02/DOC_2&format=PDF

Natural gas	0,20
Fuel oil	0,28

Table A4-11. Fuel emission factors 2010-2050

4.3. Employment factors

The following employment factors have been used for the differentiation of the employment creation in the different subsectors (manufacturers, repair, wholesale and retail).

Sector	**M€/Employee**
Employment at manufacturers (2017)	0,192
Repair (2017)	0,120
Wholesale (2016)	0,279
Retail (2016)	0,075

Table A4-12. Employment factors

4.4. Mark-up factors

The mark-up factors are used to determine the distribution of turnover based on purchase prices. They are based on the 2015 Impact Assessment and agreed to by stakeholders during the review study.

Sector	%
Manufacturer selling price of final product price excl. VAT	76 %
Wholesale margin	15 %
Retail margin	15 %

Table A4-13. Mark-up factors

4.5. Use hours

The purpose of the space to be heated has a strong influence on how the local space heaters are used in terms of temperature settings and use hours, the main differentiation being between service and residential purposes. Furthermore, the temperature settings and the number or hours of functioning of a local space heater are influenced by its use as either a primary or a secondary heating source.

This study considers that for the service sector, gas and liquid fuelled (luminous and tube) heaters are used 8 hours a day at a temperature setting of 18 °C and 16 hours a day at a temperature setting of 14 °C, as they are usually the primary heat source in the places where they are installed[58].

As far as residential use, it is considered that heaters are used an average of 1,5 hours/day in the heating season when installed as secondary heat source, while when installed as primary heat source they are used an average of 8 hours at a setting of 20 °C and 16 hours at a setting of 15,5 °C, as recommended in standard EN 12831.

The Heating degree days (HDD)[59] are provided by Eurostat[60] for the years 1990 to 2020. The projection until 2050 is carried out by a linear forecast in HDD development from 2020 to 2050, yielding a slightly decreasing (-0,00365 HHD/year) heat demand due to global warming.

[58] For further details on use patterns see the review study

Furthermore, the heat demand is adjusted to the increasing building renovation rate of the building stock. The Annual renovation rate (as % of entire housing stock) and the Energy Savings from renovation (as % of space-heating average) is drawn from REF2020.

	2015	2020	2025	2030	2035	2040	2045	2050
Annual renovation rate (as % of entire housing stock)	0,68%	0,95%	1,03%	1,16%	0,86%	0,87%	0,81%	0,87%
Energy Savings from Renovation (% of space-heating, average)	10,6%	28,2%	39,5%	45,4%	37,3%	30,7%	26,9%	24,4%
Combined effect on heat demand (% heat demand reductions)	0,07%	0,27%	0,41%	0,53%	0,32%	0,27%	0,22%	0,21%

Table A4-14. Data on renovation rates and impact from the EU Reference 2020 scenario

Based on HDD data, building renovation and use patterns, the development of total annual use hours were calculated.

Fuel type	Full load hour equivalents	Heating hours per year, 2018
Gas and liquid	Luminous	1.408
	Tube	1.408

Table A4-15. Annual use hours in the service sector

Energy source	Local space heater type	Type of use pattern	Heating hours per year, 2020
Electric	Portable Electric	Secondary	457
	Fixed electric >250 W	Primary	1.479
	Fixed electric <250 W	Primary	1.479
	Electric storage heaters	Primary	1.479
	Electric underfloor	Equal secondary and primary	926
	Visibly glowing >1,2	Equal secondary and primary	878
	Visibly glowing <1,2	Equal secondary and primary	878
	Towel heaters	Equal secondary and primary	878
Gas and liquid	Open fronted	Equal secondary and primary	878
	Closed fronted	Equal secondary and primary	878
	Balanced	Equal secondary and primary	878
	Flueless	Secondary	277
	Open to chimney	Secondary	277

Table A4-16. Annual use hours in residential settings for primary and secondary heating

[59] HHDs is a measure of heat demand in a building. HDDs are derived from outside temperature measurements, and its deviation from the "base temperature", which in the EU is 15.5 ˚C. This means that the number of days/year where the temperature is lower than the base temperature is multiplied with how much colder it was on those days (as a sum-product) (see: https://www.eea.europa.eu/data-and-maps/indicators/heating-degree-days-2)

[60] Heating degree days by country - annual data

4.6. Average heat output

The average heat output (as kW capacity) for each local space heater type is displayed in Table A4-17[61].

Product	Capacity (kW)
Electric portable	1,00
Electric fixed heat output >250 W	1,00
Electric fixed heat output <250 W	0,25
Electric storage	2,75
Electric underfloor	1,00
Electric visibly glowing l heat output >1.2 kW	2,00
Electric visibly glowing heat output < 1.2 kW	1,00
Towel heaters	0,60
Gas luminous	20,00
Gas tube	30,00
Gas tube 120-300 kW	200,00
Gas open combustion, open fronted	4,20
Gas open combustion/closed fronted	4,20
Gas balanced flue	4,20
Gas flueless	2,50
Gas open to chimney	4,20
Liquid tube	30,00
Liquid tube 120-300 kW	200,00
Liquid open combustion, open fronted	2,00
Liquid open combustion, closed fronted	4,00
Liquid balanced flue	4,00
Liquid flueless	3,00
Liquid open to chimney	4,00

Table A4-17. Average capacity (kW) of local space heaters

4.7. Average efficiency

The evolution under BAU of the average efficiency for each type of local space heater are based on the review study and displayed on the table below.

Average efficiency	2010	2015	2020	2025	2030	2035	2040	2045	2050
Electric portable	80%	86%	92%	92%	92%	92%	92%	92%	92%
Electric fixed heat output >250 W	80%	86%	92%	92%	92%	92%	92%	92%	92%
Electric fixed heat output <250 W	80%	82%	85%	85%	85%	85%	85%	85%	85%
Electric storage	80%	89%	98%	98%	98%	98%	98%	98%	98%
Electric underfloor	80%	81%	82%	82%	83%	83%	83%	83%	83%

[61] Based on the review study (table 49, p. 184), and that again is based on the capacity ranges given for each base case technology in the 2012 preparatory study task 4 report.

Electric visibly glowing l heat output >1.2 kW	80%	85%	90%	90%	90%	90%	90%	90%	90%
Electric visibly glowing heat output < 1.2 kW	80%	85%	90%	90%	90%	90%	90%	90%	90%
Towel heaters	80%	81%	82%	82%	83%	83%	83%	83%	83%
Gas luminous	73%	82%	91%	91%	91%	91%	91%	91%	91%
Gas tube	64%	70%	75%	75%	75%	75%	75%	75%	75%
Gas tube 120-300 kW	64%	67%	70%	70%	70%	70%	70%	70%	70%
Gas open combustion, open fronted	42%	43%	44%	45%	46%	46%	46%	46%	46%
Gas open combustion/closed fronted	52%	56%	60%	60%	60%	60%	60%	60%	60%
Gas balanced flue	64%	69%	74%	74%	74%	74%	74%	74%	74%
Gas flueless	18%	18%	18%	18%	18%	18%	18%	18%	18%
Gas open to chimney	18%	18%	18%	18%	18%	18%	18%	18%	18%
Liquid tube	64%	70%	75%	75%	75%	75%	75%	75%	75%
Liquid tube 120-300 kW	64%	67%	70%	70%	70%	70%	70%	70%	70%
Liquid open combustion, open fronted	42%	43%	44%	45%	46%	46%	46%	46%	46%
Liquid open combustion, closed fronted	52%	56%	60%	60%	60%	60%	60%	60%	60%
Liquid balanced flue	64%	69%	74%	74%	74%	74%	74%	74%	74%
Liquid flueless	18%	18%	18%	18%	18%	18%	18%	18%	18%
Liquid open to chimney	18%	18%	18%	18%	18%	18%	18%	18%	18%

Table A4-18. Average efficiencies for the baseline BAU

4.8. Material input

The material inputs are used to estimate the embedded environmental effects of each product. They have been calculated with the EcoReport tool, which includes a range of background data for calculating environmental impacts from different materials and is currently being revised in order to provide more granularity and better reflect latest data on environmental impacts of materials.

The environmental impacts modelled for this study which are related to material input are:

- Primary embedded energy (MJ)
- GWP_{100}[62] (kg CO2-eq.)
- Acidification (g SO2-eq.)

The phases considered are production, distribution and end-of-life of the material. The environmental impacts related to the use phase have been calculated outside the EcoReport tool as the latter is not detailed enough to reflect efficiency changes and economic (cost-related) parameters.

The material composition of local space heaters has already been determined in the preparatory study and is displayed in the following tables.

[62] GWP effect calculated over a 100-year period. This is the most prevalent way of calculating GWP, however a 20 or 200 year perspective can also be used. See e.g.: https://www.ghgprotocol.org/sites/default/files/ghgp/Global-Warming-Potential-Values%20%28Feb%2016%202016%29_1.pdf GWP100 is also used on the Ecoreport tool, which this analysis is based on.

	Portable	Fixed > 250W	Fixed ≤ 250W	Storage	Underfloor	Visible glowing > 1,2 kW	Visible glowing ≤ 1,2 kW	Towel heaters
Bulk plastic (e.g. ABS) g	195	300	75	0	560	390	195	300
TEC plastic (e.g. PA6) g	650	300	75	100	2,503	1,300	650	300
Ferro (cast iron) g	195	3750	938	30,000	50	390	195	3,750
Non-ferro (CU) g	98	307	77	75	280	195	98	307
Non-ferro (Al) g	98	307	77	300	280	195	98	307
Coating g	0	311	78	100	0	0	0	311
Electronics (PCBs) g	65	75	75	100	0	130	65	75
Miscellaneous	0	0	0	109,000	300	0	0	0
Total	**1,300**	**5,275**	**1,319**	**139,675**	**3,973**	**2,600**	**1,300**	**5,275**

Table 4-19. Material composition of electric local space heaters

	Open combustion/open fronted with exhaust restriction	Open combustion/closed fronted	Balanced flue	Flueless	Open to chimney
Bulk plastic (e.g. ABS) g	25	25	25	13	13
TEC plastic (e.g. PA6) g	25	25	25	13	13
Ferro (cast iron) g	17,820	17,820	17,820	8,910	8,910
Non-ferro (CU) g	396	396	396	198	198
Non-ferro (Al) g	396	396	396	198	198
Coating g	732	732	732	366	366
Electronics (PCBs) g	25	25	25	13	13
Miscellaneous g	346	346	346	173	173
Total	**19,765**	**19,765**	**19,765**	**9,883**	**9,883**

Table A4-20. Material composition of gas/liquid heaters

	Luminous local space heaters (non-residential)	Tube local space heaters[63] (non-residential)
Bulk plastic (e.g. ABS) g	0	0
TEC plastic (e.g. PA6) g	0	0
Ferro (cast iron) g	17,000	76,500
Non-ferro (CU) g	0	307
Non-ferro (Al) g	0	307
Coating g	800	3,600
Electronics (PCBs) g	400	1,800

[63] Large tube heaters (120-300 kW) have been scaled up based on an average capacity of 200 kW.

Miscellaneous g	1,800	8,100
Total	**20,000**	**90,614**

Table A4-21. Material composition of commercial heaters

5. MAIN ASSUMPTIONS FOR THE MODELLING

Due to the lack of available data, some assumptions have been made in order to complete the calculation of the impacts, according to stakeholders input during the review study and notably from the industry. Table A4-22 summarises those assumptions.

	Common to all the new product sub-groups falling in the scope
	The efficiency of the new product sub-groups coming in the scope rises from 78% (BAU) to 81,75%
Policy option 2	**Removing the exemption for slave heaters**
	- Electric fixed heaters: the share of electric fixed slave heaters that do not fulfil the ecodesign requirements and will need to fulfil them due to the application of the measure has been estimated in 30% of the existing stock. This share is based on stakeholders input, mainly from industry. - Electric underfloor heaters: the share of electric underfloor heaters that do not fulfil the ecodesign requirements and will need to fulfil them due to the application of the measure has been estimated in 90% of the existing stock. This share is based on stakeholders input, mainly from industry.
	Including in the scope of the LSH Regulation electric heaters for drying and heating towels
	- The share of towel heaters that do not fulfil the ecodesign requirements and will need to fulfil them due to the application of the measure has been estimated in 90% of the existing stock. This share is based on stakeholders input, mainly from industry. - The 10 % of towel heaters that were already on scope of the LSH Regulation are assumed to be above 250 W and therefore feature the same energy efficiency as fixed heaters (i.e. 95% efficiency).
	Including in the scope of the LSH Regulation tube heaters between 120 and 300 kW
	Tube heaters of 120 - 300 kW increase energy efficiency from 83 % to 85 % to match the efficiencies of tube heaters < 120 kW
	Changing the definition of gas and liquid heaters
	Gas open combustion/closed fronted increases efficiency from 60 % to 64 %
Policy option 3	**Removing factor F(4) for standby and replacing it by suitable low power modes**
	All electric products received a 0,5 % increase in energy efficiency
Policy option 4	**Software updates for apps**
	All electric products get a 0,5 % increase in energy efficiency

	Availability of spare parts, repair and maintenance information requirement and EOL information requirement
	Repair costs - 20% of the products are repaired only once over their lifetime - Additional repair rate of 20% of the product price Lifetime electric - No lifetime increase for: electric portable, fixed < 250 W and electric underfloor heaters. - 5% lifetime increase for: electric fixed > 250 W, storage heaters, visibly glowing, towel heaters Lifetime gas and liquid - No lifetime increase for: flueless and open to chimney - 5% lifetime increase for all other gas and liquid types

Table A4-22: Overview of each measure and how their quantitative effects are modelled through changes to PO1, i.e. the BAU scenario

ANNEX 5: ADDITIONAL INFORMATION ON THE IMPACTS OF THE MEASURES

This section provides additional information on the impacts of the measures proposed. It consists of two parts: the first part provides additional graphic information about the impacts already assessed in section 6 of the report. The second part assesses the impacts of the policy options on two new parameters related to resource efficiency: water consumption and waste generation.

1. FURTHER INSIGHT ON THE ANALYSIS OF THE IMPACTS ASSESSED IN SECTION 6 OF THE IA

1.1. Environmental impacts

1.1.1. In-use energy consumption

Under the BAU scenario, the final energy consumption of local space heaters is expected to decrease 19% in 2030 with respect to 2020 levels, from 172 TWh/year down to 140 Twh/h. This saving comes from the increase in energy efficiency due to technological improvements, but also from a decrease in the heating demand due to the global warming and a better insulation of buildings.

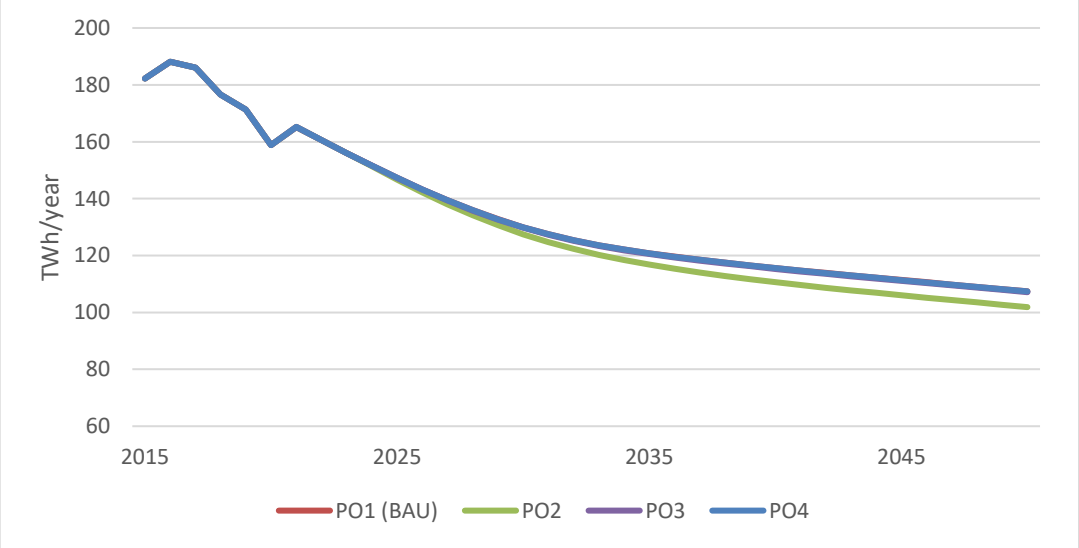


Figure A5-1. EU energy consumption 2020-2050 during the use phase for each PO

Most savings (in particular until 2040) come from measures that apply exclusively to **electric heaters** (PO2a, PO2b and PO3). Removing the exemption for slave heaters (measure PO2a) accounts for the highest electricity savings, with around 72% of the total energy savings for 2030.

Measures related to circular economy (PO4a and PO4b) do not yield significant results on in-use energy savings. Only modularity is able to reduce energy consumption by a very small quantity due to the upgrading of the controllers and the subsequent improvement of energy efficiency.

1.1.2. Primary energy consumption

In addition to the use phase, primary energy includes embedded energy from production, distribution and end-of-life phases, i.e. all life cycle phases other than the use phase. The primary energy consumption follows a similar trend as the final energy consumption seen above, due to the embedded energy being less than 1% (0,4%) of the total energy consumption.

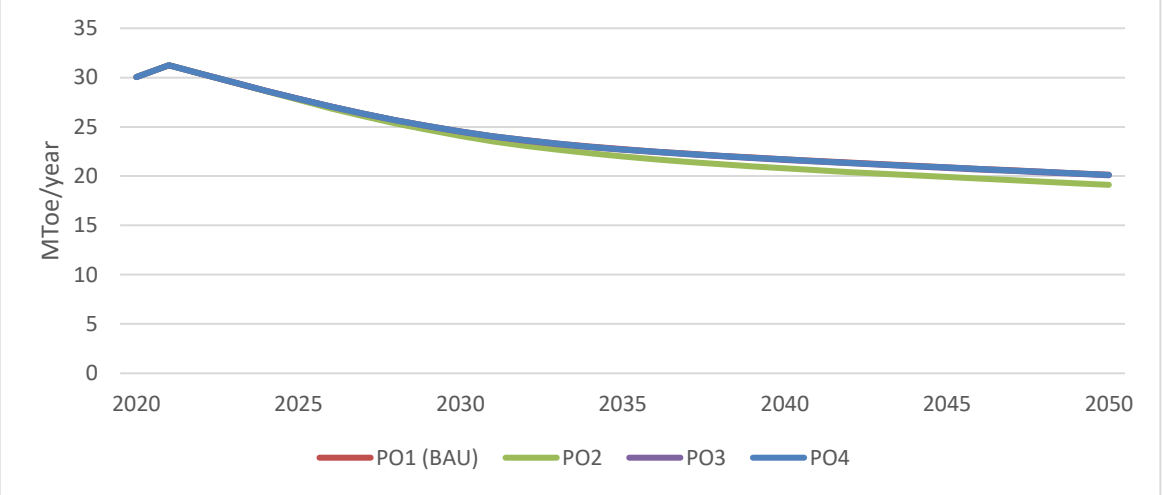


Figure A5-2. EU primary energy consumption 2020-2050 for embedded energy and during use phase for each PO

Under BAU, primary energy consumption is expected to decrease by 18% in 2030, namely 5,5 Mtoe/year.

Measure PO2a provides again the highest savings compared to BAU. The contribution of circular economy measures is still very low, although it can be noticed a slight effect from PO4a on the embedded energy due to the extension of the lifetime of certain types of heaters. This effect is only noticeable as from 2035 as the products purchased in 2025 are getting older and need to be repaired.

1.1.3. Greenhouse gas emissions reduction

Alike primary energy, GHG emissions also include embedded emissions from production, distribution and end-of-life. Under BAU, GHG emissions are reduced by 43% from 2020 to 2030 and by 79% in 2050, due to energy savings enhanced by the expected reduction in carbon intensity of the electricity mix.

Measure PO2a brings about the highest GHG emission savings, approximately a 70% of the total. Same as for the final energy consumption, PO4a has a slight influence on GHG emissions caused by the reduction of embedded energy that is noticed as from 2035.

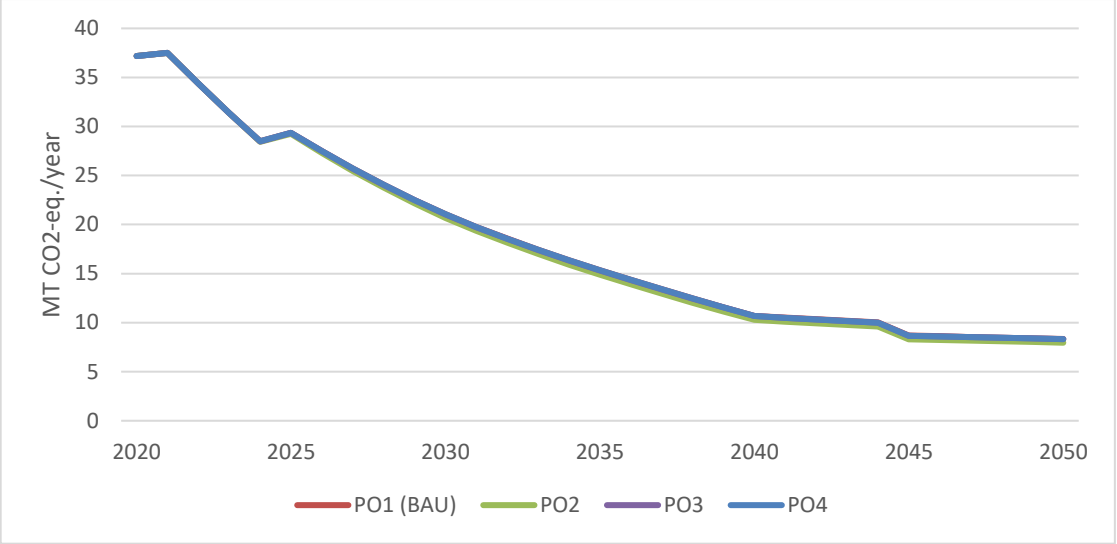


Figure A5-3. EU GHG emissions 2020-2050 from production, distribution, use and end-of-life for each PO

1.2. Economic impacts

1.2.1. Turnover

The total turnover of the industry decreased between 2010 and 2015 due to dwindling sales but increased again between 2015 and 2020 due to the better energy efficiencies brought about by the current LSH Regulation. The application of the new policy options, in particular the expansion of the regulatory scope to slave heaters, will lead to higher efficiencies and subsequently higher sales prices and further increase of the turnover. In the longer term, prices are expected to level off and the slight, continuous decrease in sales of oil and gas-fuelled local space heaters, will also slowly reduce turnover towards 2050 for all options.

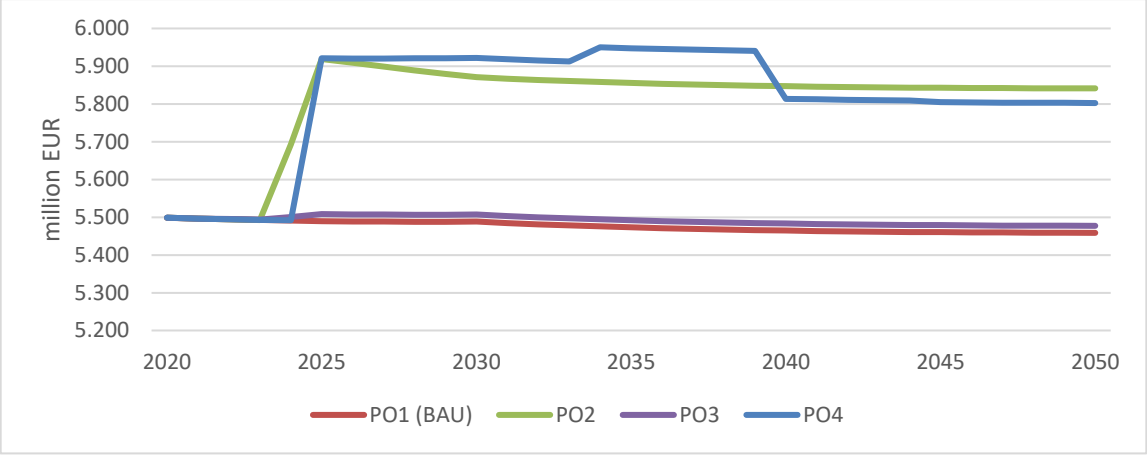


Figure A5-4. Total turnover within the EU 2020-2050 for each policy option

Measure PO4b (modularity) leads to the greatest increase in turnover due to the need to redesign the products in order to support modularity, which is reflected in higher prices. PO4a on the contrary brings about a reduction of turnover in the long term due to a longer lifetime of the products and the subsequent contraction of sales per year. PO2a, as expected, raises also significantly the turnover following the achievement of better efficiencies for slave heaters which leads to an increase of the prices.

1.2.2. Consumer expenditure

Consumer expenditure includes purchase costs (which in turn incorporates end-of-life costs that are normally paid by the user through a levy at purchase), energy spent during use, and repair and maintenance. The consumer expenditure under BAU is expected to decrease from € 34.161 million in 2020 to € 30.657 million in 2030 (10% reduction) due to energy efficiency improvements, rise of outdoor temperatures and better insulation of buildings.

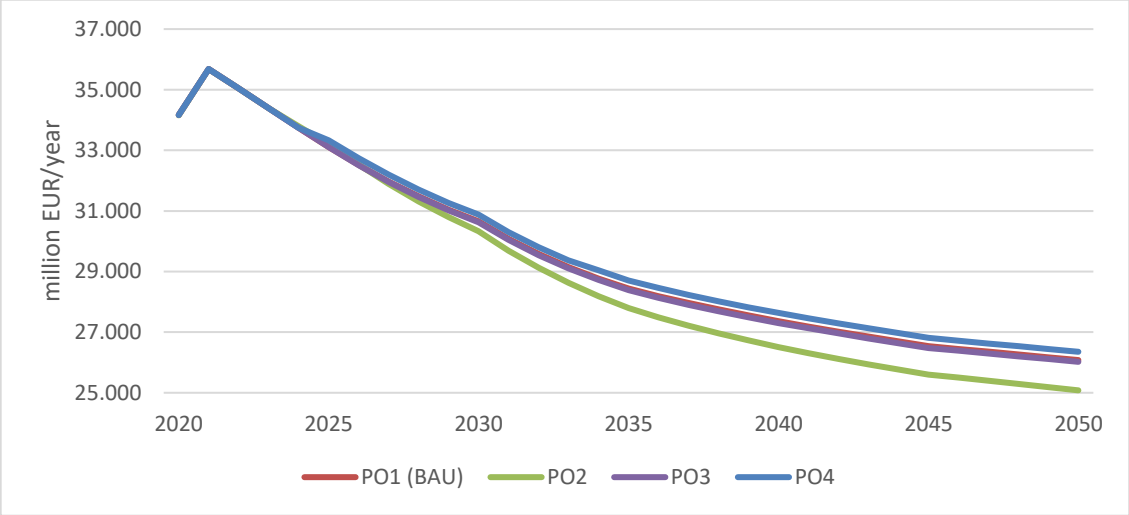


Figure A5-5. Consumer expenditure including purchase costs, energy costs and repair & maintenance costs for 2020-2050, for each policy option

PO2a yields the highest benefits. Apart from PO4b which cost increase cannot be offset by better energy efficiencies, the lowest benefit for consumer comes from PO2d. This is because gas prices in the baseline scenario is only around 25% of electricity prices, what makes that the increasing purchase prices cannot be fully offset through less fuel consumption. Even if gas and electricity prices information is outdated in the light of the current developments, the effect of PO2d remains nevertheless very small in the overall context.

2. RESOURCE EFFICIENCY ANALYSIS

From the EcoReport tool, two impacts are considered to be relevant from the point of view of resource efficiency, on top of the analysis already made as regards fuel and material consumption: water use and generation of waste. Water is mostly used during the production phase, whereas waste is related to production and use. EoL waste has been discounted as a negative contribution to total waste to account for recyclability.

2.1. Water consumption savings

Cumulative water consumption savings for each product subgroup have been calculated as follows: first, the cumulative water consumption under BAU is obtained by multiplying the water consumption per unit and product subgroup provided by the EcoReport Tool, by the cumulative sales for the periods considered (2025-2030, 2025-2040 and 2025-2050). Summing the subtotals for each product subgroup gives the total water consumption under BAU, which is then multiplied by the ratio *cumulative embedded energy savings/embedded energy consumption under BAU* for the same periods, which gives the cumulative water savings. It has been assumed that water

savings are proportional to embedded energy savings, since the use of water for local space heaters is linked to its production, according to the EcoReport tool.

	Water consumption per unit	Cumulative water consumption (Mlitres)			Ratio cumulative energy savings/BAU embedded energy consumption			Total water saved (Mlitres)		
		2025-2030	2025-2040	2025-2050	2025-2030	2025-2040	2025-2050	2025-2030	2025-2040	2025-2050
Electric portable	32	1.372	3.658	5.943						
Electric fixed nominal heat output > 250 W	39	1.174	3.131	5.088						
Electric fixed nominal heat output =< 250 W	29	218	582	946						
Electric storage	729	1.175	3.134	5.092						
Electric underfloor	39	298	794	1.290						
Electric visibly glowing radiant nominal heat output >1,2 kW	65	59	157	256						
Electric visibly glowing radiant nominal heat output =< 1,2 kW	32	12	33	53.937						
Towel heaters	39	44	1.176	1.912						
Subtotal electric		**4.749**	**12.665**	**20.581**	**0**	**0,002**	**0,012**	**0**	**22**	**247**
Gaseous luminous	0	0	0	0						
Gaseous tube	0	0	0	0						
Gaseous fuel open combustion/open fronted with exhaust restriction	0	0	0	0						
Gaseous fuel open combustion/closed fronted	0	0	0	0						
Gaseous fuel balanced flue	0	0	0	0						
Subtotal gas fuel[64]		**0**	**0**	**0**	**0**	**0,002**	**0,012**	**0**	**0**	**0**
Liquid tube > 300	0	0	0	0						
Liquid tube 120-300	0	0	0	0						
Liquid fuel open combustion/open fronted with exhaust restriction	0	0	0	0						
Liquid fuel open combustion/closed fronted	0	0	0	0						
Liquid fuel balanced flue	0	0	0	0						
Subtotal liquid fuel		**0**	**0**	**0**	**0**	**0,002**	**0,012**	**0**	**0**	**0**
Total cumulative water savings (Mlitres)								**0**	**22**	**247**

Table A5-1. Cumulative water consumption savings for the preferred policy option

2.2. Waste reduction

Cumulative waste reduction has been calculated in a similar way as cumulative water savings. The cumulative waste under BAU for each product subgroup has been obtained by multiplying the waste per unit in each product subgroup with the respective cumulative sales over the periods considered (2025-2030, 2025-2040 and 2025-2050). The cumulative waste for each product subgroup is then summed in two subtotals, one for electric heaters and one for gas/liquid fuelled heaters.

From the EcoReport Tool, it can be observed that for electric heaters, the vast majority of the waste (more than 99% of the total) is produced during the use phase, whereas for non-electric heaters almost all the waste (around 98,5% of the total) is generated during the production phase. This is because, for electric heaters, the waste is due to the

[64] Due to the lack of information in the EcoReport tool about the water consumed in the manufacturing of gas and liquid fuel local space heaters, the water consumption per unit has been set at zero. This has very little influence in the final results, as the sales of gas and fuelled local space heaters are anyway residual.

transformation of primary energy into electric power, whereas for non-electric heaters this process does not take place and therefore the manufacturing phase is the major generator of waste. For this reason, it is possible to simplify and consider that for electric heaters all waste savings are generated during the in-use phase and for gas and liquid-fuelled heaters all waste savings are generated during the production phase. Following this approach, the cumulative waste subtotal for electric local space heaters is multiplied by the ratio *cumulative in-use energy savings/in-use energy consumption under BAU*, and the cumulative waste subtotal for gas/liquid fuelled heaters is multiplied by the ratio *cumulative embedded energy savings/embedded energy consumption under BAU*.

	Waste per unit (Kg)	**Cumulative waste (Ton)**			**Ratio cumulative energy savings/BAU embedded energy consumption**			**Total waste saved (Ton)**		
		2025-2030	**2025-2040**	**2025-2050**	**2025-2030**	**2025-2040**	**2025-2050**	**2025-2030**	**2025-2040**	**2025-2050**
Electric portable	9,5	405.126	1.080.337	1.755.547						
Electric fixed nominal heat output > 250 W	49,7	1.495.627	3.988.338	6.481.049						
Electric fixed nominal heat output =< 250 W	13,4	101.047	269.458	437.869						
Electric storage	54,0	87.081	232.215	377.350						
Electric underfloor	86,3	658.905	1.757.079	2.855.254						
Electric visibly glowing radiant nominal heat output >1,2 kW	33,7	30.586	81.563	132.540						
Electric visibly glowing radiant nominal heat output =< 1,2 kW	20,1	7.822	20.858	33.895						
Towel heaters	20,8	235.112	626.964	1.019						
Subtotal electric		**3.021.305**	**8.056.813**	**12.074.523**	**0,0120**	**0,0258**	**0,0347**	**36.387**	**207.555**	**453.719**
Gaseous luminous	4,8	103	263	423						
Gaseous tube	21,3	496	1.265	2.034						
Gaseous fuel open combustion/open fronted	4,3	204	520	835						
Gaseous fuel open combustion/closed fronted	4,3	408	1.039	1.670						
Gaseous fuel balanced flue	4,3	587	1.497	2.406						
Liquid tube > 300	21,3	46	117	188						
Liquid fuel open combustion/open fronted	4,3	20	52	84						
Liquid fuel open combustion/closed fronted	4,3	41	104	167						
Liquid fuel balanced flue	4,3	59	150	241						
Subtotal gas/liquid fuel		**1.928**	**4.912**	**7.897**	**0**	**0,002**	**0,012**	**0**	**9**	**94**
Total waste savings (tons)								**36.387**	**207.564**	**453.813**

Table A5-2. Cumulative waste reduction for the preferred policy option

ANNEX 6: ANALYSIS OF CURRENT ECODESIGN REQUIREMENTS FOR LOCAL SPACE HEATERS

In the context of the Better Regulation policy[65], the Commission is engaged to evaluate all EU activities intended to impact society or the economy. For the review of the Ecodesign Regulation on local space heaters, three out of the five standard evaluation criteria foreseen by Better Regulation need to be addressed, namely the effectiveness, efficiency and relevance of the proposed measures[66].

The other two evaluation criteria, coherence and EU added-value, have already been addressed at the framework level, i.e. in 2012, when the Ecodesign Directive was reviewed[67]. Later, a joint evaluation of the Ecodesign and Energy Labelling Directives[68] was carried out by the Commission in 2015, the main outcome being that the ecodesign and energy labelling measures in place are effective and bring tangible and substantial energy and cost savings. The implementation of the two Directives is estimated to save 175 Mtoe primary energy per year by 2020, which corresponds to 19% savings with respect to business-as-usual energy use for those products. These policies will deliver almost half of the 20% energy efficiency target by 2020. Dependency on imports of energy would be reduced by 23% and 37% for natural gas and coal, respectively. In total, the ecodesign and energy labelling measures in place to date are estimated to have saved end-users of products 100 billion euro per year in 2020 through lower utility bills (translated into roughly 500 euros yearly savings in each household).

This annex presents the relevant findings of the review of the current LSH Regulation focussing on its effectiveness, efficiency and relevance. The assessment builds on the information collected during the 2019 review study and any data updates on development collected during the present impact assessment.

1. EFFECTIVENESS

The impacts on energy consumption (and subsequently GHG emissions, which are directly dependent on energy) and NOx reflect the **effectiveness** of the LSH Regulation, namely its capacity to achieve the objectives. The period analysed is 2000-2020.

[65] https://ec.europa.eu/info/law/law-making-process/overview-law-making-process/evaluating-and-improving-existing-laws/reducing-burdens-and-simplifying-law/refit-making-eu-law-simpler-and-less-costly_en

[66] https://ec.europa.eu/info/law/law-making-process/planning-and-proposing-law/better-regulation-why-and-how_en

[67] COM(2012) 765 final, REPORT FROM THE COMMISSION TO THE EUROPEAN PARLIAMENT AND THE COUNCIL, Review of Directive 2009/125/EC establishing a framework for the setting of ecodesign requirements for energy-related products (recast)

[68] COM(2015) 345 final, Report from the Commission to the European Parliament and the Council Review of Directive 2010/30/EU of the European Parliament and of the Council of 19 May 2010 on the indication of labelling and standard product information of the consumption of energy and other resources by energy-related products

1.1. Energy savings

In the 2015 Impact Assessment[69] it was estimated under its BAU (BAU 0)[70]that the stock of local space heaters would continue to grow and that the energy consumption in 2020 would rise to 2.362 PJ/year (656,1 TWh/year). The most optimistic policy option casted around 8% energy savings by 2020[71].

These numbers were however not fully accurate, because they included solid fuels local space heaters and were based on a different stock for the different types of local space heaters than what was modelled in the 2019 review study based on updated stakeholder inputs[72]. The major difference between the stock calculations in the 2015 impact assessment and the 2019 review study were the assumption on lifetime of fixed local space heaters, set to 9 years in the 2015 impact assessment and 15 years in the review study, based on stakeholder inputs during the review process. The resulting difference in stock (which is otherwise based on the same sales data from 1990 to 2010) is shown in Figure A6-1[73]. The large difference in total stock (80 million by 2020) is due to the longer lifetime of fixed electric heaters, which is by far the largest segment of local space heaters.

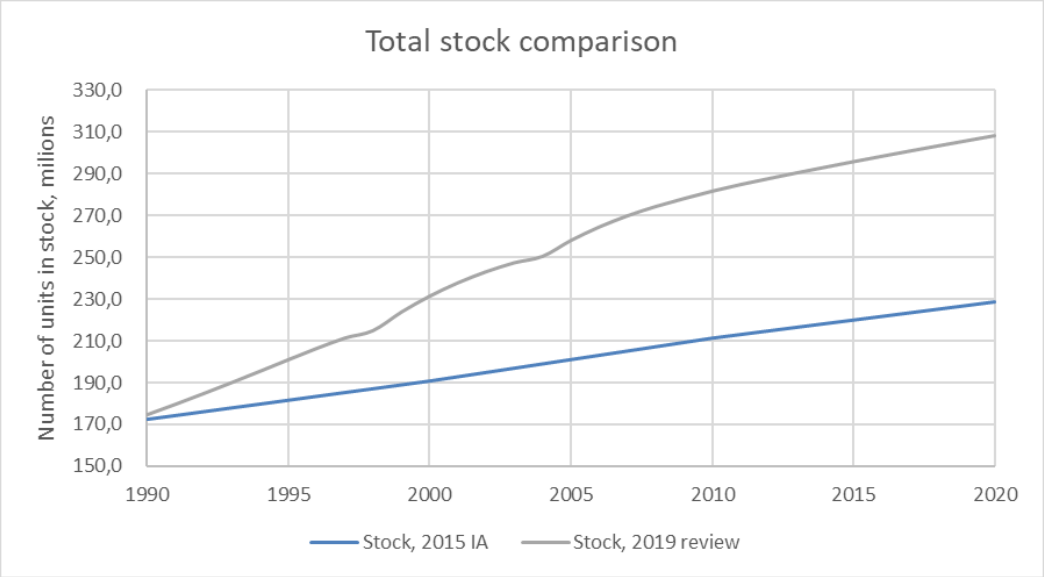


Figure A6-1. Comparison of stock data in the 2015 impact assessment and the 2019 review study

[69] COMMISSION STAFF WORKING DOCUMENT IMPACT ASSESSMENT Accompanying the documents Commission Regulation implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to ecodesign requirements for local space heaters, and Commission Delegated Regulation implementing Directive 2010/30/EU of the European Parliament and of the Council with regard to energy labelling for local space heaters https://ec.europa.eu/smart-regulation/impact/ia_carried_out/cia_2015_en.htm

[70] This is also called the "BAU 0" scenario, and represents a scenario where no regulation was in place for local space heaters.

[71] This correspond to Policy Option E from the 2015 Impact Assessment, which is also the one coming closest to the actual regulation implemented.

[72] Review study on Local Space Heaters, Final Report, Viegand Maagøe and Danish Technical Institute, May 2019 for the European Commission. https://www.eco-localspaceheaters.eu/documents

[73] Numbers adjusted to account only for the scope of the LSH Regulation, i.e. without solid fuel local space heaters, which were included in the 2015 Impact Assessment.

To assess whether the LSH Regulation has been effective, it is then necessary to compare the BAU scenario (henceforth BAU0) in the absence of the LSH Regulation as provided in the 2015 IA but adapted to the abovementioned corrections in terms of stock, with the BAU scenario of the review study (henceforth BAU1).

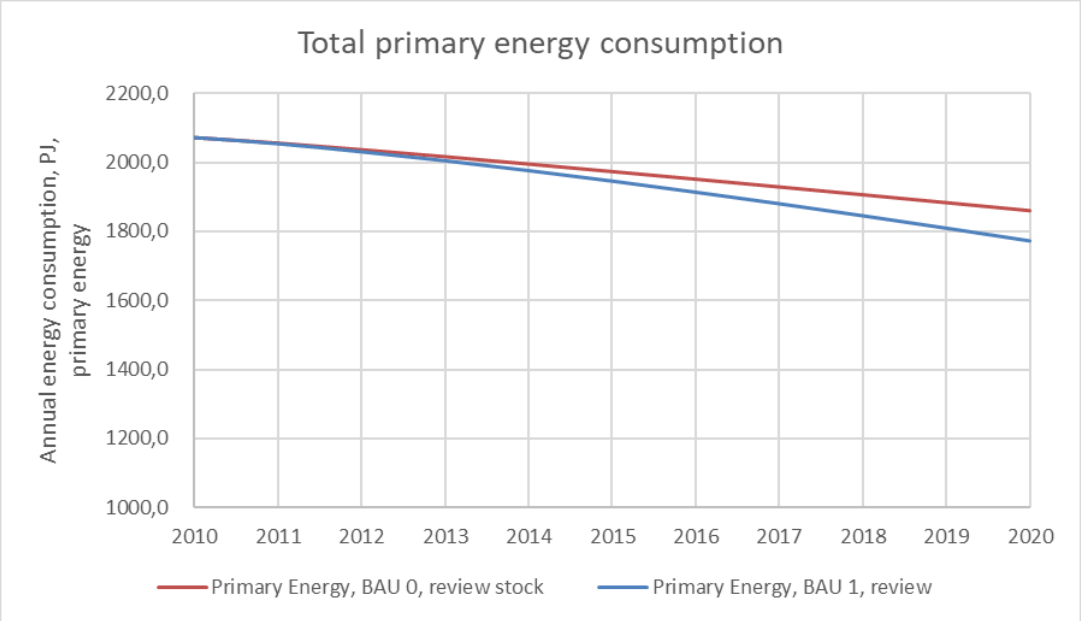


Figure A6-2. Total primary energy consumption in BAU0 and BAU1 from2010-2020

When comparing the relative energy savings in the two studies, the savings expected in 2020 in the impact assessment were 158 PJ (44,2TWh or 10%), while they were 88 PJ (24,7 TWh or 5%) in the review study. Hence, overall the expected savings have only been around half of what was expected, despite the larger stock. This is mainly due to two reasons:

- The expected savings in the 2015 Impact Assessment included the effect of energy labelling, and can thus not be attributed to the LSH Regulation.
- The 2015 IA assumed that efficiencies started to increase compared to BAU already by 2010, while the review study set the origin of the efficiency improvement in 2013.

This means that the savings calculated in the review study occur around 3 years later than expected in the 2015 IA. Comparing the 2015 IA savings in 2017 with the review study savings in 2020 is therefore more accurate. This yields a saving of 110 PJ (30.9 TWh, or 7%) per year in 2017 in the 2015 IA, still higher than the 2020 savings calculated in the review study, even though the difference is smaller.

Overall, it can be concluded that the LSH Regulation has been effective in achieving energy savings, even though not as effective as anticipated in the 2015 IA. According to stakeholders consulted during the review study, this is not because the measures in the LSH Regulation are not technically ambitious, but because the scope and definitions results in **unforeseen loopholes** and fewer products than anticipated comply with the requirements. This is precisely the problem that is addressed in this IA.

1.2. NOx emissions

As it was the case for the 2015 IA, there is still very little data regarding NOx emissions from local space heaters in the EU, and it has not been therefore possible to quantify the impact of the LSH Regulation on NOx emissions. However, the 2015 impact assessment recommended that a limit value for NOx emissions is set in order to avoid the increase of these emissions in the future. This limit was set at 130 mg/kWh.

During the review study, stakeholders reported that they had to discontinue production of some of the products in their portfolio due to the NOx limit, and to modify a majority of the remaining products to make them compliant. Consequently, even though the exact NOx emission savings cannot be quantified, it is evident that the LSH Regulation has led to the decrease of NOx emissions for many of the products on the market, and that the products with the highest NOx emissions have been entirely removed from the EU market. We can therefore conclude that the LSH Regulation has been effective in reducing NOx emissions.

2. EFFICIENCY

The impacts on user expenditure, manufacturer turnover and dealers turnover relate to the **efficiency** of the policy options, namely their capacity to generate energy savings while not generating additional costs to users, manufacturers and dealers, respectively. In general it can be said that a regulation that is directly applicable in all Member States ensure that there are not costs for national administrations related to transposition of the legislation into national law. The use of a regulation also provides a level playing field for the industry, as the measure comes into force simultaneously and identically across all Member States.

The specific questions answered regarding efficiency are:

- To what extent has the LSH Regulation been cost efficient?
- What have been the costs and benefits of the LSH Regulation and to which stakeholders they apply?
- To what extent are the costs of the LSH Regulation justified given the impact achieved?

The total consumer expenditure is calculated as the total amount spent by all users in Europe each year on purchasing, using and maintaining their local space heaters. The LSH Regulation has resulted in lower total consumer expenditure compared to no regulation (BAU0) as seen in Figure A6-3, despite increasing product prices. The consumer expenditure under both BAU0 and BAU1 scenarios increases with time due to the rise of fuel prices, which is decoupled from the LSH Regulation.

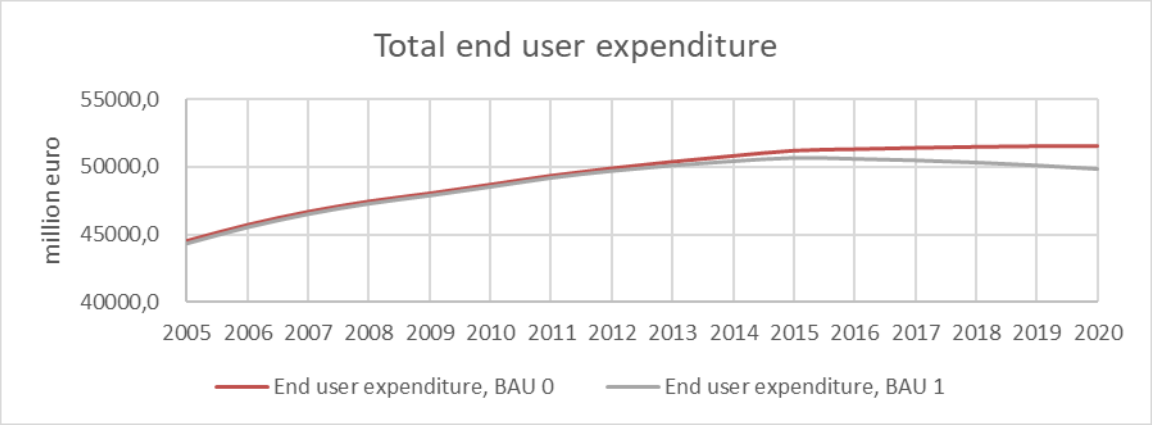


Figure A6-3. Development in total user expenditure over time for BAU0 (No regulation) and BAU 1(current Regulation) scenarios

Two aspects result from the analysis:

- The total consumer expenditure increased since 2010 for both scenarios, but mostly in the BAU0 scenario
- The far majority of consumer expenditure is comprised by fuel costs. This is because the lifetime of the products is generally long. Also, this means that the driver for lower total costs in the BAU1 scenario is the fuel savings (2.029 M€/year lower in 2020 compared to BAU0), which outweigh the increased expenditure for product purchase (505 M€/year higher in 2020 compared to BAU0).

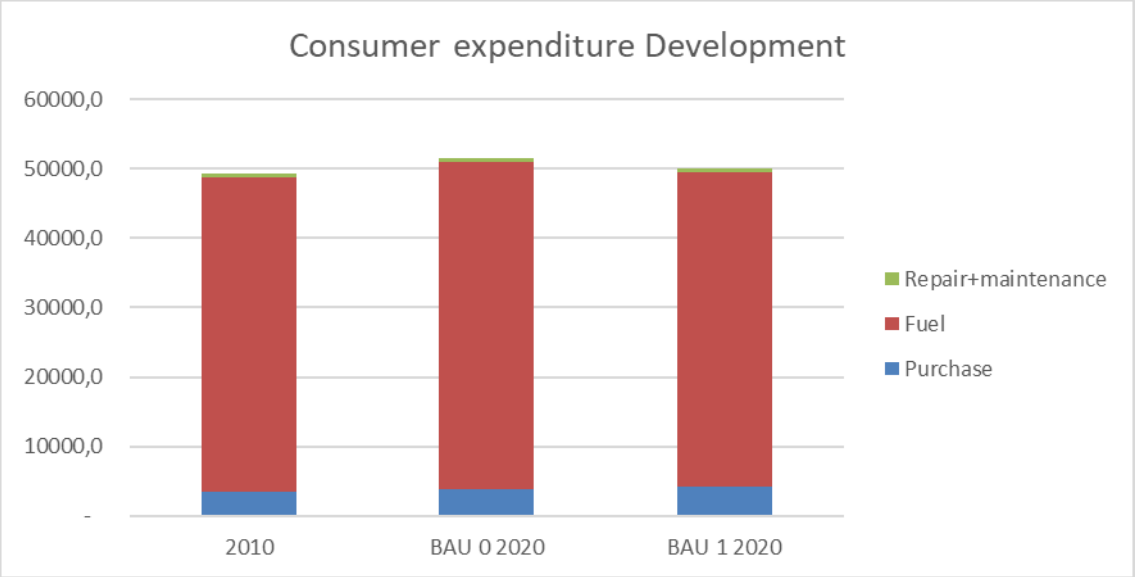


Figure A6-4. Split of total consumer expenditure in 2020 in the BAU 0 (no regulation) and BAU 1 (current regulation) scenarios

Figure A6-5 show that the more expensive products reaching the market under BAU1, entail an increase of the manufacturer turnover which, in turn, leads to a pace of job creation which is proportional to the turnover.

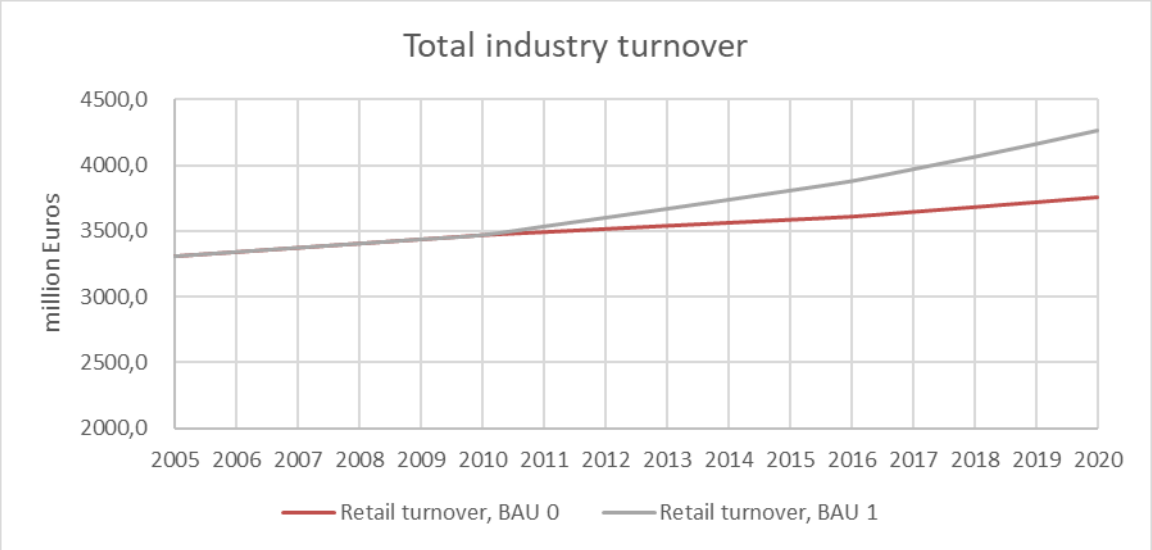


Figure A6-5. total industry turnover in BAU 0 (no regulation) and BAU 1 (current regulation) in million euros

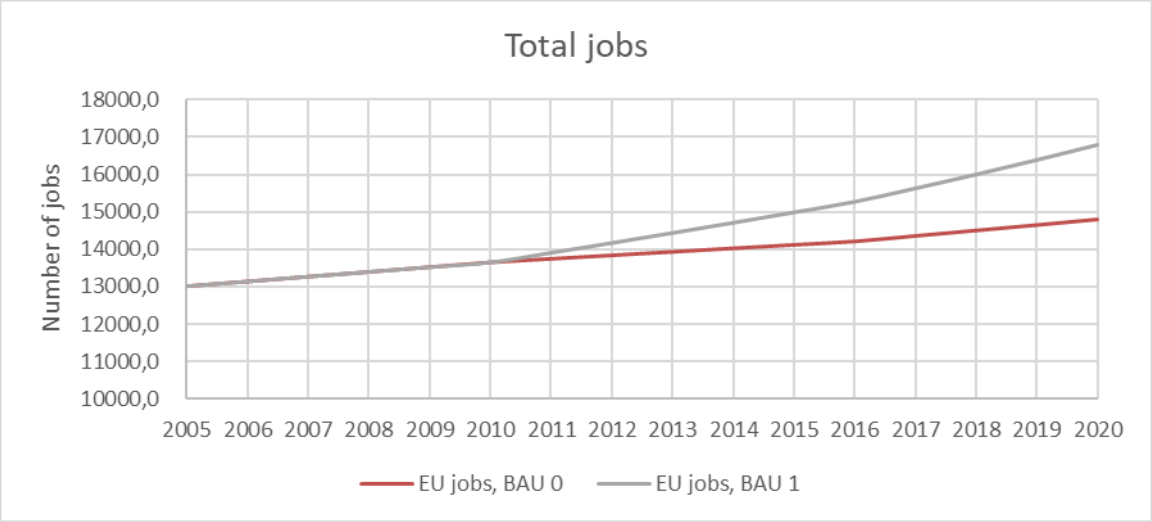


Figure A6-6. number jobs in the local space heaters industry in the BAU0 (no regulation) and BAU1 (current regulation) scenarios

3. RELEVANCE

The specific questions to be answered regarding the relevance of the LSH Regulation are:

- Are the objectives of the current regulation still relevant?
- Can higher savings be obtained with more requirements?

The objective of the LSH Regulation is to reduce the energy consumption and NOx emissions of local space heaters. Since the LSH Regulation only entered into force recently, the market is still adjusting to the requirements and the overall goal of the LSH Regulation of reducing CO_2 and NOx emissions remains highly relevant for the protection of the environment and human health.

As far as possible higher savings, the technology of local space heaters is very mature and no or very slow change is seen in the market. The majority of the development in energy efficiency happens through the development of control functions to better regulate

the heat emitted without decreasing comfort. A significant trend for local space heaters is thus more advanced controllers, which is specific to the electric local space heaters rather than to gas of oil fuelled heaters, and where there is still margin for improvement.

Another path for increasing the energy efficiency from local space heaters is the widening of the scope of the LSH Regulation for instance to slave heaters, which are currently excluded.

ANNEX 7: SOME MEASURES THAT DO NOT ENTAIL ANY DIRECT ENVIRONMENTAL OR ECONOMIC IMPACT

The amendment to the LSH Regulation will incorporate some modifications that do not entail any environmental or economic impact and have therefore not been included in the policy options. Nevertheless, it is convenient to mention them in a separate annex. Those modifications are the following:

1. REPHRASING THE OVERALL DEFINITION OF LOCAL SPACE HEATERS

During the review study, multiple stakeholders made different suggestions for a new definition of local space heaters in order to be able to be more precise about the type of product included in that definition. Some suggested removing the term "human thermal comfort" as this wording could create a loophole as regards of decorative heaters or towel heaters. However, removing those words would raise doubts about the inclusion in the scope of other devices such as those used in e.g. agriculture, industrial facilities and other sectors, which should not be affected by the LSH Regulation. It has been agreed nevertheless to replace the term "reach and maintain a certain level of human thermal comfort" by "contribute to a certain level of human thermal comfort", to make clear that products that are not the only source of heat are also affected. In addition, the definition has been further simplified by removing redundancies such as the term "in combination with heat transfer to a fluid" which is equivalent to "combined with a heat output to other spaces". The sentence "through use of the Joule effect or combustion" is unnecessary and has also been removed.

With all the above, the definition of local space heater will finally be simplified as follows:

Current definition: *local space heater* means a space heating device that emits heat by direct heat transfer or by direct heat transfer in combination with heat transfer to a fluid, in order to reach and maintain a certain level of human thermal comfort within an enclosed space in which the product is situated, possibly combined with a heat output to other spaces and is equipped with one or more heat generators that convert electricity or gaseous, liquid or solid fuels directly into heat, through use of the Joule effect or combustion of fuels respectively.

New definition: *local space heater* means a device that is equipped with one or more heat generators with the purpose of converting electricity or gaseous or liquid fuels directly into heat to contribute to a certain level of human thermal comfort in the enclosed space in which it is situated by direct heat transfer, possibly combined with a heat output to other spaces or with heat transfer to a fluid.

Stakeholder views: stakeholders are in favour of the suggested changes and especially the wording 'contribute' instead of 'reach and maintain'.

2. CHANGING THE DEFINITION OF AIR-HEATING APPLIANCES

It is suggested to change the definition of **air-heating appliances** as the definition of the same product should always be aligned across different regulations to avoid confusion. In

the current definition, it is not clear what is meant with the words "only to an air-based heating system; or that can only be ducted". Hence, it is suggested to incorporate the definition from Regulation 2016/2281[74] as follows:

Current definition: *air heating product* means a product providing heat to an air-based heating system only that can be ducted and is designed to be used while fastened or secured in a specific location or wall mounted which distributes the air by means of an air moving device in order to reach and maintain a certain level of human thermal comfort within an enclosed space in which the product is situated.

New definition, aligned with Regulation 2016/2281: *air heating product* means a device that incorporates or provides heat to an air-based heating system; is equipped with one or more heat generators; and may include an air-based heating system for supplying heated air directly into the heated space by means of an air-moving device. A heat generator designed for an air heating product and an air heating product housing designed to be equipped with such a heat generator shall, together, be considered as an air heating product.

Stakeholder views: No stakeholders opposed to the new definition.

3. CHANGE THE DEFINITION OF "ELECTRIC PORTABLE HEATERS"

A new definition for portable electric heaters shall ensure that this definition is restricted to heaters which can be only used in a portable mode, and which have no option to be used as fixed heaters. Hence, if any accessories are made available for fixing the heater, either together or separately from it, the heater will have to comply with the requirements for fixed heaters.

Current definition: **electric portable local space heater** means an electric local space heater which is not an electric fixed local space heater, electric storage local space heater, electric underfloor local space heater, electric radiant local space heater, electric visibly glowing local space heater or slave heater.

New proposed definition: **electric portable local space heater** means an electric local space heater that is designed to be portable in order to be operated in different rooms or locations. A portable appliance with features which can be used to fix it to a wall and/or floor is considered an electric fixed local space heater.

Stakeholder views: No stakeholders opposed to the new definition.

4. REMOVE THE "ELECTRIC RADIANT HEATER" CATEGORY

The term **radiant heaters** has created confusion, since most electric heaters emit radiant heat and they do not belong to this category. Some stakeholders have suggested renaming

[74] Commission Regulation (EU) 2016/2281 of 30 November 2016 implementing Directive 2009/125/EC of the European Parliament and of the Council establishing a framework for the setting of ecodesign requirements for energy-related products, with regard to ecodesign requirements for air heating products, cooling products, high temperature process chillers and fan coil units (OJ L 346/1, 20.12.2016, p 1)

the category. Other stakeholders have pointed out that they are not aware of any radiant electric heaters with a surface temperature rise of 130 degrees, which means that the category today is empty and other categories cover radiant heaters. Hence, the current definition is considered to be obsolete and only causing confusion among producers, reason why it should be removed.

Stakeholder views: no stakeholder has opposed to the removal of the electric radiant heater category.

5. CHANGE THE DEFINITIONS FOR LUMINOUS LOCAL SPACE HEATER AND TUBE LOCAL SPACE HEATER

During the review study, stakeholders have suggested adding the word **commercial** to the definitions of **luminous local space heater** and **tube local space heater** in order to make clear that these types of devices are only commercial heaters and not domestic.

In addition, liquid heaters are removed from the definition of luminous commercial local space heater as these heaters do not exist.

Stakeholder views: no stakeholder has opposed to the proposed changes.

6. UPDATE OF THE CC-FACTOR TO 2,1

Following the modification of the correction coefficient (CC) set out in the EED from CC = 2,5 to CC = 2,1, the limit efficiency values in the LSH Regulation need to be modified accordingly.

Product category	Existing regulation requirements with CC=2.5	Revised regulation requirements with CC=2.1
Portable	36%	43,6%
Fixed >250 W	38%	45,6%
Fixed <250 W	34%	41,6%
Storage	38.5%	46,1%
Underfloor	38%	45,6%
Visibly glowing >1,2 kW	35%	42,6%
Visibly glowing <1,2 kW	31%	38,6%

Table A7-1. Energy efficiency values for CC=2,5 and CC = 2,1

ANNEX 8: SENSITIVITY ANALYSIS

While the calculations and scenario modelling in this impact assessment has been carried out with energy prices, carbon intensity and renovation rates from the EU 2020 Reference scenario, a sensitivity assessment is made to cater for future fluctuations of the different parameters. For this sensitivity analysis the MIX scenario (in line with the fit-for-55 objectives)[75] has been used, with lower carbon intensity, higher energy prices and higher building renovation rates than REF2020.

The input parameters and the corresponding output parameters for which the sensitivity analysis has been analysed are listed in Table A8-1.

Parameter changed	**Affected output parameters**
Carbon intensity of electricity production	GHG emissions
Fuel prices	End user expenditure
Renovation rates of buildings	GHG emissions Energy consumption End user expenditure

Table A8-1. sensitivity parameters

The sensitivity analysis is only carried out for the preferred option, but the same effects on end results are applicable to all scenarios.

1. DATA INPUTS FROM THE MIX SCENARIO

The sensitivity analysis is made for the difference between the REF2020 and the MIX Scenario. Table A8-2 to Table A8-6 show the input parameters from the MIX scenario that are used for the sensitivity analysis.

Electricity	**2010**	**2015**	**2020**	**2025**	**2030**	**2035**	**2040**	**2045**	**2050**
Households	168,5	175,9	191,4	203,8	210,6	216,9	224,8	229,2	224,6
Services	146,8	157,1	176,4	186,5	191,0	193,9	200,9	206,4	203,9

Table A8-2. Electricity prices from MIX scenario, €/MWh

Natural gas	**2010**	**2015**	**2020**	**2025**	**2030**	**2035**	**2040**	**2045**	**2050**
Households	67,8	74,1	50,8	72,6	85,9	103,9	137,3	149,6	143,5
Services	54,6	157,1	176,4	186,5	191,0	193,9	200,9	206,4	203,9
Industry	37,7	35,5	22,9	29,1	35,9	39,7	54,7	104,8	107,0

Table A8-3. Natural gas prices from EU MIX Scenario in €/MWh

LPG	**2010**	**2015**	**2020**	**2025**	**2030**	**2035**	**2040**	**2045**	**2050**
Households	86,2	77,0	72,2	94,4	107,1	136,0	180,4	211,3	223,6
Services	72,5	62,4	60,4	84,9	98,4	115,3	138,1	153,2	161,4
Industry	84,0	70,7	71,5	89,9	100,5	103,3	104,2	104,2	102,0

Table A8-4. LPG (for liquid local space heaters) prices from EU MIX Scenario in €/MWh

[75] MIX Scenario

MT CO2-eq./TWh	2010	2015	2020	2025	2030	2035	2040	2045	2050
Electricity	0,318	0,305	0,213	0,149	0,100	0,037	-0,001	-0,004	-0,008
Natural gas	0,20								
Fuel oil	0,28								

Table A8-5 Fuel emission factors 2010-2050. EU MIX Scenario. NB: negative carbon intensity values in the years 2040-2050 are all set to zero instead of the negative values.

As seen in table A8-5 the carbon intensity of electricity becomes negative in 2040 and continues to become even more negative towards 2050. In order not to get an analysis showing that the more electricity is used, the better for the climate, these values are instead set to 0,00 in the sensitivity analysis.

	2015	2020	2025	2030	2035	2040	2045	2050
Annual renovation rate (as % of entire housing stock)	0,7%	0,9%	1,6%	2,1%	2,0%	2,0%	1,9%	2,0%
Energy Savings from Renovation (% of space-heating, average)	10,6%	28,2%	49,6%	61,1%	60,2%	55,2%	50,7%	45,0%
Combined effect on heat demand (% heat demand reductions)	0,07%	0,27%	0,79%	1,26%	1,20%	1,08%	0,94%	0,89%

Table A8-6. Data on renovation rates and impact from the EU MIX scenario

2. EFFECT OF FUEL PRICES: CONSUMER EXPENDITURE

In this section only the fuels costs are changed, while all other parameters are kept constant. The fuel costs in the MIX scenario are higher than in REF2020 for all years after 2020, the kick-off year for both scenarios. This results in overall higher consumer cost, and thus lower consumer expenditure savings from 2020 to 2050. Both REF2020 and MIX give the same energy savings, but in the MIX scenario the rise in electricity price counteracts those savings, which is not the case in REF2020.

However, the savings between BAU and the preferred policy option (called PO4 in the figures in this Annex) will be higher in 2050 using the MIX fuel costs (around €1.700 million) than the REF2020 fuel costs (around € 1.350 million). This effect can be seen Figure A8-1: on the one hand, consumer expenditure is higher for the two MIX scenarios, but nevertheless the difference between BAU and the preferred policy option is larger for MIX than for REF2020. This indicates that higher energy prices enhance the benefits of energy saving measures.

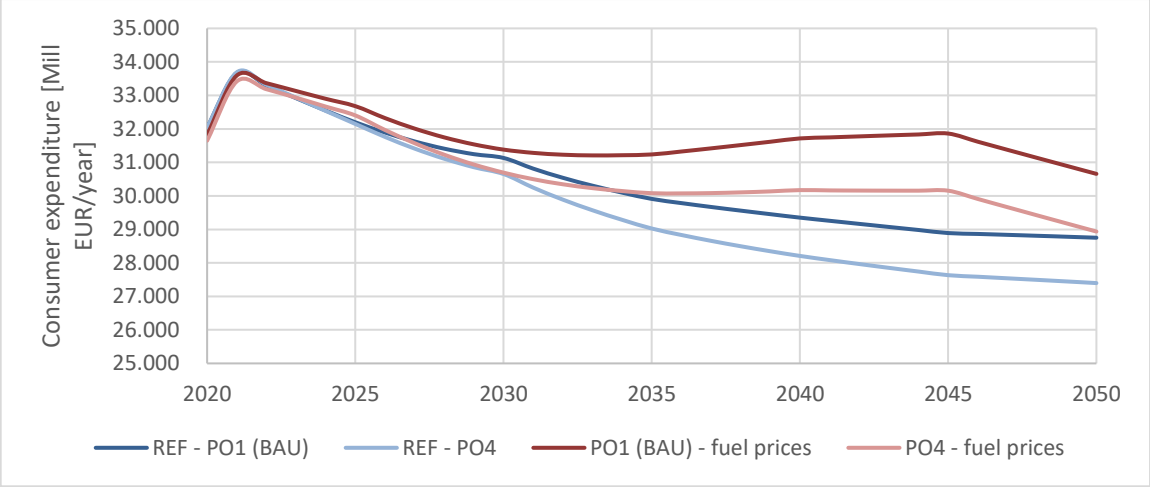


Figure A8-1. Consumer expenditure for BAU and the preferred policy option using REF2020 (labelled "REF") and MIX (labelled "carbon intensity").

3. EFFECT OF LOW CARBON ELECTRICITY: GHG EMISSIONS

In this section only the carbon intensity of electricity is changed, while all other parameters are kept constant. The carbon intensity of electricity in the MIX scenario is lower than in REF2020, resulting in lower GHG emission savings from the preferred policy option.

As seen in Figure A8-2 the lower carbon intensities with the MIX scenario result in much lower GHG emissions for both BAU and the preferred policy option, being 0 the difference between BAU and the preferred policy option in 2050 (as the carbon intensity is 0 kg CO2/kWh), while REF2020 shows a difference between the preferred policy option and BAU of 0,4 MT CO2/year by 2050.

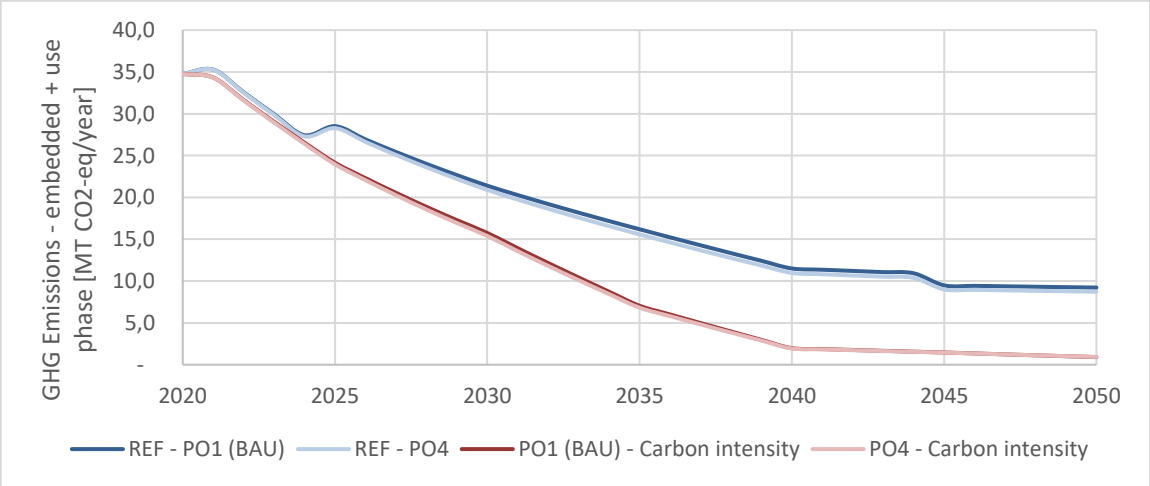


Figure A8-2: Greenhouse gas emissions for BAU and the preferred policy option using REF2020 (labelled "REF") and MIX (labelled "carbon intensity").

4. EFFECT OF RENOVATION RATES: ENERGY CONSUMPTION, GHG EMISSIONS, AND END USER EXPENDITURE

In this analysis only the renovation rates and the input parameters influenced by the former are changed, while all other parameters are kept constant. The annual renovation rates and the effect of the energy renovations on heat demand, are both higher in the MIX scenario than in REF2020, resulting in an overall lower heat demand from local space heaters and a consequent lower energy consumption, GHG emissions and consumer expenditure. However, as seen in the following three figures, the differences between REF2020 and MIX are very small, and it can be therefore concluded that the energy savings achieved by the measures proposed in this IA are not significantly affected by a change in scenario.

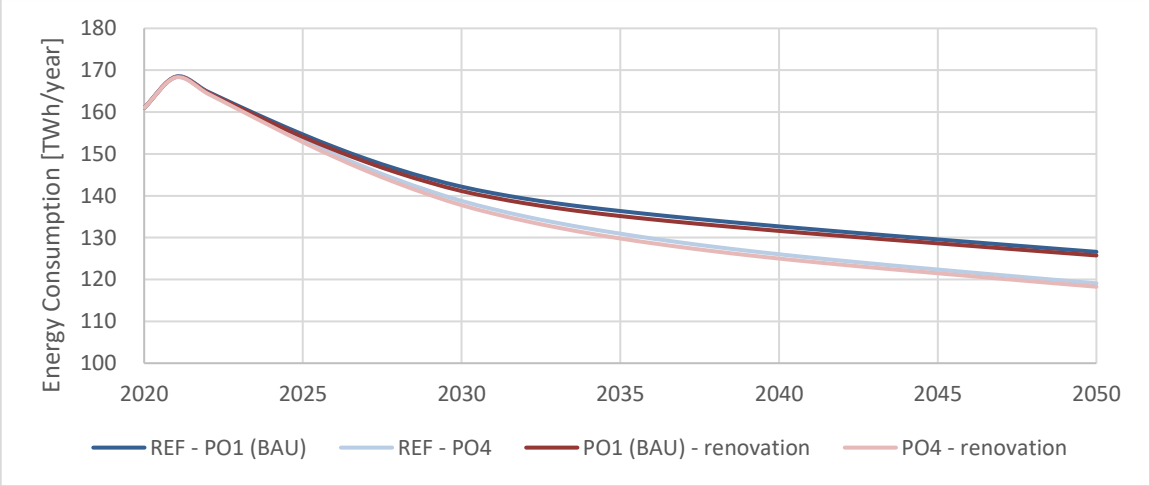


Figure A8-3. Effect on total energy consumption of higher renovation rates in the MIX scenario (labelled "renovation") compared to the REF2020 scenario (labelled "REF")

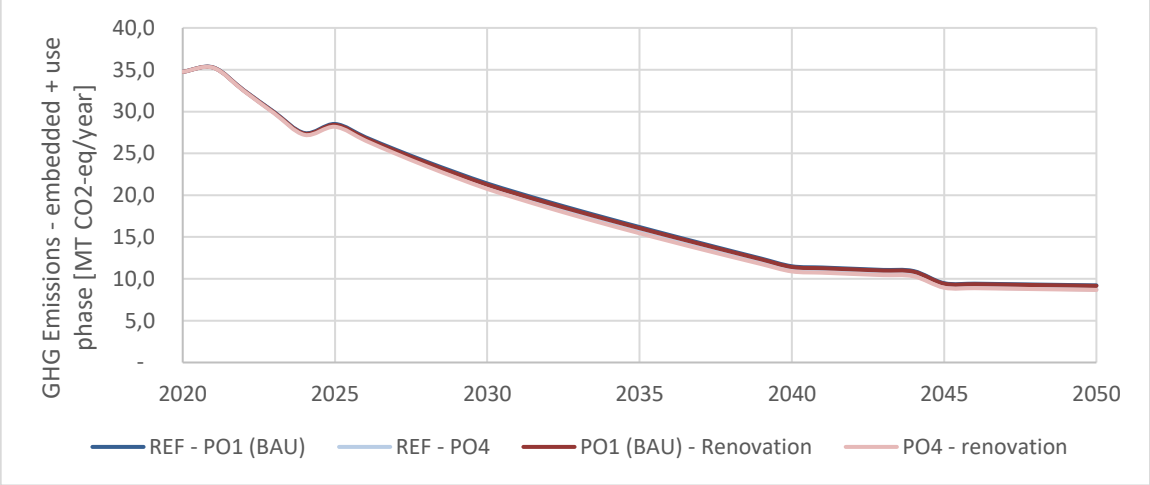


Figure A8-4. Effect on GHG emissions of higher renovation rates in the MIX scenario (labelled "renovation") compared to the REF2020 scenario (labelled "REF")

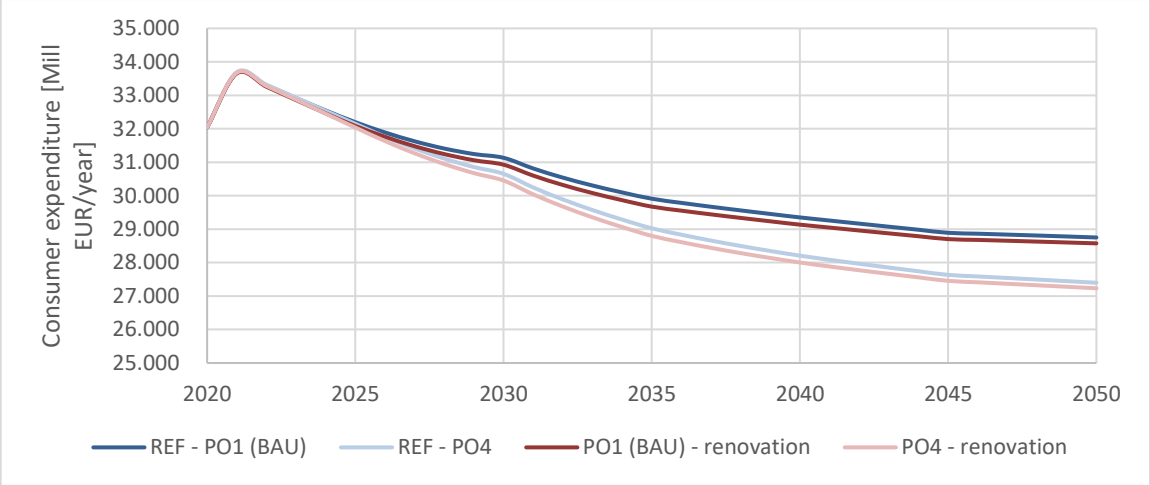


Figure A8-5. Effect on end user expenditure of higher renovation rates in the MIX scenario (labelled "renovation") compared to the REF2020 scenario (labelled "REF")

ANNEX 9: SME TEST

The SME test is carried out according to Tool#23 of the Better Regulation Toolbox.

Step (1) of the SME test refers to the identification of the affected businesses and addresses the following threshold question: to what extent is the initiative relevant for SMEs?

1. IDENTIFICATION OF THE AFFECTED BUSINESSES

In the absence of specific data on the number of manufacturers of local space heaters, including SMEs, the share of SMEs in the sector has been worked out from data from the Eurostat Structural Business Statistics. Eurostat provides turnover and number of SMEs for the entire appliance sector (NACE 275). The share of SMEs that are manufacturers of local space heaters is calculated by multiplying the number of SMEs for the appliance sector by the ratio *turnover of the local space heaters industry (calculated in section 6.2.1)/turnover for the entire appliance sector* also from Eurostat. The ratio is 1.043M€/54.336M€ = 0,019, which multiplied by 2.983 SMEs for the entire appliance sector, gives 57 SMEs in the local space heaters sector, which will be affected by the measures. Assuming that the share SMEs/total companies remains the same for local space heaters as for the entire appliance sector, 57 SMEs would be equivalent to 96% of the manufacturers.

2. THRESHOLD QUESTION: TO WHAT EXTENT IS THE INITIATIVE RELEVANT FOR SMES?

Although SMEs are in the scope of the initiative, the latter is not specifically targeting SMEs. The initiative will apply to the whole sector of local space heaters, with no distinction between big companies and SMEs.

As already addressed in sections 6 and 7, the measures proposed will not have significant negative impacts on the companies affected. SMEs will not be negatively affected either because measures that could entail a significant effort of adaptation and therefore an increase of costs that could not be offset with lower operating costs have been discarded. This concerns two specific measures. First, the possibility to set stricter ecodesign limits, which has been ruled out at an early stage due to the priority given to modify the scope of the LSH Regulation. Second, measure PO4b on modularity has also been ruled out because it would result in a steep increase of costs that would not be offset by its benefits.

In addition, after a thorough revision of the feedback provided by stakeholders during the process of legislative review, and in particular the relevant industry representatives mentioned in section 5.4.1, no particular concerns have been noted about SMEs.

Finally, the SME filter list, which screens all the initiatives published in the “have your say” portal after 1 January 2022, does not mention ecodesign for local space heaters as one of the initiatives that could be relevant for SMEs.

Given the background above, it can be concluded that the initiative does not entail any negative impact on SMEs and subsequently there is no need to carry out the next steps of the SME test as laid down in the Better Regulation Toolbox.

ANNEX 10: MULTI-CRITERIA ANALYSIS

1. MULTI-CRITERIA DECISION ANALYSIS METHODOLOGY: BASIC DESCRIPTION OF SOCIAL MULTI-CRITERIA EVALUATION (SMCE) METHODOLOGICAL FRAMEWORK AND SOFTWARE TOOL

This section describes the analysis of the impacts carried out with the software SOCRATES (**SO**cial multi **CR**iteria **A**ssessmenT of **E**uropean policie**S**) developed by the Joint Research Centre. SOCRATES has been used to validate the choice of the preferred policy option according to a mathematical model.

SOCRATES is a new multiple criteria assessment software tool, explicitly designed for *ex-ante* Impact Assessment (IA) problems[76].

Quantitative evidence plays an important role in many IAs, but also qualitative data such as stakeholder input, conclusions of evaluations, as well as scientific and expert advice are frequently used. This generates a multitude of criteria, which should be consistently integrated and evaluated when comparing policy options. The most widespread multidimensional approach to ex-ante IAs is multi-criteria decision analysis, which forms the basis for social multi-criteria evaluation (SMCE[77]), which has been explicitly designed for public policy. SMCE allows taking into account a wide range of assessment criteria, such as the impact on SMEs, the degree of protection of fundamental rights, consumer protection, etc. while all the multidimensional profiles of the problem remain in their original scales of measurement. Indeed, the latter is the main difference with traditional cost-benefit analysis (CBA), which grounds on steps like monetizing all social, environmental, and human rights aspects. In this respect, CBA and SMCE are not conflictual but complementary, as CBA can be utilised as component of a SMCE framework, dealing with the economic dimension.

Overall, the objective of SOCRATES and the underlying SMCE methodology is not to substitute policy-makers through a mathematical model, but to improve their understanding of the main features of the problem at hand, such as key assumptions, degree of uncertainty, robustness of results and overall technical and social defensibility of options chosen. While SMCE has already been applied in a multitude of policy problems, its recent technical implementation through SOCRATES is now applied for this Impact Assessment.

SMCE proceeds on the basis of following main concepts: dimensions, objectives, criteria, weights, criterion scores, impact matrix and compromise solution.

---

[76] Ownership details: The software has been developed in the context of the European Commission's Competence Centre on Modelling (Non free license).

[77] Munda, G., A social multi-criteria framework for ex-ante impact assessment: Operational Issues, EUR 28752 EN, Publications Office of the European Union, Luxembourg, 2017, ISBN 978-92-79-72293-6, doi:10.2760/909528, JRC107899.; Munda, G., Dealing with Fairness in Public Policy Analysis: A Methodological Framework , EUR 28751 EN, Publications Office of the European Union, Luxembourg, 2017, ISBN 978-92-79-72292-9, doi:10.2760/75185, JRC107843.; Munda, G., On the use of Cost-Benefit Analysis and Multi-Criteria Evaluation in ex-ante Impact Assessment, EUR 28768 EN, Publications Office of the European Union, Luxembourg, 2017, ISBN 978-92-79-73213-3, doi:10.2760/311199, JRC107900.

- **Dimension** is the highest hierarchical level of analysis and indicates the scope of objectives, criteria and criterion scores.
- **Objectives** indicate the direction of change desired, e.g. growth has to be maximized, social exclusion has to be minimized, and carbon dioxide emissions have to be reduced.
- A **criterion** is a function that associates alternative actions with a variable indicating its desirability.
- **Weights** are often used to represent the relative importance attached to dimensions, objectives and criteria. The idea behind this practice is very intuitive and easy, that is, to place the greatest number in the position corresponding to the most important factor.
- A **criterion score** is an assessment of the impact consistent with a given criterion with reference to a policy option. Criterion scores can be both qualitative and quantitative.
- The **impact matrix** presents in a structured way, the information on the various criterion scores, i.e. each element of the matrix represents the performance of each option according to each criterion.

In general, in a multi-criterion problem, there is no solution (ideal or utopia solution) optimizing all the criteria at the same time, and therefore **"compromise solutions"** have to be found.

In summary a SMCE approach can supply a methodological framework where the hierarchical structure of the option comparison step of a typical ex-ante IA (including dimensions, objectives and evaluation criteria) is clarified as much as possible by means of well-established concepts in the decision theory literature. This might help in increasing the degree of homogeneity across IA studies. The SOCRATES software helps structuring such a methodological framework.

A typical SOCRATES input requires the definition of policy options (called alternatives) dimensions, objectives and criteria. This information leads to the construction of an impact matrix, which may include crisp, stochastic or fuzzy measurements of the performance of an alternative with respect to an evaluation criterion. Qualitative information can be introduced too (in the form of linguistic or ordinal criterion scores). Weights as importance coefficients, may also be introduced. They can be attached to dimensions or criteria. Indifference and preference thresholds can also be introduced when needed. Generally a social conflict matrix is also constructed, where the impacts of each policy option on each social group are presented in a transparent way.

In this IA, for better granularity the analysis is made with respect to the individual measures instead of for each of the policy options as a whole, so that it is possible to combine those individual measures in a single and preferred policy option. The time horizon taken for the analysis is 2050 because the impact of the measures related to circular economy mostly take place when the stock has aged, being therefore 2030 a too short term to assess PO4, in particular for PO4a. The rest of the policy options keep a similar behaviour throughout the whole period under study.

A number of criteria have been chosen that correspond to the three main objectives of an IA, which is to assess **effectiveness**, **efficiency** and **coherence** of the policy options.

For **effectiveness** two criteria have been included: *reduction of energy consumption during the use phase* and *reduction of embedded energy consumption (production, distribution and EoL)*.

For **efficiency**, the following criteria have been included: *increase of business turnover*, *consumer savings* and *administrative costs*.

As far as **coherence**, the following concepts have been included: *harmonisation with other EU legislation/standards* already existing for other products, *reduction of legal loopholes/reduction of regulatory confusion* and *level of stakeholders support*.

The multi-criteria analysis is made in two steps:

- First step: multi-criteria analysis for PO2, which will result in a preferred combination of measures ($PO2_f$).

- Second step: multi-criteria analysis for a preferred policy option combining $PO2_f$, PO3 and PO4a.

2. FIRST STEP: MULTI-CRITERIA ANALYSIS FOR PO2

PO2 consists of the following four measures:

PO2a: including slave heaters in the scope.

PO2b: including towel heaters in the scope.

PO2c: including tube heaters between 120 and 300 W.

PO2d: modifying some definitions corresponding to gas and oil fuelled heaters.

PO2a is, according to section 6, the measure that generates by far the highest share of positive impacts in terms of effectiveness and efficiency, and considering that any alternative would be significantly weakened if 2a is not included it has been decided to include 2a in all combinations. With this simplification, the number of possible combinations is reduced from 15 to 8.

The following impact matrix displays the score of the different PO2 configurations against the chosen criteria in order to choose a final combination that will be called $PO2_f$.

Criteria	**PO2a**	**PO2a+ PO2b**	**PO2a+PO2c**	**PO2a+PO2d**	**PO2a+PO2b+PO2c**	**PO2a+PO2b+PO2d**	**PO2a+PO2c+PO2d**	**PO2a+PO2b+PO2c+PO2d**
Effectiveness								
C1. Energy savings during the use phase (Twh/year)	4,69	5,57	4,76	4,83	5,64	5,71	4,90	5,78
C2. Reduction of embedded energy consumption (production, distribution and EoL) (Twh/year)	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
Efficiency								
C3. Increase of business turnover	280,24	356,39	280,68	305,79	356,83	381,94	306,23	382,38

(M€/year)								
C4. Consumer savings (M€/year)	849,10	996,76	852,37	847,98	1000,03	995,64	851,25	998,91
C5. Administrative costs	+	+	+	+	+	+	+	+
Coherence								
C6. Harmonisation with other EU legislation /standards	=	=	=	+	=	+	+	+
C7. Reduction of legal loopholes/reduction of regulatory confusion	+	++	+	+	++	++	+	++
C8. Level of stakeholders support	+	+	++	++	++	++	++	++

Legend: (-) decrease/bad; (=) indifferent; (+) increase/good; (++) significant increase/very good

Table A10-1. Multi-criteria impact matrix for PO2

2.1. Justification of the score for the qualitative criteria

- On *administrative costs*, measures that entail a broadening of the scope (PO2a, PO2b and PO2c) have been scored with a +, since some administrative costs will apply to the new products in the scope. Since the effect of PO2a is much larger than PO2b and PO2c (also on new models coming in the scope), it has been considered that the administrative costs brought about by PO2b and PO2c are small compared to PO2a, and this is why the same score has been given to the various measure combinations where PO2a is present.

- On *harmonisation with other EU legislation/standards*, PO2d adds a "+" to the combinations in which it is present due to harmonisation with existing standards. The rest of measures are indifferent in this respect.

- On *reduction of legal loopholes/regulatory confusion*, again PO2a and PO2b add a "+" each, combinations with PO2a are scored with a "+", and when they include PO2b they get an additional "+". This is because PO2a and PO2b are expected to simplify the judgement of market surveillance authorities and companies on whether a product should or not be considered in the scope, aspect which is currently a source of frequent disputes between authorities and industry, as already pointed out.

- On *level of stakeholders support*, all combinations have been scored with at least a "+" due to the general agreement of stakeholders on the convenience of the measures. Combinations with PO2c or PO2d feature an additional "+" because the support to those measures is not only general but also unanimous.

2.2. Result of the multi-criteria analysis

The multi-criteria analysis run by SOCRATES shows that the preferred option is PO2a+PO2b+PO2c+PO2d. Therefore, $PO2_f$ will include the four measures under assessment. This option is the strongest one in terms of **effectiveness**, as the energy savings and therefore GHG emissions reduction are the highest compared to the rest of the options. On **efficiency**, PO2a+PO2b+PO2c and PO2a+PO2b+PO2c+PO2d are in a similar situation, the latter encountering slightly higher turnover but also somewhat lower consumer savings. As far as **coherence**, both combinations are alike.

The importance of mathematical approaches in SMCE is their ability to allow a consistent aggregation of the diverse information. Otherwise, even if everybody would

agree on the multidimensional nature of an IA study, the implementation in a real-world assessment exercise would be impossible. The standard objection might be that the aggregation of apples and oranges is impossible. Multi-criteria mathematics does answer to this objection in a definitive way. When using mathematical rules, consistency between the problem structuring and the ranking of policy options is guaranteed, this makes the overall IA study much more defensible.

SOCRATES makes all required computations very quick. From a mathematical point of view, the information contained in the impact matrix useful for solving the so-called multi-criterion problem is:

- Intensity of preference (when quantitative criterion scores are present).
- Number of criteria in favour of a given alternative.
- Weight attached to each single criterion.
- Relationship of each single alternative with all the other alternatives.

Combinations of this information generate different aggregation conventions, i.e. manipulation rules of the available information to arrive at a preference structure. The aggregation of several criteria implies taking a position on the fundamental issue of compensability. For example, in evaluating a policy option that presents a very bad environmental impact and a very good economic impact, it is clear that allowing or not for compensability and to which degree is the key assumption.

An aggregation rule that is simple, non-compensatory and minimises the rank reversal phenomena is the Kemeny rule. Its basic idea is that the maximum likelihood ranking of policy options is the ranking supported by the maximum number of criteria (or criterion weights) for each pair-wise comparison, summed over all pairs of options considered. There is agreement in the literature that the Kemeny method is “the correct method” for ranking options, and that the only drawback of this aggregation method is the difficulty in computing it when the number of options grows. A numerical algorithm solving this computational drawback in an efficient way has been developed recently at JRC and has been implemented in SOCRATES[78].

Various authors have argued that the presence of qualitative information in evaluation problems concerning socio-economic issues is a rule, rather than an exception. Thus there is a clear need for methods that are able to take into account information of a "mixed" type (both qualitative and quantitative criterion scores). Moreover, ideally, this information should be precise, certain, exhaustive and unequivocal. Nevertheless, in reality, it is often necessary to use information which does not have those characteristics so that one has to face the uncertainty of a stochastic and/or fuzzy nature present in the data. Therefore, multi-criteria methods able to tackle consistently the widest types of mixed information should be considered as desirable ones in the IA framework.

From a mathematical point of view, SOCRATES deals with two main issues:

[78] Azzini I. and Munda G. (2020) – A New Approach for Identifying the Kemeny Median Ranking, European Journal of Operational Research, Vol. 281, pp. 388-401 https://doi.org/10.1016/j.ejor.2019.08.033.

1. The problem of equivalence of the procedures used in order to standardize the mixed criterion scores;
2. The problem of comparison of fuzzy numbers typical of all fuzzy multi-criteria methods.

These two issues are dealt with a new semantic distance that is useful in the case of continuous, convex membership functions also allowing a definite integration[79].

Overall, the objective of SOCRATES is NOT substitution of policy-makers through a mathematical model, on the contrary, the objective is to improve their understanding of the main features of the problem at hand, such as key assumptions, degree of uncertainty, robustness of results and overall technical and social defensibility of options chosen. The philosopher Socrates said: "*I cannot teach anybody anything. I can only make them think.*" This is the main inspiring principle of the SOCRATES software too.

The SOCRATES software offers a measurement framework where the various criterion scores can assess impacts by using both quantitative (e.g. as result of simulation models) and qualitative (e.g. results of participatory techniques) information, and the mathematical aggregation rule guarantees consistency and transparency of results.

Three main components constitute the core of SOCRATES: multi-criteria, equity and sensitivity analyses. Multi-criteria analysis requires the definition of relevant dimensions, objectives and criteria. It uses weights as importance coefficients and clarify their role in the hierarchical structure. The impact matrix may include quantitative (including also stochastic and/or fuzzy uncertainty) and qualitative (ordinal and/or linguistic) measurements of the performance of an alternative with respect to an evaluation criterion. It supplies a ranking of the alternatives according to the set of evaluation criteria (i.e. the technical compromise solution/s).

By applying SOCRATES to the information contained in the impact matrix, the following ranking, described in Figure A10-1 is obtained (under the assumption that all criteria have the same weight, see Figure A10-2).

Ranking

Rank	1°	2°	3°	4°	5°	6°	7°	8°
20.4	2a+2b+2c+2d	2a+2b+2d	2a+2b+2c	2a+2b	2a+2c+2d	2a+2d	2a+2c	2a

Figure A10-1. preferred option

The ranking is very clear: option ***PO2a+PO2b+PO2c+PO2d*** is the best choice followed by option ***PO2a+PO2b+PO2d.*** The set composed by options ***PO2a+PO2c*** and ***PO2a*** is the worst one.

[79] Munda G.– *Intensity of Preference and related Uncertainty in Non-Compensatory Aggregation Rules*, Theory and Decision, Volume 73, Issue 4 (2012), Pages 649-669, DOI: 10.1007/s11238-012-9317-4.

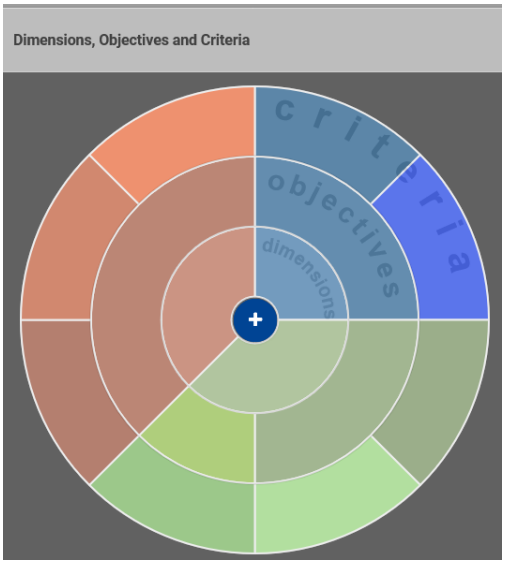


Figure A10-2. Equal criterion weighting assumption

By assuming that all dimensions have the same weight (see Figure A10-3), the ranking stays the same, thus both basic weighting schemes produce the same result. This result robustness will be further checked by means of local and global sensitivity analyses.

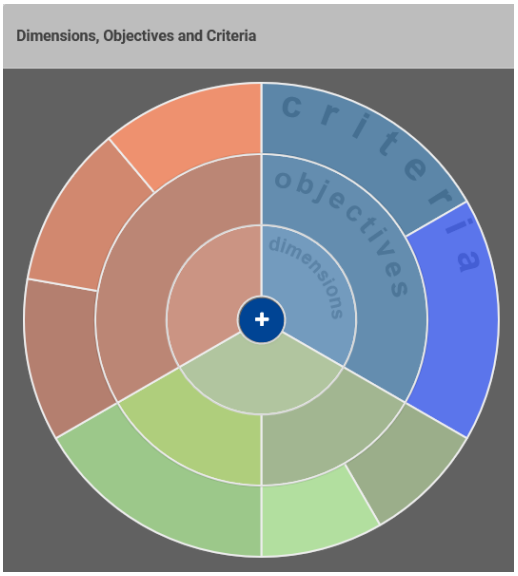


Figure A10-3. Equal dimension weighting assumption

More information can be obtained by checking the pairwise comparisons, which allow one to be fully aware of the mutual weaknesses and strengths on each single evaluation criterion. This information is summarised graphically in the following figure A10-4, where the degrees of credibility that any option is preferred or indifferent with respect to another one on each single criterion are illustrated. From this figure it is possible to deduce that options ***PO2a+PO2b+PO2c+PO2d*** and ***PO2a+PO2b+PO2d*** are indifferent on 5 criteria but 3 criteria show a clear preference towards ***PO2a+PO2b+PO2c+PO2d.*** Overall from the pairwise comparison, it can be safely concluded that option ***PO2a+PO2b+PO2c+PO2d*** is the preferred option.

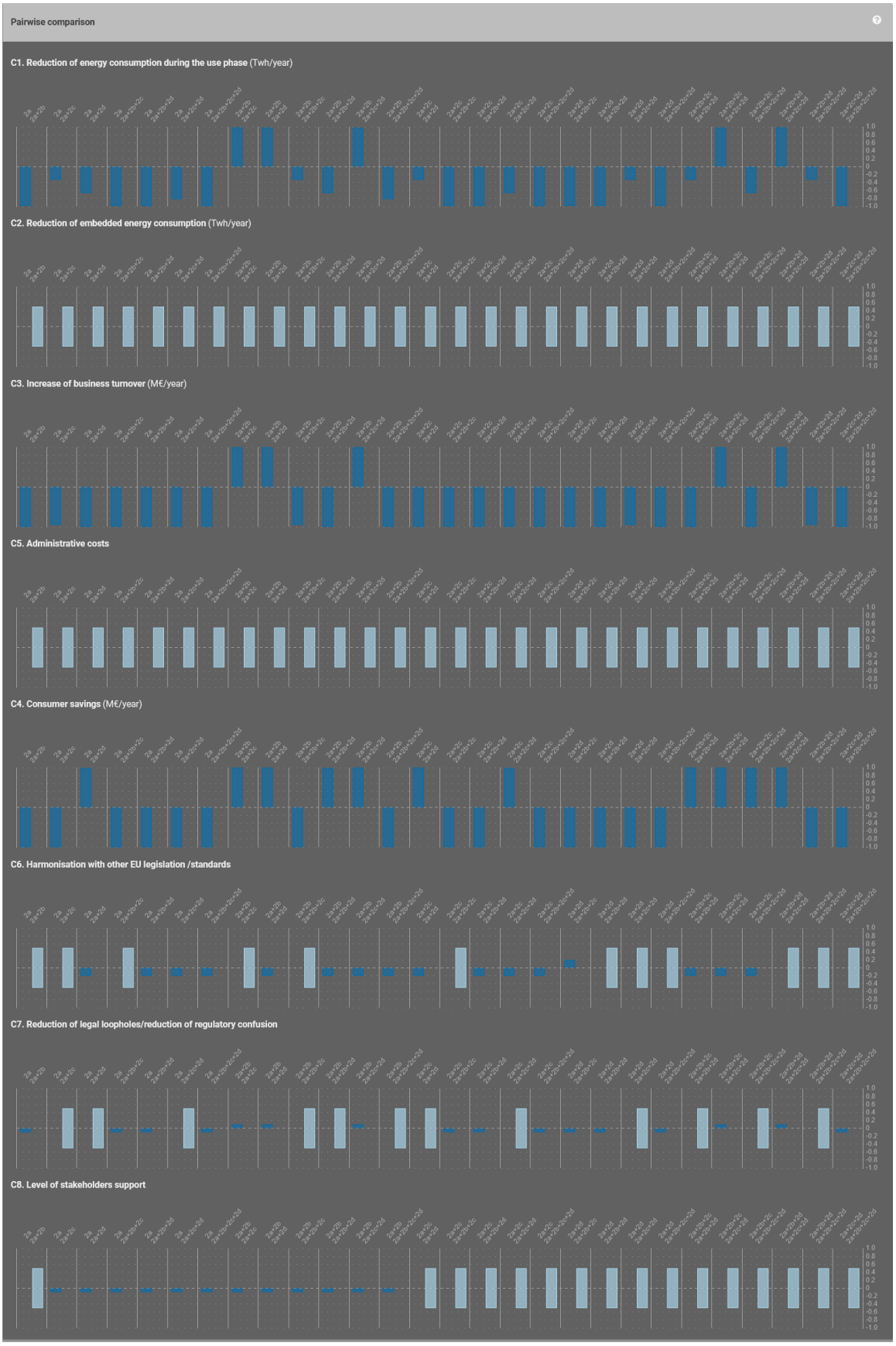


Figure A10-4. Pairwise comparison

To further clarify the preference structure, it is advisable to perform a *sensitivity analysis.* In the framework of SOCRATES, the objective of sensitivity analysis is to check if the rankings provided are stable and to determine which of the input parameters influence more the model output. Local sensitivity analysis looks at the sensitivity of results to a) the exclusion/inclusion of different criteria and dimensions and b) dimensions and criterion weights change. All parameters are changed one per time. Global sensitivity analysis focuses on all the possible combinations of criterion weights; all parameters are changed simultaneously. The whole information produced by local and global sensitivity analyses is synthesised into simple graphics.

Let us then first look at the influence of the exclusion of the various criteria and dimensions, first one criteria per time and second the subset of criteria belonging to one dimension only (i.e. first one criterion per time is eliminated and the corresponding ranking is obtained and second a whole dimension with all its criteria is eliminated and the effect on the final ranking is checked).

The results of this exercise are presented in the following Figure A10-5, where it is indicated how many times each option is present in any rank position, and the percentage each rank position is occupied by each single option. In this way, option ***PO2a+PO2b+PO2c+PO2d*** appears as the most desirable one, in fact it occupies the first position in the ***92,8 per cent*** of all the rankings obtained.



Figure A10-5: Dimensions – criteria summary

Finally, the robustness of results with respect to weights is particularly relevant. The rankings according to the equal criterion and dimension weighting assumptions have already been computed, and now the focus is set on what happens if all possible combinations of criterion weights are considered. This exercise is carried out by means of the global sensitivity analysis.

As one can see in figure A10-6, the results are very stable, in fact, whatever weight set is used option ***PO2a+PO2b+PO2c+PO2d is the top one in the 97 per cent of all the rankings obtained.***



Figure A10-6: Weights summary.

3. SECOND STEP: MULTI-CRITERIA ANALYSIS FOR A PREFERRED POLICY OPTION COMBINING $PO2_F$, PO3 AND PO4A

Criteria	$PO2_f$	PO3	PO4a	$PO2_f$+PO3	$PO2_f$+PO4a	PO3+PO4a	$PO2_f$+PO3+PO4a
Effectiveness							
C1. Energy savings during the use phase (Twh/year)	5,78	0,32	0,00	6,21	5,79	0,32	6,21
C2. Reduction of embedded energy consumption (production, distribution and EoL) (Twh/year)	0,00	0,00	≈ 0	0,00	≈ 0	≈ 0	≈ 0
Efficiency							
C3. Increase of business turnover (M€/year)	382,38	18,60	-177,38	408,50	148,99	-158,78	173,92
C4. Consumer savings (M€/year)	998,91	58,20	18,46	1.074,72	1.023,28	76,66	1.099,21
C5. Administrative costs	+	=	=	+	+	=	+
Coherence							
C6. Harmonisation with other EU legislation /standards	+	+	+	++	++	+	++
C7. Reduction of legal loopholes/reduction of regulatory confusion	+	=	=	+	+	=	+
C8. Level of stakeholders support	+	+	+	+	+	+	+

Table A10-2. Impact matrix of the options obtained by combining $PO2_f$, PO3 and $PO4_f$

By assuming that all dimensions have the same weight the ranking sets option **$PO2_f$+PO3+$PO4_f$** as the best choice followed by option **$PO2_f$+PO3** (see figure A10-7).

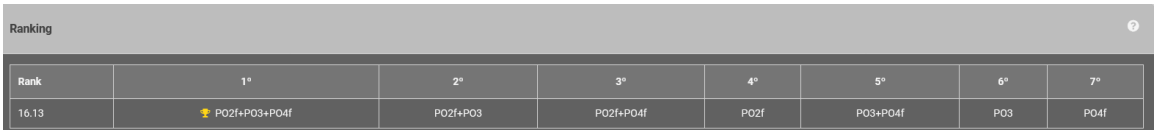


Figure A10-7. Preferred option under equal dimension weighting assumption

By applying SOCRATES under the assumption that all criteria have the same weight, the rankings show that both **$PO2_f$+PO3+$PO4_f$** and **$PO2_f$+PO3** can be top options (see Figures A10-8 and A10-9).

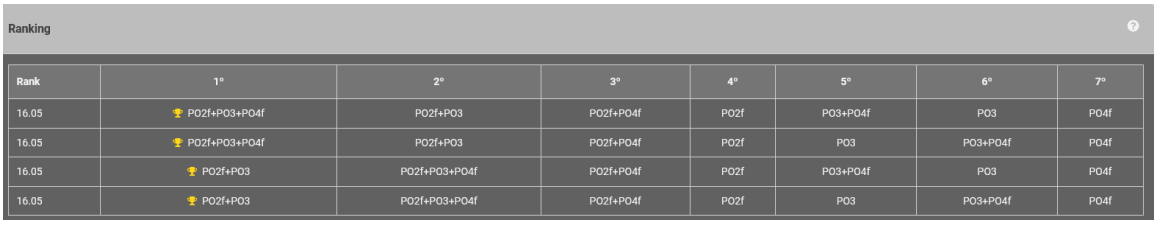


Figure A10-8. Preferred option under equal criterion weighting assumption

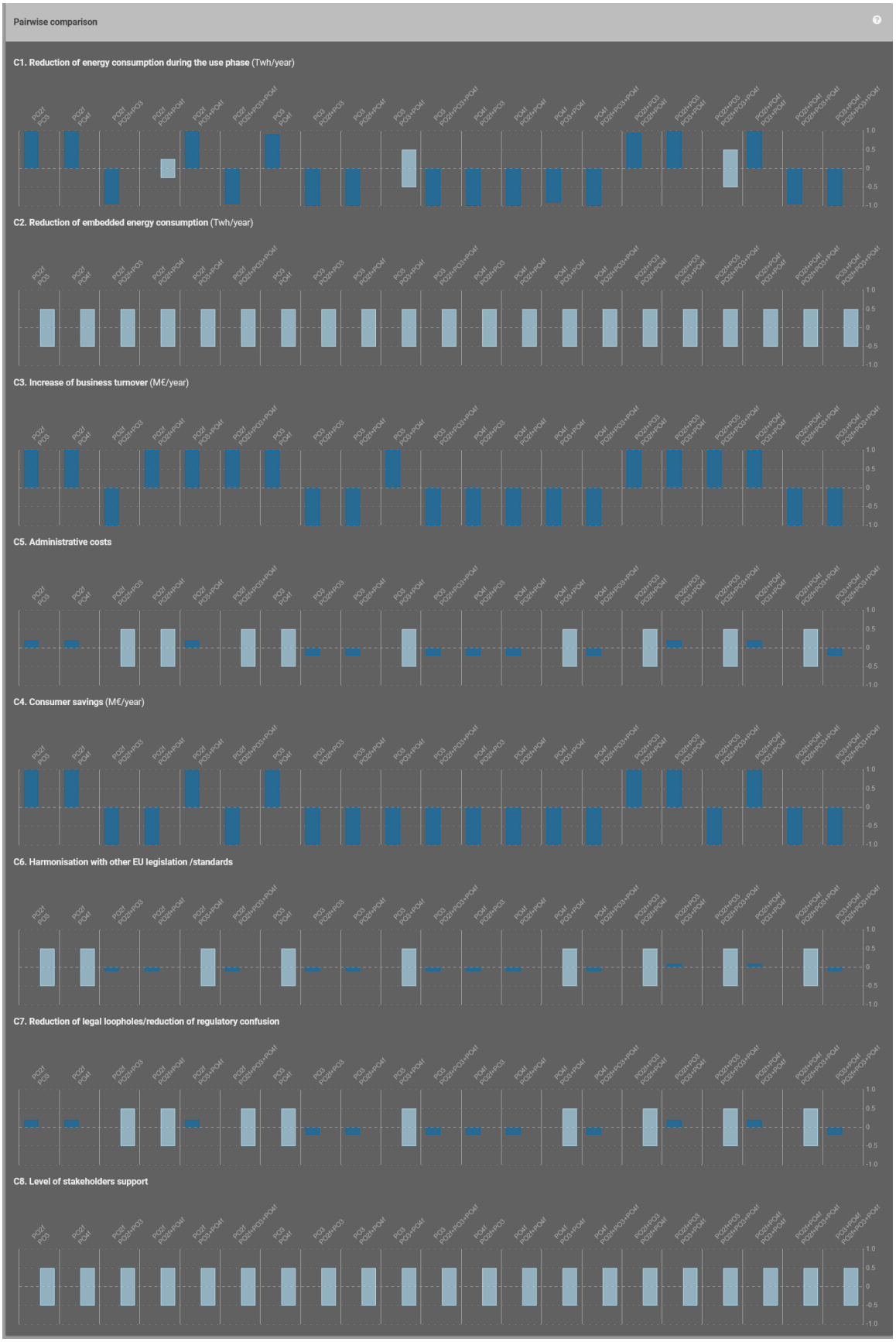


Figure A10-9. Pairwise comparison

Given this equality, the analysis of the influence of dimensions/criteria more in depth by means of sensitivity analysis becomes particularly important.

Let us then first look at the influence of the exclusion of the various criteria and dimensions, first one criteria per time and second the subset of criteria belonging to one dimension only. The results of this exercise are presented in the following Figure A10-10. **PO2$_f$+PO3+PO4$_f$** is the preferred option, in fact, it occupies the first position in the ***93 per cent*** of all the rankings obtained. The preference towards **PO2$_f$+PO3+PO4$_f$** is even clearer and more robust when exploring the weighting space of dimensions, in fact whatever weight is attached to any of the dimensions, **PO2$_f$+PO3+PO4$_f$** is always ranked first (see Figure A10-11). This result is corroborated by exploring the criterion weights too by local sensitivity analysis (see Figure A10-12). In this case, weights are changed to one criterion per time till a maximum of 50 per cent of the total importance; ***88 per cent*** of cases is in fact in favour of option **PO2$_f$+PO3+PO4$_f$** . It is finally checked the effect of changing weights for all criteria by means of the global sensitivity analysis, where all weights are changed at the same time and extreme cases are considered too (e.g. the case where only one criterion has the 100 per cent of importance). As shown in Figure A10-13 the results indicate that options **PO2$_f$+PO3+PO4$_f$** and **PO2$_f$+PO3** are closer, but still option **PO2$_f$+PO3+PO4$_f$** is slightly better than **PO2$_f$+PO3** (it is the top option in the ***54 per cent*** of all the rankings obtained). In summary, it can be concluded that by considering its overall performance, option **PO2$_f$+PO3+PO4$_f$** is the best candidate to be the preferred option.

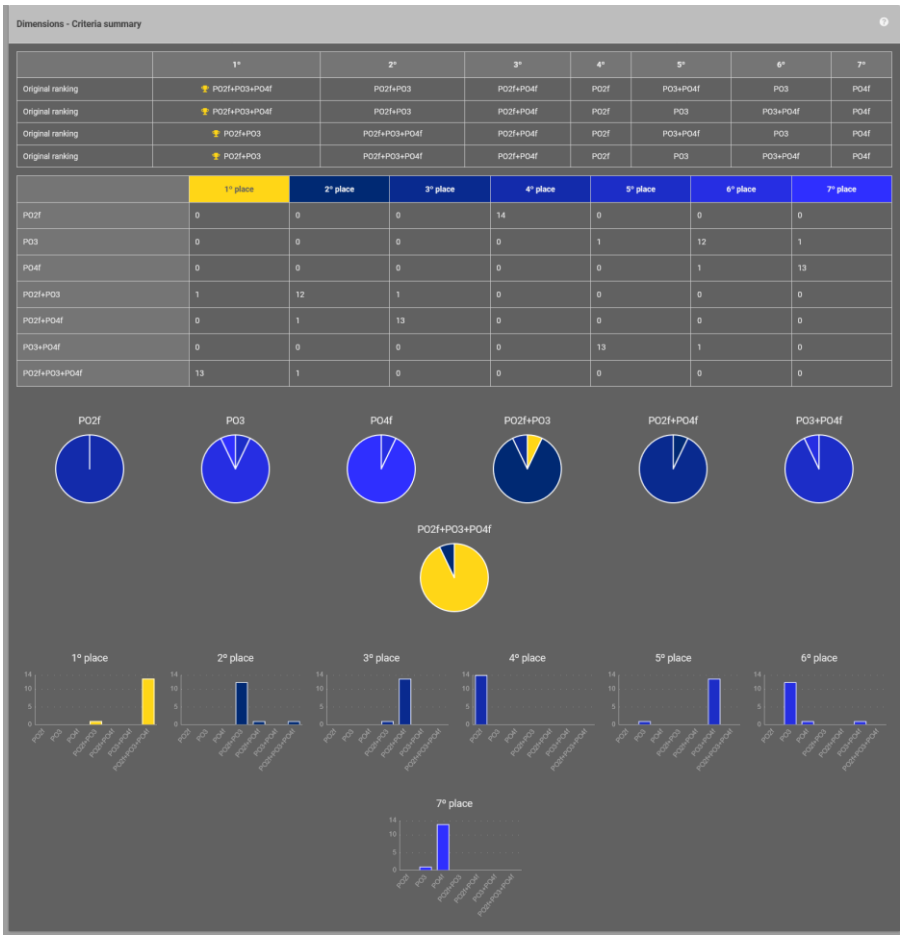


Figure A10-10: Dimensions – criteria summary

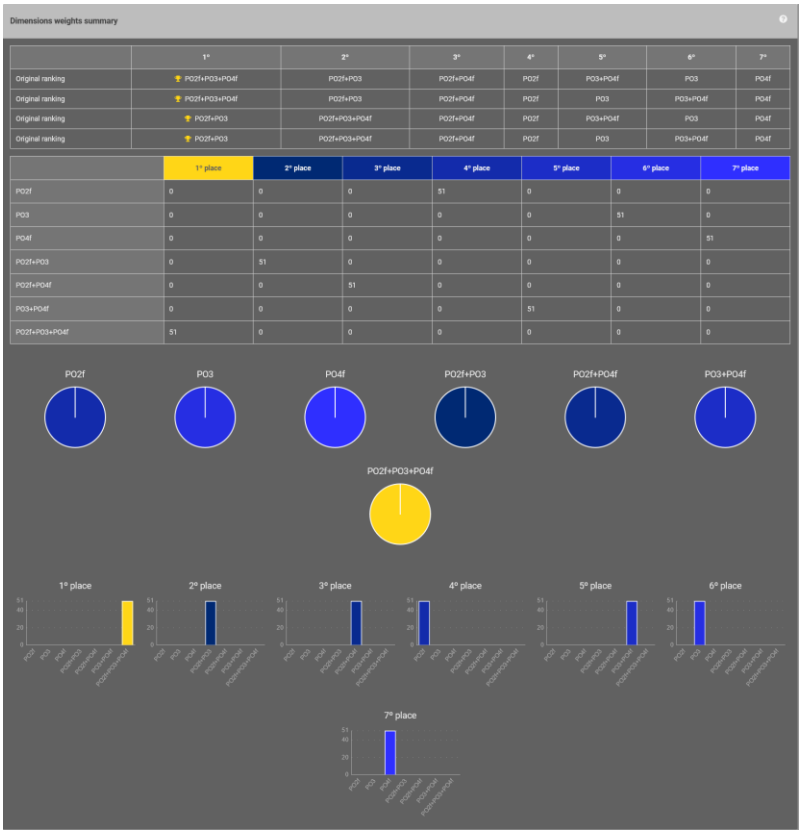


Figure A10-11. Dimension weights summary

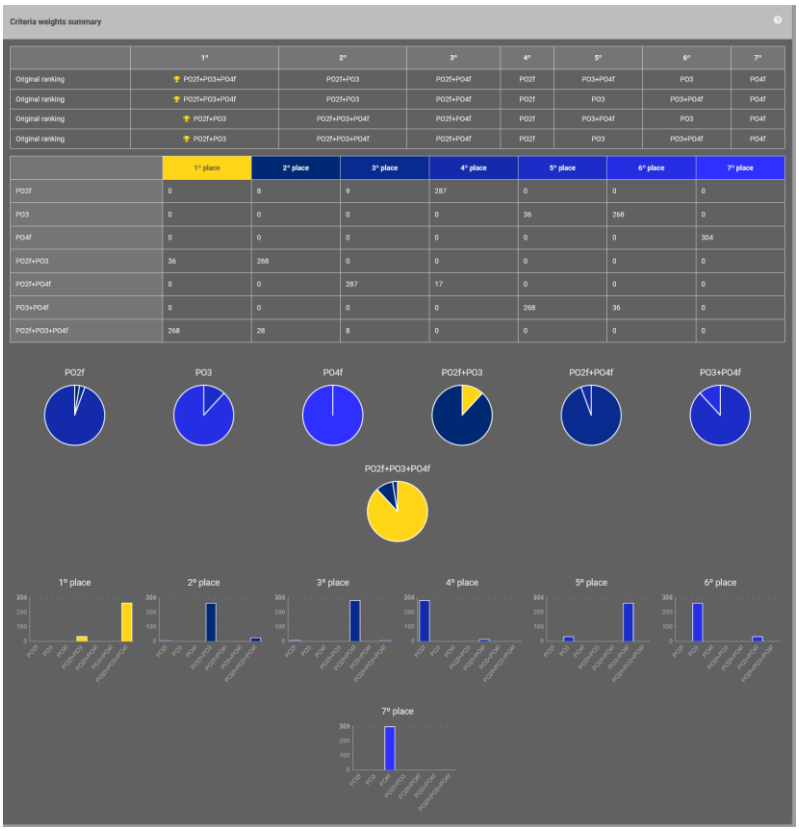


Figure A10-12. Criteria weights summary (local sensitivity analysis)

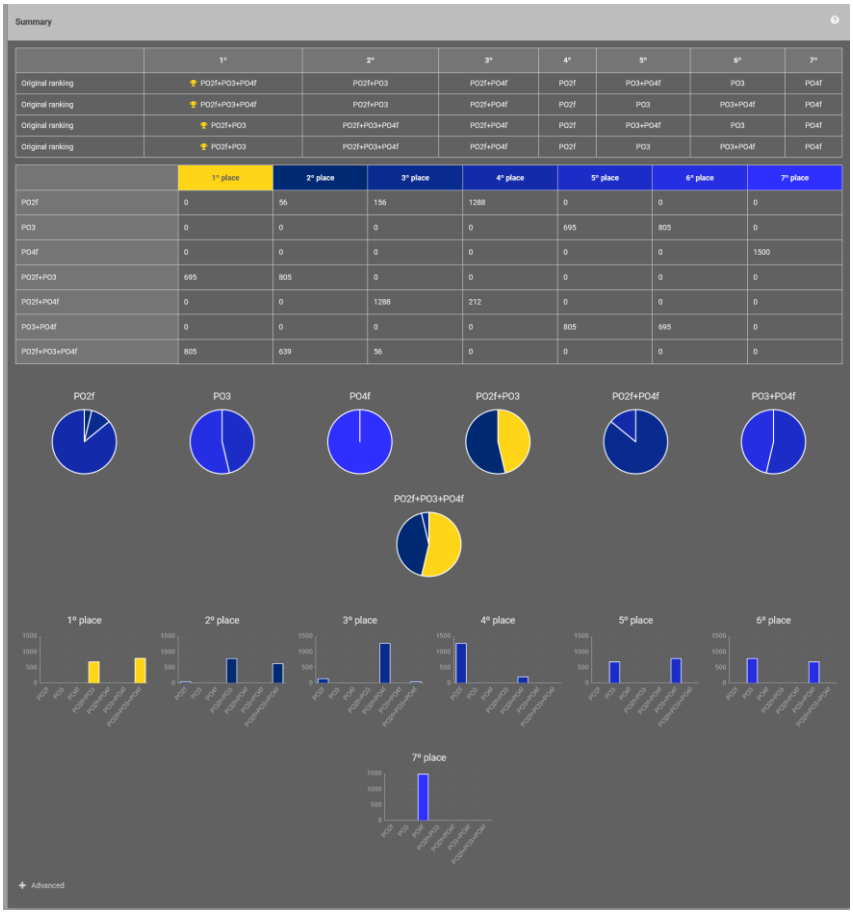


Figure A10-13. Criteria weights summary (global sensitivity analysis)

ANNEX 11: THE ECODESIGN AND ENERGY LABELLING FRAMEWORK

The Ecodesign Framework Directive and Energy Labelling Framework Regulation are framework rules, establishing conditions for laying down product-specific requirements in regulations adopted by the Commission. The Commission's role in the implementation of delegated and implementing acts is to ensure a maximum of transparency and stakeholder participation in presenting a proposal, based on generally accepted data and information, to the European Parliament and Council for scrutiny. Figure gives an overview of the procedures leading to a voluntary agreement, Ecodesign requirements or an Energy label.

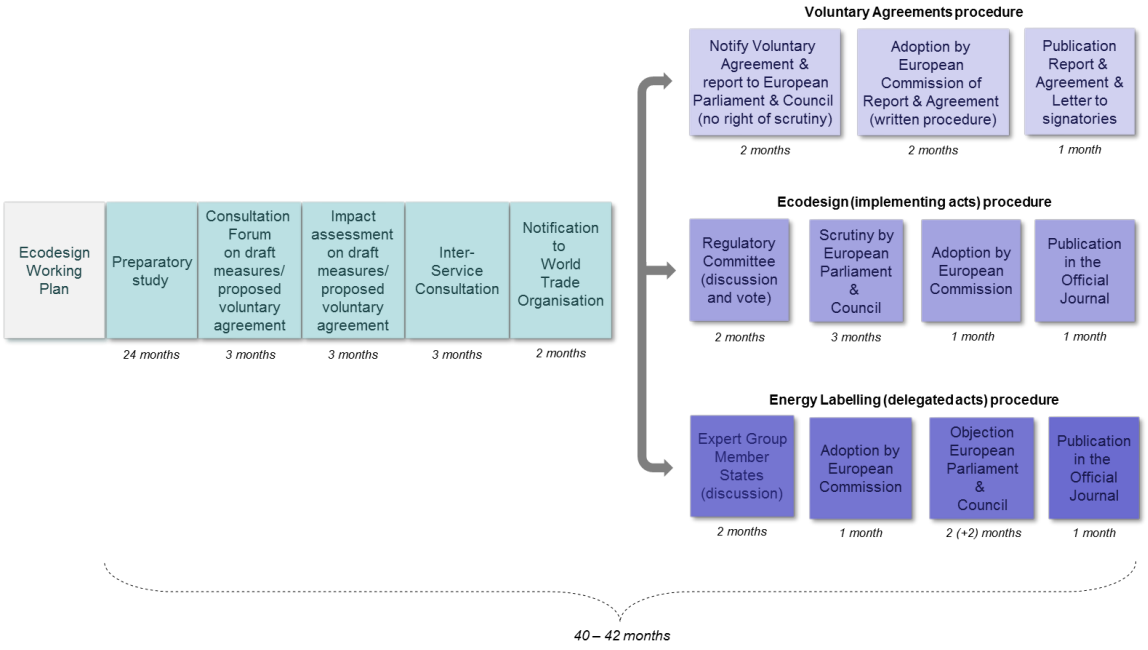


Figure A11-1. Overview of the three processes

Ecodesign and Energy labelling are complementary. Ecodesign implementing measures lay down minimum energy efficiency requirements for a product group and therefore ban worse performing products. In contrast, energy labelling provides objective comparable information to enable consumers to choose better performing products through mandatory rating. In view of the complementarity, energy labelling delegated acts are usually adopted in parallel with Ecodesign implementing acts to ensure coherence between the measures.

The process

The process starts with establishing the priorities for Union action in this area. Priority product groups are selected based on their potential for cost-effective reduction of energy use and greenhouse gas emissions and follows a transparent process culminating in a working plan that outlines the priorities for the development of implementing measures for the next years. Working plans are adopted by the Commission after consultation of the Ecodesign and Energy labelling Consultation Forum. The forum brings together representatives of Member States and relevant stakeholders.

The Ecodesign Working Plan 2016-2019 identified Local Space Heaters as a priority. A new Ecodesign and Energy labelling Working plan 2022-24 has now been adopted and retains this product.

Once the product group has been selected, a preparatory study is undertaken by an independent consultant, also involving extensive technical discussions with interested stakeholders. The preparatory study follows the standard Methodology for the Ecodesign of Energy Related Products (MEErP). Subsequently, the Commission's first drafts of Ecodesign and energy labelling measures are submitted for discussion to the Consultation Forum.

After the Consultation Forum, the Commission drafts an impact assessment, which after RSB approval accompanies the draft implementing measures in the Commission's inter-service consultation. In this and subsequent steps, the Parliament's functional mailboxes for delegated/implementing acts are copied on each message from the Commission services. After the inter-service consultation, stakeholders are alerted when the draft measures are published in the WTO notification database.

After the WTO notification phase is completed, the two procedures follow different paths. The draft energy labelling delegated act is discussed in a Member States Expert Group where opinion(s) are expressed and consensus is sought but no vote is taken. The draft Ecodesign measure is submitted for vote to the Regulatory Committee.

The European Parliament and Council have the right of scrutiny for a period of up to four months. Within this time, the co-legislators can block the adoption process by the Commission. Parliament committees sometimes discuss draft objections to measures (light bulbs and fridges in 2009) or vote to reject a measure (vacuum cleaners in 2013[80]). On one occasion, an objection was adopted in plenary, blocking the measure for televisions in 2009[81].

Today, 29 Ecodesign Regulations, 14 Energy Labelling Regulations, 3 voluntary agreements and 2 tyre labelling regulations have been implemented. An overview of the measures in force in March 2022 is available on the Europa website.

Market Surveillance

The Market Surveillance Regulation[82] establishes the rules on Union market surveillance and control of products entering the Union market. Market Surveillance Authorities (MSAs), designated by Member States, will verify the conformity of the products with the requirements laid down in the implementing measures and delegated acts. Checks can be done on the product itself or by verifying the technical documentation.

Revision of the Ecodesign framework

In March 2022 the Commission has proposed to amend the Ecodesign legislation by replacing it with a Regulation. The main effects of the proposed changes would be to

[80] This objection was defeated in ENVI committee by 43 votes against and 4 in favour.

[81] The motivation of the objection was that the EP wanted to delay the discussion of the draft labelling measure so that it would have to become a delegated act under the recast post-Lisbon Energy Labelling Directive in 2010. The measure was subsequently adopted as a delegated act.

[82] Regulation (EC) No 765/2008 of the European Parliament and of the Council of 9 July 2008 setting out the requirements for accreditation and market surveillance relating to the marketing of products and repealing Regulation (EEC) No 339/93 OJ L 218, 13.8.2008, p. 30

extend the type of products within the scope of the legislation, new potential requirements such as the possibility to introduce requirements that would apply to multiple product groups, Digital Product Passports, requirements for green public procurement and requirements relating to the destruction of unsold goods.

Ecodesign product regulations adopted under the existing Directive would continue in force until repealed. Under the proposed new Regulation, product specific measures would continue to be carried out under secondary legislation. This proposal is now undergoing the normal legislative procedure. When the legislation is eventually adopted the future amendments to existing product regulations would be carried out under its framework.